RESIDUE: Proofs, counterexamples, and unresolved obligations in the eighth package on *Les Dérivateurs*

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This text claims no priority. A status of *Open* means that we have not settled the exact stated obligation in this work; it is not a claim that the problem is open in the mathematical literature. Several principal obligations remain Open, including the adjoint-transport bifibration over the sphere monoid, the general all-vertical path comparison, and the raw-space realization over the real line. The results register and report retain their exact scope. A finite computation certifies only its explicitly recorded domain.

Conventions. We fix a universe and let Cat denote its small categories. We write $BA = |NA|$, $W_\infty = \{u : Bu \text{ is a weak homotopy equivalence}\}$, and $\mathrm{Hot} = \mathrm{Cat}[W_\infty^{-1}]$. For $u : A \to B$, the objects of A/b are $(a, u(a) \to b)$ and those of $b\backslash A$ are $(a, b \to u(a))$. A final functor in the covariant convention has every $b\backslash A$ nonempty and connected. A sieve is closed under sources; a cosieve is closed under targets. Coefficients on A are presheaves on A, with indexing category A^{op}.

Unless explicitly discussing the absolute version in Part B, a fundamental localizer satisfies the following exact relative package: it contains isomorphisms and has two-out-of-three; if $f : X \to Y$ and $g : Y \to X$ have $gf, fg \in W$, then $f, g \in W$; the projection $A \to *$ belongs to W when A has a terminal object; and, for every triangle $A \xrightarrow{u} B \to C$, the condition $u/c : A/c \to B/c \in W$ for every c implies $u \in W$. Part B distinguishes this package from the printed absolute axiom. A category is W-aspherical when its projection to $*$ lies in W. We use Fib_W for maps such that, after every strict base change, pullback preserves W. All stated equivalences of localizations specify whether ordinary or infinity categorical localization is meant. Manuscript references use chapter and manuscript page, with the supplied transcription treated as fallible.

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A Strict bifibrations and equivalence transports

A.i The two-type obstruction

A.1 [Proved]. Equivalence transports have no higher connecting maps.

Let B and C be small categories in the fixed universe. Let $p : E \to B$ be a Grothendieck opfibration whose fibre categories are equivalent to C and whose transport functors are equivalences of categories. For every basepoint b and every object x of its fibre, the connecting homomorphisms

$$\partial : \pi_{r+1}(BB, b) \longrightarrow \pi_r(BC, x)$$

of its classifying-space homotopy-fibre sequence are zero for $r \geq 2$. Consequently, if BE is weakly contractible and B is nonempty and connected, then BC is a 1-type and BB is a 2-type. In particular there is no such opfibration with weakly contractible total category over the five-element rectangular-band monoid of the question, whose classifying space is S^2.

Proof. Let $\mathrm{Eq}(C)$ be the monoidal groupoid of equivalences $C \to C$ and natural isomorphisms. Its monoidal nerve is a group-like space. Its delooping $B\,\mathrm{Eq}(C)$ is a 2-type: its loop space is the nerve of a groupoid and therefore a 1-type. A cleavage and fibre identifications give a pseudofunctor from B to the one-object bicategory with endomorphism groupoid $\mathrm{Eq}(C)$. The associativity and unit isomorphisms are precisely the coherence needed to define its classifying map $BB \to B\,\mathrm{Eq}(C)$. Applying classifying spaces to equivalences and natural isomorphisms gives a map

$$B\,\mathrm{Eq}(C) \longrightarrow B\,\mathrm{hAut}(BC).$$

The local system underlying $BE \to BB$ is the pullback of the universal family along the composite. This follows directly from the Grothendieck homotopy-colimit description: both families have fibre BC, the prescribed transports, and the same coherent transition maps. Naturality of the homotopy-fibre sequence factors the displayed connecting map through $\pi_{r+1}(B\,\mathrm{Eq}(C))$, which vanishes for $r + 1 \geq 3$.

When BE is weakly contractible, its long exact sequence makes every displayed connecting map an isomorphism for $r \geq 2$. Hence both its source and target vanish. This proves the two truncation assertions. For $BB \simeq S^2$, $\pi_3(BB) \cong \mathbb{Z}$, a contradiction. Empty fibres are irrelevant to a contractible nonempty total space over a connected nonempty base: equivalence transport and zigzags in the base make either all fibres empty or all fibres nonempty. For a disconnected base the argument applies separately on each component with a nonempty fibre. □

A.2 [Proved]. All steps of the central-cocycle argument in Lead 2.

Let $M = \{1, e_{00}, e_{01}, e_{10}, e_{11}\}$ with $e_{ij}e_{kl} = e_{il}$. If the transport functors of a Grothendieck opfibration $E \to \mathbf{B}M$ are categorical equivalences, there is a normalized central cocycle $c : M \times M \to A$, where $A = \mathrm{Aut}(1_C)$, and an isomorphism over $\mathbf{B}M$ from E to the category having objects those of C and arrows $(m, f) : x \to y$, $f : x \to y$ in C, with composition twisted by c. The action in Lead 2 is a degreewise free action of the simplicial group $N(\mathbf{B}A)$; with that precise meaning the Borel argument and its conclusion are valid.

Proof. Choose identity cocartesian lifts over 1. Each nonidentity m is idempotent, so its coherence is an isomorphism $T_mT_m \cong T_m$. Since T_m is an equivalence, cancellation gives a natural isomorphism $T_m \cong 1_C$. Choose such isomorphisms and transfer the pseudofunctor structure to the identity functors. A natural automorphism of 1_C commutes with every arrow of

C, and two such automorphisms commute componentwise by naturality. Thus A is abelian. In additive notation the transferred associativity equation is

$$c(k,m) + c(km,n) = c(m,n) + c(k,mn), \qquad c(1,m) = c(m,1) = 0.$$

Every arrow of E over m factors uniquely through the chosen cocartesian arrow; the chosen isomorphism $T_m x \cong x$ then identifies it uniquely with an arrow $f : x \to y$ of C. This is an identity-on-objects bijection on each hom-set. With the convention that the later arrow multiplies on the left, its composition law is

$$(n,g) \circ (m,f) = (nm, gf\, c(n,m)_x),$$

where the central automorphism is written multiplicatively when acting on x. It is therefore a categorical isomorphism, not merely an equivalence, after the choices have been made.

Let $M_c = M \times A$ with multiplication $(m,a)(n,b) = (mn, a + b + c(m,n))$, using the same composition-order convention throughout. The functor $C \times \mathbf{B}M_c \to E$ sends $(f,(m,a))$ to (m, fa). The simplicial group $G = N(\mathbf{B}A)$ acts on $N(C \times \mathbf{B}M_c)$: in degree q, the element $(b_1, \ldots, b_q) \in A^q$ multiplies the q successive C-arrows by b_i and subtracts b_i from their M_c-coordinates. Centrality makes this commute with composition faces; normalization makes it commute with degeneracies. The action is free because the A-coordinates in M_c recover every b_i. Its orbit simplicial set is exactly NE, degree by degree.

To justify the homotopy quotient without an unstated local-triviality hypothesis, form the bisimplicial bar construction for this action. In every fixed simplicial degree its action groupoid is a free action groupoid, and its nerve maps by a weak equivalence to the discrete orbit set. The diagonal weak-equivalence theorem for bisimplicial sets gives

$$BE \simeq (BC \times BM_c) \mathbin{//} BA.$$

The homotopy-quotient fibre sequence is $BC \times BM_c \to BE \to B^2A$. If BE is contractible, its fibre is weakly equivalent to $\Omega B^2 A = BA$. Thus $\pi_r(BC \times BM_c) = 0$ for $r \geq 2$, and projection onto the first factor makes $\pi_r(BC) = 0$. This verifies the claimed implication. An ordinary discrete A-action would be the wrong statement; the simplicial group $N(\mathbf{B}A)$ is essential. □

A.3 [Proved]. Exact classification for equivalence transports.

Fix a connected nonempty small category B and a locally constant space-valued object with fibre homotopy type X. It has a bifibration representative whose transports are equivalences of categories if and only if there are a small category C, an equivalence $BC \simeq X$ of spaces, and a homotopy lift of its classifying map

$$BB \longrightarrow B\,\mathrm{hAut}(X)$$

through $B\,\mathrm{Eq}(C) \to B\,\mathrm{hAut}(BC) \simeq B\,\mathrm{hAut}(X)$. For disconnected B, this condition is applied independently to every component, allowing a different C on different components.

Proof. Necessity is the classifying-map construction of A.1, with the chosen fibre comparison retained. Conversely the nerve of the one-object bigroupoid $\mathrm{Eq}(C)$ is a Kan 2-type. A homotopy class of maps from NB into it is represented by an actual simplicial map, because NB is cofibrant and the target is Kan. Such a map is precisely a normal pseudofunctor: its edges are equivalences $C \to C$, its triangles are their invertible composition constraints, and its tetrahedra assert the coherence equation. The higher simplices are determined by these data. Its Grothendieck construction is an opfibration. Every equivalence has a right adjoint, so the same projection is a fibration. The given homotopy between classifying maps supplies the required equivalence with the specified local system. For $X = \varnothing$, take $C = \varnothing$; the same argument applies. The condition is a lift through the indicated categorical automorphism 2-group, not mere factorization through some unspecified 2-type. □

A.ii Splitting the rectangular-band action

A.4 [Proved]. The split data and their curvature class.

Up to equivalence of coherent pseudo-actions, an action of M on a small category C is specified by a small category D, functors $A_i : D \to C$, $B_j : C \to D$, and natural isomorphisms $\beta_{ji} : B_j A_i \to 1_D$, for $i, j \in \{0, 1\}$. The corresponding nonunit functors are $T_{ij} = A_i B_j$ and their multiplication constraints are

$$(A_i B_j)(A_k B_l) \xrightarrow{A_i \beta_{jk} B_l} A_i B_l.$$

There is no additional equation on the four β_{ji}: the pseudo-action associativity equation follows by interchange for two disjoint contractions. Equality $T_{ij} = A_i B_j$ refers to this equivalent model, not to the original unmodified functors.

Proof. We give the splitting used in the comparison. Write $P = T_{00}$ and $\mu : P^2 \to P$ for its multiplication isomorphism. Associativity and cancellation of μ imply $P\mu = \mu P$. Let D have objects (x, θ) with $\theta : Px \to x$ an isomorphism satisfying $P\theta = \mu_x$, and arrows $h : (x, \theta) \to (y, \psi)$ satisfying $h\theta = \psi P h$. Let $A : D \to C$ forget θ, and let $B : C \to D$ send x to (Px, μ_x). Then $AB = P$ strictly. The arrows θ define an isomorphism $\epsilon : BA \to 1_D$, and $A\epsilon B = \mu$. Every category and functor just defined is small.

Equivalently, this extends the action to the two-object subcategory of the idempotent completion of $\mathbf{B}M$ on 1 and e_{00}. Its arrows from e_{00} to 1 are e_{i0}, and those from 1 to e_{00} are e_{0j}. Their composites satisfy $e_{0j} e_{i0} = e_{00}$, the identity of the new object. Explicit choices are $A_i = T_{i0} A$ and $B_j = B T_{0j}$, with the isomorphisms to T_{ij} and to 1_D obtained from the original multiplication constraints and ϵ. All their coherence diagrams are images of the original associativity diagrams together with $A\epsilon B = \mu$. Conversely the displayed formula defines a normal pseudo-action from the split data: on $A_i B_j A_k B_l A_r B_s$, contracting $B_j A_k$ and $B_l A_r$ in either order gives the same natural isomorphism by naturality and interchange. The split construction and this reconstruction are inverse up to the specified coherent equivalences; a different splitting is related by the functors induced from $B'A$ and BA' and their counit isomorphisms. □

For these data, the exact adjoint condition is representability of

$$x \longmapsto \mathrm{Hom}_C(A_i B_j x, y) \qquad (C^{\mathrm{op}} \to \mathbf{Set})$$

for every i, j and every $y \in C$. Representing objects $G_{ij} y$ assemble uniquely to right adjoints by Yoneda. If these right adjoints exist, $B(A_i B_j)$ is a homotopy equivalence by its adjunction. Since $B(B_j A_i) \simeq 1$, each BA_i and BB_j is a homotopy equivalence. Put $X = BD$ and write a_i, b_j for these space maps.

The four isomorphisms determine the following precise clutching element. Choose a homotopy $\eta : a_0 b_0 \simeq 1_{BC}$ completing (a_0, b_0, β_{00}) to a homotopy equivalence; choose it with its triangle coherence. Define the path of equivalences

$$u_1 : a_1 \xrightarrow{\eta^{-1} a_1} a_0 b_0 a_1 \xrightarrow{a_0 \beta_{01}} a_0.$$

Then, in traversal order, the loop

$$1_X \xrightarrow{\beta_{10}^{-1}} b_1 a_0 \xrightarrow{(b_1 u_1)^{-1}} b_1 a_1 \xrightarrow{\beta_{11}} 1_X \tag{A.4.1}$$

represents the clutching element in $\pi_1 \,\mathrm{hAut}(X)$, with the sphere orientation fixed by this formula. Indeed the two-object category above has four elementary relations $B_j A_i = 1_D$. Contracting its

A_0 edge and using the first three relations as the reference trivializations leaves precisely the fourth relation as the boundary loop (A.4.1). The corresponding normalized bar 2-cycle in BM is

$$[e_{00}|e_{00}] - [e_{00}|e_{10}] - [e_{01}|e_{00}] + [e_{01}|e_{10}].$$

In the special case where all four functors are identities, evaluation is $\beta_{00} - \beta_{01} - \beta_{10} + \beta_{11}$. This fixes the sign. Changing the fibre framing conjugates the clutching element by the chosen equivalence of X; changing the sphere orientation inverts it.

The exact contractibility condition is also expressible in this language. Let $\chi : S^2 \to B\,\mathrm{hAut}(X)$ be the class represented by (A.4.1), and choose $x \in X$. The orbit map

$$\Omega S^2 \xrightarrow{\Omega\chi} \mathrm{hAut}(X) \xrightarrow{\mathrm{ev}_x} X \tag{A.4.2}$$

must be a weak equivalence. This condition is necessary and sufficient: choosing a point in the associated total space gives a map from the based path fibration over S^2 into that family, whose fibre map is (A.4.2); the fibrewise equivalence criterion proves both directions. The path local system has $X = \Omega S^2$ and clutching loop given by left translation along a generator of $\pi_1 \Omega S^2$. Merely requiring evaluation of (A.4.1) to generate $\pi_1 X$ is weaker than (A.4.2) and is not asserted sufficient.

A.iii The complete monoid enumeration through order five

A.5 [Proved]. The unique non-1-type monoid of order at most five.

Up to monoid isomorphism, the rectangular-band monoid M is the only monoid with at most five elements whose classifying space is not a 1-type. The numbers of isomorphism classes in orders $1, 2, 3, 4, 5$ are respectively $1, 2, 7, 35, 228$; the complete multiplication tables and reduction certificates accompany the paper.

Proof. We first justify the reductions used by the computation. If e is an idempotent in a monoid L and $eLe = eL$ or $eLe = Le$, then $BL \simeq B(eLe)$. Form the two-object category K with objects $1, e$ and arrows $K(x, y) = yLx$. The inclusion of the object 1 is homotopy initial: its comma category over 1 has a terminal object, and the category over e has objects $m \in eL$ and arrows $a : m \to m'$ with $m'a = m$. The components $m : m \to e$ give a natural transformation from its identity functor to the constant functor at e, so that comma category is contractible. If $eLe = Le$, the inclusion of the object e is homotopy initial: in its comma category over 1, the object e is terminal, because every $m \in Le$ belongs to eLe and factors uniquely through e. If $eLe = eL$, use instead the outgoing comma category, in which e is initial. Quillen's Theorem A, and its opposite, give the claimed zigzag through BK. For $e \neq 1$, the corner has strictly fewer elements, since $1 \in eLe$ would imply $e = 1$.

The program fixes the identity as 0 and enumerates every multiplication table. At each node it assigns an unknown entry and repeatedly imposes all currently determined associativity equalities $(ab)c = a(bc)$; a conflict rejects that branch. No value is discarded except by one of these equalities. Hence every associative table occurs. The terminal tables are checked again against all n^3 equations and quotiented by all permutations fixing 0. The labeled counts are $1, 2, 11, 156, 4122$. The independent verification program then records every permitted corner reduction and checks its target against the same table catalogue. There remain exactly the trivial group, $C_2, C_3, C_4, C_2 \times C_2, C_5$, and M. Of the 273 total isomorphism classes, precisely one reduces to M, namely M itself. All other types reduce to groups and are 1-types.

For completeness, $\pi_1(BM)$ is trivial because every nonunit generator is idempotent. Put $M_i = \{e_{i0}, e_{i1}\}$. The right $\mathbb{Z}[M]$-modules $\mathbb{Z}[M_i]$ are projective idempotent images. The exact resolution

$$0 \to \mathbb{Z}[M_0] \oplus \mathbb{Z}[M_1] \to \mathbb{Z}[M] \xrightarrow{e_{00} - e_{01}} \mathbb{Z}[M_0] \to \mathbb{Z} \to 0$$

has the first map the two inclusions and the last the augmentation. The middle map sends 1 to $e_{00} - e_{01}$ and all nonunits to zero, proving exactness directly. Tensoring with the trivial left module gives $\mathbb{Z}^2 \xrightarrow{(1,1)} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$. Thus BM has the integral homology of S^2; simple connectivity and the homological Whitehead theorem give $BM \simeq S^2$. This is also the explicit unnumbered lemma in Fiedorowicz (2002, pp. 33–34), whose primary text was opened and checked. □

A.6 [Known]. Every local system over a 1-type base has a strict bifibration model.

If BB is a 1-type, every small space-valued locally constant object over B has a small strict bifibration representative with categorical-isomorphism transports. This includes empty values and disconnected bases.

Proof. Let $q : B \to \Pi_1 B$ be ordinary groupoid completion. The specified map Bq is a weak equivalence under the hypothesis, so restriction of local systems along it is an equivalence. On each component of $\Pi_1 B$ choose a group model G and a Kan fibration of small simplicial sets $Y \to NG$ representing the resulting family. Pull back along the standard contractible free G-simplicial set $EG \to NG$. The resulting simplicial set $\widetilde{Y}$ has a strict G-action, is weakly equivalent to the fibre, and its Borel family recovers $Y \to NG$. The simplex category $\Delta/\widetilde{Y}$ inherits a strict action by categorical isomorphisms. Its classifying space is naturally weakly equivalent to $\widetilde{Y}$. Integrating this action and pulling back along q gives the required bifibration and its specified diagramwise comparison. A small family of small simplicial sets gives small simplex categories in the fixed universe. Empty fibres yield empty simplex categories. This is the result of STRATA E.9, here proved using simplex categories in place of contractible-patch categories. □

A.7 [Proved]. Finite acyclic fibre categories cannot solve the sphere test.

Here a finite acyclic category means a finite category with no nonidentity endomorphisms and no directed cycles of nonidentity arrows. A bifibration over M with weakly contractible total category cannot have such a category as fibre; in particular its fibre cannot be a finite poset.

Proof. Adjoint transport functors are homotopy equivalences, so the Grothendieck homotopy-fibre theorem identifies the fibre with ΩS^2. The homology Serre spectral sequence of $\Omega S^2 \to PS^2 \to S^2$ has only columns 0 and 2 and contractible total space. Its d_2 therefore identifies $H_q(\Omega S^2; \mathbb{Z})$ with $H_{q+1}(\Omega S^2; \mathbb{Z})$ for every $q \geq 0$. Since ΩS^2 is connected, these groups are all $\mathbb{Z}$. A finite acyclic category has a finite-dimensional nerve: a nondegenerate simplex cannot revisit an object. Its integral homology vanishes above its number of objects minus one, a contradiction. □

A.8 [Open]. The remaining strict-adjoint construction and the requested action search.

We do not construct, or exclude, a bifibration with weakly contractible total category over M when its transports have right adjoints but are not equivalences. The same question for the seven-element monoid of CATHOT G.10 remains open for such transports; A.1 excludes equivalence transports there because $\pi_3(S^3) \neq 0$. The general-base existence question and the exhaustive pseudo-action enumeration on categories with at most five objects and twelve arrows remain unresolved here. In the proposed filtered-colimit enlargement of the fence category, a definition of the completion together with endpoint-compatible universal arrows and a fibrewise weak-equivalence comparison is still missing. Closing under all presheaves would produce contractible fibre categories and would therefore lose the required fibre ΩS^2; the smaller filtered completion is not proved to acquire the adjoints. For categories of reduced words, no specified composition and adjoint pair satisfying all rectangular-band relations is constructed here. For a bar or bisimplicial resolution,

its contraction does not by itself give both categorical lifting properties of the endpoint projection; no shift construction with both universal properties is established. The four Beck–Chevalley squares have not been proved invertible from the adjunction data. Accordingly none of these routes is recorded as a negative existence theorem: each still lacks its decisive comparison or adjunction step. A.5 completes only the requested enumeration of base monoids of order at most five. No finite pseudo-action table is asserted to be exhaustive.

B Absolute and relative Theorem A

B.1 [Proved]. The component-dependent example is valid.

For all small categories put

$$u \in W^{\dagger} \quad \Longleftrightarrow \quad \pi_0(u) \text{ is bijective and } (|\pi_0(\operatorname{dom} u)| = 1 \Longrightarrow u \in W_{\infty}).$$

This class satisfies the literal Loc (1), Loc (2), and absolute Loc (3), but not relative Loc (3 bis). It is not closed under arrow retracts. Its strong saturation is W_0.

Proof. Isomorphisms belong to the class. If two of f, g, gf belong, their three component sets are in bijection. If their common cardinality is one, use two-out-of-three for weak homotopy equivalences; otherwise the component test is the entire test. This includes cardinality zero and arbitrary infinite cardinalities. If gf and fg belong, both component maps are bijections. In the connected case their images in the homotopy category of spaces have invertible composites in both orders, so each factor is invertible. This proves the moderate saturation clause.

The $W^{\dagger}$-aspherical categories are exactly the weakly contractible categories. Consequently the terminal-object axiom and absolute Theorem A follow from contraction of a category with terminal object and Quillen's Theorem A, respectively (Quillen, 1973, §1, Theorem A, p. 93).

For the relative counterexample let $G = \mathbb{Z}/2$, let E be the chaotic groupoid on the object set G, and send its unique arrow $x \to y$ to $y - x \in G$ to obtain $p : E \to BG$. The triangle is $u : E \times BG \to E \xrightarrow{p} BG$. Both total categories are connected; their fundamental groups are G and 1, so $u \notin W^{\dagger}$. An object of $E/*$ is $(x, t) \in G^2$. An arrow from (x, t) to (y, t') exists precisely when $t' + y - x = t$. Hence $t + x$ labels two chaotic connected components. Moreover $(E \times BG)/* = (E/*) \times BG$. The comma comparison is a bijection on its two components and therefore belongs to $W^{\dagger}$. This verifies every premise of the relative axiom and refutes its conclusion.

The arrow $BG \to e$, which is not weak, is an arrow retract of $(BG \amalg e) \to (e \amalg e)$. The latter belongs to $W^{\dagger}$; the sections select the first components, while the retractions send the added point to the unique object on either side. This proves failure of retract closure. More generally, if $f : A \to B$ is bijective on components and $A \neq \varnothing$, choose an object $a \in A$. Then f is an arrow retract of $f \amalg 1_e$, using the constant maps at a and $f(a)$ on the added components. The latter belongs to $W^{\dagger}$. The empty case is an isomorphism. Thus the strong saturation contains W_0. Conversely $W^{\dagger} \subset W_0$, and W_0 is strongly saturated as the inverse image of isomorphisms under π_0. This proves equality. □

B.2 [Proved]. An elementary quotient of the homotopy category.

Let $\mathscr{K}$ be the homotopy category of CW homotopy types, and fix $K = S^1$. Let $I(X, Y)$ be the set of homotopy classes $X \to Y$ factoring through K. There is a category $\mathscr{Q}$ with the same objects and

$$\mathscr{Q}(X, Y) = (\mathscr{K}(X, Y) \amalg \{0\})/(I(X, Y) \sim 0).$$

For the quotient functor $q : \mathscr{K} \to \mathscr{Q}$, the morphism $q(f)$ is invertible exactly when f is invertible in $\mathscr{K}$, or both its source and target are homotopy retracts of S^1. Those retracts have homotopy type e or S^1.

Proof. The family I is a two-sided ideal: precomposition or postcomposition preserves a factorization through K. Declare the adjoined zeros absorbing. Composition therefore descends, is associative, and preserves the quotient identities. We adjoin a zero even when the original hom-set is empty. In particular the empty space causes no exception.

The equation $q(1_X) = 0$ holds precisely when 1_X factors through K, which is exactly the assertion that X is a homotopy retract of K. Such an object is a zero object of $\mathscr{Q}$: every morphism to or from it factors through K, and hence every corresponding hom-set is a singleton. Conversely, suppose $q(f) : X \to Y$ is invertible and neither identity is zero. An inverse cannot be the zero morphism. Represent it by $g : Y \to X$ in $\mathscr{K}$. Every nonzero equivalence class in the quotient is a singleton, so $q(gf) = q(1_X)$ and $q(fg) = q(1_Y)$ imply the actual equalities $gf = 1_X$ and $fg = 1_Y$ in $\mathscr{K}$. If exactly one identity is zero, invertibility is impossible; if both are zero it is automatic. These observations prove the criterion.

If X is a homotopy retract of S^1, it is nonempty and connected, its higher homotopy groups vanish, and $\pi_1 X$ is a retract of $\mathbb{Z}$. An idempotent endomorphism of $\mathbb{Z}$ is multiplication by 0 or 1. Thus $\pi_1 X$ is 0 or $\mathbb{Z}$. The CW Whitehead theorem identifies X with e or S^1. Both types are indeed retracts. The empty space is not one, since there is no map $S^1 \to \varnothing$. □

B.3 [Refuted]. Retracts and strong saturation do not repair the absolute axiom.

Neither Loc (1 bis), Loc (2), Loc (3) nor Loc (1 ter), Loc (2), Loc (3) implies Loc (3 bis). A counterexample on all small categories is

$$W_{\mathrm{circ}} = W_\infty \ \cup \ \{u : A \to B \mid BA \simeq S^1 \simeq BB,\ H_1(Bu; \mathbb{Q}) \text{ is invertible}\}.$$

In the second alternative, the induced integer degree is any nonzero integer. The W_{circ}-aspherical categories are exactly the weakly contractible categories.

Proof. Consider the ordinary functor

$$\Phi : \mathscr{K} \longrightarrow \mathscr{Q} \times \mathrm{Vect}_{\mathbb{Q}}, \qquad X \longmapsto (qX, H_1(X; \mathbb{Q})).$$

By B.2, the inverse image of isomorphisms consists of homotopy equivalences, together with maps between retracts of S^1 which are isomorphisms on rational first homology. Among those retracts, a map from the contractible type to the circle type, or conversely, fails the latter test; between contractible types it is already a homotopy equivalence; between circles the test is exactly nonzero degree. Classifying spaces of categories are CW complexes, so the inverse image of isomorphisms under ΦB is exactly the displayed class.

An inverse image of isomorphisms under any functor is strongly saturated: the functor factors through localization at that inverse image, and any arrow inverted by the localization must consequently have invertible image. It is therefore closed under retracts and satisfies all clauses of Loc (1 ter), including the literal moderate saturation. A single universe enlargement suffices for the homotopy category and its quotient; the formula defining W_{circ} itself quantifies over the original small categories.

Since the target of $A \to e$ is not circle type, this arrow belongs to W_{circ} exactly when BA is weakly contractible. Loc (2) follows. If every A/b is W_{circ}-aspherical, Quillen's Theorem A makes $A \to B$ a weak homotopy equivalence, hence a member of W_{circ}. This verifies absolute Loc (3) on all small categories. The finite relative counterexample is B.4. □

B.4 [Proved]. A finite certificate for the failure of relative Theorem A.

The class in B.3 violates relative Theorem A on a functor from a ten-object poset to a six-object poset, over a three-object poset. The failure is detected integrally by multiplication by 2 *on* H_2.

Proof. For even $m \geq 4$ let C_m have objects $0, \dots, m-1$, with the nonidentity comparisons

$$2i < 2i-1, \qquad 2i < 2i+1 \pmod{m}.$$

There are no composable nonidentity arrows. Its nerve is the polygon with m edges. The order map $v : C_8 \to C_4$, $i \mapsto i \bmod 4$, wraps this polygon twice around the target, and belongs to W_{circ}. We can check its degree without choosing a topological parametrization: the cycle

$$z_m = \sum_{i \text{ even}} \left([i, i+1] - [i, i-1]\right)$$

generates $H_1(NC_m; \mathbb{Z})$, and $v_{\#} z_8 = 2z_4$.

Let $\widehat{C}_m$ be obtained by adjoining two incomparable objects t_+ and t_- above every object of C_m. Its nerve is the union of two simplicial cones on NC_m, hence is a triangulated S^2. Let $S = \{o < +, o < -\}$ and map C_m to o and $t_\pm$ to $\pm$. The order map $\widehat{v} : \widehat{C}_8 \to \widehat{C}_4$ extends v and preserves the two added objects. At o, its overcomma comparison is v. At each of $+$ and $-$, the overcomma category is the corresponding single cone, with terminal vertex $t_\pm$. Those two comparisons are weak homotopy equivalences. Thus every relative premise holds.

The top cycle obtained by coning z_m to t_+ and subtracting its cone to t_- is primitive and generates $H_2(N\widehat{C}_m; \mathbb{Z})$. The induced map sends it to twice the target generator. Thus $\widehat{v}$ is not a weak equivalence, and its endpoints are sphere types rather than circle types; it is not in W_{circ}.

The independent program `code/BC/verify_counterexamples.py` constructs these orders and chain maps, checks $d^2 = 0$, computes all boundary Smith normal forms through degree four, verifies the two displayed degree equations, and computes fundamental-group presentations from maximal trees and triangle relations. Its Tietze eliminations leave one free generator for each crown and none for either suspension or cone. The output contains the relations and each elimination. The simplicial description already proves the asserted homotopy types; the program is an independent exact certificate, not a homology-only inference of them. □

B.5 [Known]. Literal homotopy invariance and Rappel 1.2.

Under literal Loc (1), Loc (2), and absolute Loc (3), naturally homotopic functors have the same image in Hot_W *and either both or neither belong to* W. *Every category-homotopy equivalence belongs to* W. *In particular both functors in an adjunction belong to* W, *and a category with initial object is* W*-aspherical.*

Proof. For every A, the overcomma categories of $p : A \times [1] \to A$ are $(A/a) \times [1]$ and have terminal objects. Absolute Theorem A therefore gives $p \in W$. Both endpoint sections i_0, i_1 lie in W by two-out-of-three, and their images in the localization equal the same inverse of p. A natural transformation $f \Rightarrow g$ is a functor $H : A \times [1] \to B$ with $Hi_0 = f$, $Hi_1 = g$. These identities prove both assertions about f, g; a zigzag of transformations gives the same conclusion. If f, g are category-homotopy inverse functors, gf and fg belong to W by comparison with the identities. The moderate clause gives $f, g \in W$. An adjunction supplies the required homotopies by its unit and counit. For an initial object the natural transformation from its constant functor to the identity contracts the category. This is the elementary argument underlying the homotopy results in Maltsiniotis (2020, §1.1); no relative Theorem A is used. □

B.6 [Known]. Rappel 1 holds for the literal axioms.

Literal Loc (1), Loc (2), Loc (3) imply $W = W^\circ$.

Proof. Let $\mathrm{Tw}(A)$ have objects arrows $f : x \to y$ and morphisms $f \to f'$ pairs $u : x' \to x$, $v : y \to y'$ with $f' = vfu$. Its source and target projections are

$$A^{\mathrm{op}} \xleftarrow{s_A} \mathrm{Tw}(A) \xrightarrow{t_A} A.$$

The category t_A/a consists of composable pairs $x \xrightarrow{f} y \xrightarrow{t} a$. Composition gives a functor to $(A/a)^{\mathrm{op}}$; its section sends $h : x \to a$ to $(h, 1_a)$. The arrows $(1_x, t)$ give a natural transformation from the identity to section followed by composition. Hence this is a category-homotopy equivalence. The target has initial object 1_a, so B.5 and two-out-of-three make t_A/a W-aspherical.

Similarly, an object of s_A/a, with the latter comma taken in A^{op}, is $a \xrightarrow{h} x \xrightarrow{f} y$. Composition maps it to $a\backslash A$, with section $a \xrightarrow{1_a} a \to y$. The arrows $(h, 1_y)$ again give a natural transformation from the identity to section followed by composition. The category $a\backslash A$ has initial object. Absolute Theorem A gives $s_A, t_A \in W$. The construction is functorial in A. For $u : A \to B$, the squares with s, t and $\mathrm{Tw}(u)$ commute. Two applications of two-out-of-three therefore give $u \in W \iff \mathrm{Tw}(u) \in W \iff u^{\mathrm{op}} \in W$. This is the twisted-arrow proof of Maltsiniotis (2020, Proposition 1.1.21 in the March 2020 version); the same construction occurs in Quillen (1973, §1, p. 94). □

B.7 [Proved]. The two equivalences in Proposition 1.3.

Both equivalences in XVI Proposition 1.3 hold when its category $\mathcal{M}$ is read as Cat*. The weak equivalence uses absolute Theorem A; the strong equivalence uses relative Theorem A.*

Proof. For a Grothendieck fibration $p : X \to S$, inclusion of the strict fibre X_s in $s\backslash X$ has a right adjoint, given by cartesian transport. B.5 makes that inclusion weak. The equivalence of absolute Theorem A with its opposite form follows from B.6. Consequently aspherical fibres imply $p \in W$.

The split replacement $\Phi_S(X)$ has objects $(s, x, \alpha : s \to p(x))$. A morphism is a pair $(h : s \to t, k : x \to y)$ with $\beta h = p(k)\alpha$. It is the split Grothendieck fibration of $s \mapsto s\backslash X$. The comparison $i_X(x) = (p(x), x, 1)$ has a left adjoint $(s, x, \alpha) \mapsto x$, and therefore belongs to W. This proves the forward weak implication. For a map between two fibrations, the fibre inclusions give a commutative square comparing fibre maps with the maps on $s\backslash X$. Two-out-of-three and the opposite form of relative Theorem A prove the forward strong implication. No preservation of chosen cartesian arrows is needed for this last square.

Conversely, assume the segment axiom and the split-fibration clauses. The projection $A \times [1] \to A$ is a split fibration with aspherical fibres, hence lies in W. The proof of B.5 now applies verbatim, without assuming Theorem A, and proves the terminal- and initial-object axioms. If all $s\backslash X$ are aspherical, the split replacement has aspherical fibres, so $\Phi_S(X) \to S$ and i_X are weak. Thus the opposite absolute axiom holds. Apply B.6 to W°, which now satisfies the absolute axioms: this gives $W = W^\circ$ and hence the absolute axiom for W itself. This step removes the apparent circularity in the manuscript's backward proof. For the strong clause use $\Phi_S(f)$ and the natural weak comparisons i_X, i_Y; the same fibrewise argument gives the opposite relative axiom, hence the relative axiom. □

B.8 [Refuted]. Products are not consequences of the literal definition.

Even under Loc (1 ter), Loc (2), Loc (3), finite products of weak arrows need not be weak. Thus XVI Rappel 2.3, and the finite-product assertion of Proposition 2.4 under the printed definition alone, require correction.

Proof. For W_{circ}, take degree two on the circle and multiply it by the identity of a second circle. The endpoints have torus type; the map is not a weak equivalence, since its map on π_1 has image of index two. It therefore does not belong to W_{circ}. If localization preserved binary products, the product of an inverted arrow and an identity would be inverted; strong saturation would put this map in W_{circ}, a contradiction. The same example adjoined to an isolated point proves failure of finite-coproduct closure under the literal definition. The editorial qualification at Rappel 2.3 correctly supplies relative Loc (3 bis). □

B.9 [Known]. The exact literature distinction.

Every class satisfying the literal package is a weak fundamental localizer in the sense of Cisinski 6.1.12 and Maltsiniotis 1.1.2. Their saturation terminology uses a split-idempotent clause implied by the literal moderate clause; their Theorem A clause is exactly absolute. The sources inspected do not supply the counterexample of B.3.

Proof. We checked Cisinski (2006, §6.1.12): its clauses are weak saturation, the terminal-object axiom, and the assertion that every aspherical functor is weak. Here weak saturation means FS1–FS3 of Cisinski 1.4.12: identities, two-out-of-three, and $ri = 1,\ ir \in W \Rightarrow i \in W$. The literal moderate clause implies FS3; we do not identify these two formulations in an arbitrary category. In particular FS3 is not closure under arbitrary arrow retracts. The March 2020 version of Maltsiniotis (2020, §1.1.2) writes the third premise as $A/b \to B/b \in W$; because B/b has terminal object this is precisely the absolute axiom. There is no additional product or descent clause. Maltsiniotis (2012, Remark 1.3) states: "On ne connaît pas d'exemple de localisateur fondamental faible qui ne soit pas un localisateur fondamental." The original author's MathOverflow answer of December 2019 states the same absence of a known example and points to Cisinski 6.1.11, 6.1.17, and 6.1.18 (Cisinski, 2019a). These are historical reports, not a claim about everything published through the present date. We searched the exact French and English terms, their conjunctions with absolute and relative Theorem A, and the cited primary texts; no priority is claimed for B.3. □

B.10 [Open]. The uncompleted statement-by-statement historical audit.

The complete audit of every assertion in XVI ms. 1–27 and XV §9, including all implications among the additional Mayer–Vietoris axioms, is not completed here. The complete list of settled and unresolved named assertions is `sections/BC_historical_audit.csv`*. B.11–B.14 below settle the formal decomposition statements, finite localization statements, the two discrete-set adjunctions, and the elementary cube implications. The remaining unresolved assertions include the infinite-product generality of Lemma 2.5 after adding identities, the variants with the full decomposition hypothesis in XV 9.7 and 9.11, and the full Mayer–Vietoris implications in XVI 3.2–3.7. B.17 refutes several of the other nontriviality equivalences under the literal absolute axioms. Replacing absolute Loc (3) in Définition 1.1 by relative Loc (3 bis) repairs the product argument in B.8; the separate correction in B.11 is still needed. Optional axioms Loc (4)–(8) retain their stated hypotheses.*

The quotient example proves the missing descent condition very concretely: Φ in B.3 inverts degree two on S^1, but does not invert its suspension. Thus an argument through homotopy colimits cannot follow from strong saturation alone. The group-dependent universal-cover

homology proposal is unnecessary for the negative answer. We have not asserted that every possible version of that proposal has been classified.

B.11 [Refuted]. The parenthesis in XVI Corollary 2.2 needs a separate correction.

Closure under finite coproducts and filtered colimits does not imply the full decomposition axiom Loc (4), even for a strongly saturated relative fundamental localizer. The parenthesis in Corollary 2.2 must say Loc (4 a), or must separately assume Loc (4 b).

Proof. Let $\epsilon : \mathrm{Cat} \to [1]$ send the empty category to 0 and every nonempty category to 1, and put

$$W_{\mathrm{ne}} = \epsilon^{-1}(\mathrm{Iso}[1]).$$

This is the coarse localizer of Maltsiniotis (2020, Example 1.1.30). An existing functor is weak precisely when its source and target are both nonempty or both empty. This inverse-image description proves strong saturation. The terminal-object axiom is immediate. For relative Theorem A, suppose $u : A \to B$ lies over S and every u/s is weak. If B is empty, so is A. If B is nonempty, choose $b \in B$ and put $s = p(b)$. The comma category B/s contains $(b, 1_s)$, hence A/s is nonempty and therefore A is nonempty. In either case u is weak. This verifies the axiom on all small categories.

A coproduct is nonempty exactly when at least one summand is nonempty, so arbitrary coproducts preserve this weak class. A filtered colimit of categories is nonempty exactly when one of its terms is nonempty; this follows already on object sets. Thus filtered colimits preserve the class too. Nevertheless the inclusion $e \to e \amalg e$ is weak, whereas its restriction to the missed summand is $\varnothing \to e$, which is not. This is a finite counterexample to Loc (4 b). Its category tables, the inclusion, and its strict pullback to the other summand are checked in `verify_counterexamples.py`. □

B.12 [Proved]. The formal decomposition and finite localization assertions.

XVI Proposition 2.1, the principal equivalence of Corollary 2.2, the finite-product and finite-coproduct versions of Lemma 2.5 when identities belong to W, and Lemma 2.6 hold without relative Theorem A. The identity hypothesis is automatic for every localizer considered here; B.15 treats the lemma's more general literal wording. The formal equivalences in XV Proposition 9.4, Corollary 9.5, and Corollary 9.6 also hold, except that this statement does not certify 9.4's parenthetical deduction from earlier axioms. XVI Corollary 2.9 at ms. 13 follows from summand restriction alone.

Proof. Write Σ for closure of W under the indicated coproducts and D for restriction of a weak arrow to a summand of its codomain. Every arrow over a coproduct of categories is the coproduct of its restrictions. Hence the assertion that a coproduct is weak exactly when each summand is weak is equivalent to $\Sigma + D$. Binary closure is equivalent to preservation under $f \mapsto f \amalg 1_Z$: one direction uses the identity arrow; the other factors $f \amalg g = (1 \amalg g)(f \amalg 1)$. Induction gives finite closure. A coproduct of an arbitrary family is the filtered colimit of its finite subcoproducts. Thus filtered-colimit closure upgrades finite Σ to arbitrary Σ, but does not provide D, as B.11 shows. Restriction to any one summand follows already by writing a coproduct as that summand plus the sum of the others, even for an infinite family. Under finite closure, restriction to the finitely many components of a summand implies restriction to the summand; filtered-colimit closure gives the same conclusion with infinitely many components. These observations prove the stated formal decomposition equivalences.

For the component test, let $f : X \to Y$ be weak and restrict to each connected component Y_j. By D its restriction is weak. If weak arrows into connected categories have connected, nonempty

sources, each inverse image X_j is connected and nonempty. Consequently $\pi_0 f$ is a bijection. The converse is immediate. This proves both versions of the stated connectivity criterion.

If $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ is an adjunction and both functors preserve the specified weak arrows, its unit and counit descend to the localizations. Their naturality on inverse weak arrows follows by multiplying the original naturality squares by inverses. The triangle identities remain identities, so the descended functors are adjoint. This proves XVI 2.6.

For two relative categories, localization of their product at the componentwise weak arrows is the product of their localizations. Indeed a functor inverting those arrows inverts weak arrows in each coordinate separately. Extend first in one coordinate and then in the other; the commutation identities persist when an arrow is inverted, which constructs the unique extension to the product localization. The same argument applies to natural transformations. Now descend the adjunction between the diagonal and binary product, assuming product closure of W. It proves the binary case of XVI 2.5; induction proves the finite case. A terminal object survives any localization: the unique ordinary maps to it define a natural transformation from the identity to its constant functor, whose naturality extends across inverted arrows; at the terminal object its component is the identity. Hence every localized map into it is that prescribed component. Taking opposites gives the finite-coproduct and initial-object assertions. These arguments also prove the finite-coproduct clause of Proposition 2.4 when its explicit closure hypothesis holds. No identification of an infinite product of localizations with localization of an infinite product has been used. □

B.13 [Known]. The two adjunctions with discrete sets need only the literal axioms.

Under literal Loc (1), Loc (2), Loc (3) and $W \subseteq W_0$, the functor $\pi_0 : \mathrm{Hot}_W \to \mathrm{Set}$ is both left and right adjoint to the discrete-set functor. Thus the second Proposition 2.8, the second Corollary 2.9, and Corollaries 2.11–2.13 of XVI hold as printed, for arbitrary small sets. Lemma 2.14 also holds in its stated abstract generality.

Proof. Cisinski's minimality theorem gives $W_\infty \subseteq W$ (Cisinski, 2006, Theorem 6.1.18). Its saturation hypothesis is precisely FS1–FS3 of §1.4.12; FS3 says that $ri = 1$ and $ir \in W$ imply $i \in W$, and is implied by the literal moderate clause. Thus no arrow-retract axiom has been inserted here. Localize first at W_∞, obtaining the homotopy category of spaces (Cisinski, 2006, Example 4.1.29), and then at the images of W. For every discrete small set I and every space X one has

$$[I, X] = \mathrm{Set}(I, \pi_0 X), \qquad [X, I] = \mathrm{Set}(\pi_0 X, I).$$

Both represented functors invert W, since $W \subseteq W_0$.

We spell out why a represented functor which already inverts W keeps its representing hom-sets after localization. If $\mathcal{A}(c, w)$ is bijective for every $w \in W$, a zigzag starting at c can be shortened inductively: replace $c \xrightarrow{h} y \xleftarrow{w} z$ by the unique $k : c \to z$ with $wk = h$. Thus every localized arrow from c is represented in $\mathcal{A}$. Injectivity follows by extending $\mathcal{A}(c, -)$ to the localization and evaluating a localized arrow at 1_c. The dual argument treats arrows into c. Apply this twice to the discrete object I in the homotopy category of spaces. The two displayed formulas survive the second localization and are precisely the two adjunctions. In particular the discrete-set functor is fully faithful, $\mathrm{Hot}_W(e, X) = \pi_0 X$, every arrow from e is represented by an object of X, and there is no arrow from a nonempty X to the empty category.

For the abstract Lemma 2.14, the representable functor $\mathcal{M}(\varnothing, -)$ is constantly a singleton and inverts every weak arrow. The preceding representable argument shows that the initial object remains initial. If no weak arrow has empty source and nonempty target, a zigzag ending at the strict initial object must remain at that object, by backwards induction: an ordinary arrow into it has initial source, and an inverse weak arrow into it has initial source by the hypothesis

on W. Hence it remains strict. To prove the asserted surjectivity from the terminal object e, shorten each segment $e \xrightarrow{u_1} X_1 \xleftarrow{u_2} X_2$, $u_2 \in W$, using the stipulated $s : e \to X_2$ with $u_2 s = u_1$ in the localization. Repeating removes every backward arrow. The result is represented by one ordinary arrow $e \to X$, exactly as claimed. □

B.14 [Proved]. The elementary cube implications are formal.

The three implications of XVI Proposition 3.1 hold under Loc (1). They require neither absolute nor relative Theorem A.

Proof. Start with a Mayer–Vietoris square having maps $i_1 : X_0 \to X_1$ and $j_1 : X_2 \to X_3$ on opposite sides. Map the degenerate Mayer–Vietoris square on $X_0 \to X_2$ to this square. Its four comparison maps are $(1_{X_0}, i_1, 1_{X_2}, j_1)$. Loc (6 a) gives $i_1 \in W \Rightarrow j_1 \in W$, and Loc (6 c) gives the reverse implication. Next map the given square to the degenerate square on $X_1 \to X_3$. The four maps are $(i_1, 1_{X_1}, j_1, 1_{X_3})$. Loc (6 b) gives the reverse implication, and Loc (6 c) gives the forward implication. Every degenerate square is still a union-of-two-sieves square, with one sieve equal to the whole category. This proves the three implications, including both consequences of Loc (6 c). The second cube's repeated label f_1 in the transcription must read $f_2 = j_1$ at its third vertex. □

B.15 [Refuted]. The arbitrary-arrow wording of XVI Lemma 2.5 is too broad.

If W is literally an arbitrary class of arrows, as specified in XVI Lemma 2.5, even its closure under arbitrary products does not imply that localization preserves binary products. The finite version becomes true after requiring identities to belong to W, as in B.12.

Proof. Let $\mathcal{M}$ be the four-object diamond poset $0 < a < 1$, $0 < b < 1$, with a, b incomparable, and put $W = \{f : a \to 1, 1_1\}$. Arbitrary products are meets. A product of arrows in W is f if any factor is f, and is 1_1 otherwise, including the nullary case. Thus every literal product-closure hypothesis holds. But $f \times 1_b$ is $g : 0 \to b$. The order map to the three-object chain $0 < 1 < 2$ which sends 0 to 0, b to 1, and $a, 1$ to 2 sends f to an identity and g to the noninvertible arrow $0 \to 1$. Consequently g is not inverted by localization at W. If that localization preserved binary products, it would invert $f \times 1_b$, since it inverts f. This contradiction proves the claim. The program enumerates the diamond's complete composition table and its sixteen binary meets, checks the detecting functor, and checks the four binary products of weak arrows and the displayed product with 1_b. □

B.16 [Proved]. A weak localizer with near-bijective infinite component maps.

Call a map of sets $f : I \to J$ a near bijection if there are finite subsets $I_0 \subset I$ and $J_0 \subset J$ such that its restriction $I \setminus I_0 \to J \setminus J_0$ is a bijection. Define, on all small categories, W_{nb} by the following two alternatives:

$$u : A \to B \in W_{\mathrm{nb}} \iff \begin{array}{l} (\pi_0 A, \pi_0 B \text{ finite and } u \in W_\infty) \\ \quad \text{or } (\pi_0 A, \pi_0 B \text{ infinite and } \pi_0 u \text{ a near bijection}). \end{array}$$

This class satisfies literal Loc (1), Loc (2), absolute Loc (3), and closure under finite coproducts. Its aspherical categories are exactly the weakly contractible categories. There are distinct parallel arrows in its localization even though it contains maps which are not bijective on components.

Proof. We first construct a detecting category. It has the small sets as objects. Adjoin an absorbing zero to every hom-set of Set, identify every finite-image function with zero, and identify two functions $f, g : I \to J$ when

$$\{f(i), g(i) \mid i \in I,\ f(i) \neq g(i)\} \quad \text{is finite.} \tag{B.16.1}$$

Condition (B.16.1) is an equivalence relation. For transitivity, the unequal values of f, h lie in the union of the unequal-value sets for f, g and g, h. Precomposition only restricts the set of unequal values, and postcomposition sends it to the image of a finite set. Finite-image functions form a two-sided ideal. The zero identifications are therefore compatible with this congruence and with composition. We obtain a category $\mathscr{R}$ and a functor $q : \mathrm{Set} \to \mathscr{R}$.

The identity of a finite set is zero, so every finite set is a zero object of $\mathscr{R}$, including the empty set. The identity of an infinite set is nonzero. Thus finite and infinite sets cannot become isomorphic. For infinite I, J, the arrow $q(f)$ is an isomorphism if and only if f is a near bijection. To prove necessity, a nonzero inverse is represented by a function $g : J \to I$, and (B.16.1) gives $gf = 1_I$ and $fg = 1_J$ outside finite subsets. Consequently f is injective outside a finite set, every fibre of f is finite, and its image is cofinite. If I_0 is a finite exceptional set for injectivity, remove $J_0 = (J \setminus f(I)) \cup f(I_0)$ and then its finite inverse image. The remaining restriction is bijective. Conversely, invert a near bijection off its two finite exceptional sets and extend the inverse arbitrarily on the exceptional codomain. The two composites differ from the identities on finite subsets, so they equal the identities in $\mathscr{R}$.

We verify the saturation clauses without any implicit finiteness assumption. If two of f, g, gf lie in W_{nb}, all three categories have finite component sets, or all three have infinite component sets. In the first case use two-out-of-three for W_∞; in the second use the isomorphism criterion for $q\pi_0$. If gf and fg lie in W_{nb}, their images under $q\pi_0$ are isomorphisms, so the images of f, g are isomorphisms. Their component sets therefore are either both finite or both infinite. In the finite case the moderate clause for weak homotopy equivalences applies; in the infinite case the near-bijection criterion applies. Identities and isomorphisms satisfy the definition. This proves every clause of literal Loc (1), including empty categories.

The class contains W_∞, and an arrow to the point belongs to it precisely when it is a weak homotopy equivalence. The terminal axiom and absolute Theorem A follow, the latter from Quillen's Theorem A. For a finite family of weak arrows, either all its categories have finitely many components and ordinary weak equivalences give the conclusion, or a summand has infinitely many components. In that case the coproduct component map is a near bijection: take the union of the finitely many exceptional finite sets; the finite-component summands have bijective component maps. The empty coproduct is the identity of the empty category.

Regard $\mathbb{N}$ as a discrete category. The shift $s(n) = n + 1$ is in W_{nb}. On the other hand, the functor $q\pi_0 : \mathrm{Cat} \to \mathscr{R}$ inverts every member of W_{nb}, and distinguishes $1_{\mathbb{N}}$ from the constant-zero endofunctor: the former has nonzero identity image, whereas the latter has finite image and is zero. It factors through the localization and proves that these parallel arrows remain distinct. All categories, arrows, and compositions in this counterexample are explicitly discrete; the infinity of $\mathbb{N}$ is essential to the proof and is not inferred from a finite simulation. The program's exact affine-tail representation checks the shift, its near inverse, and their finite exceptional sets as functions on all natural numbers. □

B.17 [Refuted]. Several nontriviality equivalences fail for the printed definition.

Under literal Loc (1), Loc (2), Loc (3), XVI Proposition 2.7 and XV Proposition 9.9, Lemma 9.10, Proposition 9.12, and Corollary 9.13 are false. The extra finite-coproduct hypothesis of XV 9.9 does not repair it. Relative Loc (3 bis) repairs these assertions.

Proof. Use W_{nb}. Its shift $s : \mathbb{N} \to \mathbb{N}$ fails the component-bijection axiom. Nevertheless its localization and its full subcategory of nonempty categories have distinct parallel endomorphisms of $\mathbb{N}$, by B.16. Neither category is equivalent to a point, and the full localization is not equivalent to [1]. This directly contradicts XVI 2.7 and XV 9.9; the latter's finite-coproduct hypothesis was proved in B.16. The map $t : \mathbb{N} \to \mathbb{N}$ defined by $t(0) = t(1) = 0$ and $t(n) = n - 1$ for $n \geq 2$ is a near bijection. It is inverted but is not injective on components, disproving Lemma 9.10. The distinct arrows $1_{\mathbb{N}}$ and the constant endofunctor directly disprove Corollary 9.13.

For Proposition 9.12, its conditions (c), (c'), and (c") hold, whereas (a) fails. Its condition (d) also holds: infinite sets of different cardinalities cannot be isomorphic in $\mathscr{R}$, because a near bijection preserves infinite cardinality. The detecting functor therefore separates arbitrarily many infinite cardinalities already in $\mathrm{Hot}_{W_{\mathrm{nb}}}$. Its condition (b) fails explicitly. Let $g_0, g_1 : \mathbb{N} \to \{0, 1\}$ be respectively the constant-zero map and the map with value 1 only at 0. Then $g_0 s = g_1 s$, and inversion of s makes $g_0 = g_1$ in the localization, although their component maps differ.

We also give the corrected implication. Relative Theorem A implies $f \times 1_X \in W$ for $f \in W$: apply it over X, whose comma comparisons are $f \times 1_{X/x}$. The projections from the source and target of the latter comparisons are category-homotopy equivalences, since X/x has terminal object. Thus those comparisons belong to W by B.5 and two-out-of-three. B.12 now proves that localization preserves finite products. If a weak $f : I \to J$ identifies two distinct components, select i_0, i_1 in them with images in one component of J. Any two functors $u_0, u_1 : X \to Y$ extend to a functor $I \times X \to Y$ using the decomposition of I separating i_0, i_1. Inverting $f \times 1_X$ transfers this map to the localization of $J \times X$. The two sections $X \to J \times X$ at $f(i_0), f(i_1)$ are category-homotopic, hence $u_0 = u_1$ in the localization. If instead a weak f misses a component, the two maps $J \to \{0, 1\}$ which agree on its image and differ on that component become equal. Evaluating at a point of the missed component makes the two point inclusions into $\{0, 1\}$ equal. Product preservation then makes the two inclusions $X \to X \amalg X$ equal for every X, and again every pair u_0, u_1 becomes equal.

Once all ordinary parallel functors are identified, all nonempty categories become canonically isomorphic by their constant functors. The category obtained by identifying all parallel ordinary functors has one initial empty object and one isomorphism class of nonempty objects, hence is equivalent to [1]. Localizing this category either keeps it or also inverts the arrow from empty to nonempty, giving a point. This proves the corrected collapse conclusion. Conversely, the component-bijection axiom factors π_0 through localization and separates all discrete cardinalities. These two directions give precisely the corrected equivalences. The printed proofs invoke product preservation at the step isolated above; B.8 and B.16 show why the relative hypothesis cannot simply be omitted from that step. □

C The fibre axioms with enough intrinsic fibrations

C.1 [Proved]. Universal weakness and fibrewise invariance.

Suppose $\mathcal{M}$ has pullbacks, W contains isomorphisms and satisfies two-out-of-three, and every map factors as a map in W followed by an intrinsic fibration. Then

$$W \cap \mathrm{Fib}_W = W^u.$$

Moreover every W-map between intrinsic fibrations over S remains weak after arbitrary change of base. Thus XII axiom (b) follows.

Proof. Let $u : X \to Y$ be weak over S, with both structural maps intrinsic, and factor an arbitrary $T \to S$ as $T \xrightarrow{i} T_0 \xrightarrow{p} S$, with $i \in W$ and p intrinsic. Pullback of u over T_0 is weak by the definition of p. The comparison maps from pullback over T to pullback over T_0 are weak

because they are pullbacks of i along base changes of $X \to S$ and $Y \to S$. The commutative square and two-out-of-three show that the pullback of u over T is weak. This proves (b).

Apply the assertion to the arrow $q \to 1_S$ in the slice when $q \in W \cap \mathrm{Fib}_W$; it says precisely that q is universally weak. Conversely, if $q : X \to S$ is universally weak, then for a weak arrow $T' \to T$ over S both vertical arrows in the square with upper edge $X \times_S T' \to X \times_S T$ are weak. Two-out-of-three makes that upper edge weak, proving intrinsic membership; the identity base change gives $q \in W$. □

C.2 [Proved]. Lead 3 without functorial factorization.

Under C.1's hypotheses, for every full subcategory

$$\mathcal{F}_S := \mathrm{Fib}_W S \;\subseteq\; \mathcal{H} \;\subseteq\; \mathcal{M}/S$$

the inclusions induce equivalences on ordinary localizations and on ∞-categorical localizations at the restricted W_S. Consequently factorization alone implies all three axioms (a), (b), (c). No global terminal object, functorial factorization, retract closure, or strong saturation of W is required.

Proof. We give all hypotheses of the localization theorem being used. For an ordinary category $\mathcal{A}$, the nerve is an ∞-category with weak equivalences and fibrations in the sense of Cisinski (2019b, Definitions 7.4.6 and 7.4.12) if it has a terminal object, the weak class contains identities and has two-out-of-three, and a class F containing identities and closed under composition has the following properties. Pullbacks of F-maps between fibrant objects along maps from fibrant objects exist and stay in F; the same pullbacks preserve $F \cap W$; every map with fibrant codomain factors as W followed by F. Here fibrant means that the terminal map belongs to F. Cisinski's Theorem 7.5.18 then states that inclusion of the full subcategory of fibrant objects induces an equivalence of ∞-categorical localizations.

Take $\mathcal{A} = \mathcal{H}$, and let F consist of its morphisms whose underlying map in $\mathcal{M}$ is intrinsic. The object 1_S is terminal in $\mathcal{H}$ and belongs to $\mathcal{F}_S$. Identities and compositions are in F, by the corresponding unconditional properties of intrinsic fibrations. The fibrant objects of $\mathcal{H}$ are exactly $\mathcal{F}_S$. Suppose $X \to Y$ is in F, with X, Y fibrant over S, and $Z \to Y$ has Z fibrant. Its pullback $P \to Z$ is intrinsic, so $P \to S$ is intrinsic by composition. Thus P belongs to $\mathcal{F}_S$, hence to $\mathcal{H}$. Since $\mathcal{H}$ is full, this is also a pullback in $\mathcal{H}$. If $X \to Y$ is weak, C.1 makes its pullback weak. Finally, for a map $B \to A$ in $\mathcal{H}$ with A fibrant, factor its underlying map as $B \xrightarrow{w} A' \xrightarrow{q} A$, where $w \in W$ and q is intrinsic. Composition with $A \to S$ makes A' fibrant, so the entire factorization lies in $\mathcal{H}$.

All of Cisinski's hypotheses have now been verified within $\mathcal{H}$. Importantly, his definition does not require arbitrary pullbacks in $\mathcal{H}$. The required pullbacks have fibrant vertices and have just been shown to remain in the smaller subcategory. The theorem gives $L(\mathcal{F}_S) \simeq L(\mathcal{H})$. Apply the same argument to $\mathcal{M}/S$ to obtain the composite equivalence $L(\mathcal{F}_S) \simeq L(\mathcal{M}/S)$; the second inclusion is consequently an equivalence as well. Taking homotopy categories gives the ordinary Gabriel–Zisman localizations. C.1 identifies W_S with W_S^u on $\mathcal{F}_S$, giving axiom (a) with exactly its stated weak class. □

C.3 [Refuted]. Five objects suffice without factorization.

Axioms (a) and (b), even at every base and for strongly saturated W, do not imply (c) or the intrinsic factorization hypothesis. A counterexample has five objects and sixteen arrows. Thus the six-object poset example is not minimal among finite categories.

Proof. Let $\mathcal{M}$ have objects $0, A, B, C, S$. The object 0 is initial, with its unique arrows to the other objects. Besides these arrows and the five identities, there are

$$\alpha, \beta : A \rightrightarrows B, \quad \gamma : A \to C, \quad b : B \to S, \quad c : C \to S, \quad x, y : A \rightrightarrows S,$$

with the only nonidentity composition relations not involving 0 given by $b\alpha = c\gamma = x$ and $b\beta = y$. Put

$$W = \{1_0, 1_A, 1_B, 1_C, 1_S, \alpha, \beta, \gamma\}.$$

These formulas define the complete composition law; associativity involves only identities or arrows from 0 and follows directly. The functor to the three-object chain $0 < m < s$ that sends A, B, C to m and S to s has W as the inverse image of identities. It proves two-out-of-three, and also strong saturation: any arrow inverted in $\mathcal{M}[W^{-1}]$ must map to an identity in this chain, hence must already belong to W.

Each slice is a poset. Over S its six objects are $0_S, x, y, b, c, 1_S$, with Hasse relations

$$0_S < x, y, \qquad x < b, c, \qquad y < b, \qquad b, c < 1_S.$$

The slice over B is the diamond $0_B < \alpha, \beta < 1_B$; over C it is the chain $0_C < \gamma < 1_C$; over A it is $0_A < 1_A$; over 0 it is terminal. Each has all binary meets, so every cospan of $\mathcal{M}$ has a pullback. In particular α and β pull back to 0, while b and c pull back to A with projections α, γ.

The intrinsic fibrant objects in the slices are exactly

base	slice objects	intrinsic fibrant objects
0	1_0	1_0
A	$0_A, 1_A$	$0_A, 1_A$
B	$0_B, \alpha, \beta, 1_B$	$0_B, 1_B$
C	$0_C, \gamma, 1_C$	$0_C, \gamma, 1_C$
S	$0_S, x, y, b, c, 1_S$	$0_S, b, 1_S$

To check the only nontrivial row, pull back the three weak arrows $x \to b$, $y \to b$, $x \to c$ in the slice over S. The object b preserves all three, producing $\alpha, \beta, 1_A$. The candidates x, y, c fail, respectively, on $y \to b$, $x \to b$, and $y \to b$, producing $0 \to A$ in each case. The bottom object and the identity always pass. The other rows follow from the same diamond and chain tests. These tests include all possible structure maps to the base, not only weak arrows whose codomain is the base.

On the fibrant subcategory the weak arrows are only identities, except at base C, where γ is also weak. Its pullbacks over C are $1_0, 1_A, \gamma$. This proves (b) at all bases. For (a), only B and S need replacement. Over B send α, β to 1_B and fix $0_B, 1_B$; the identity has a weak natural transformation to this retraction. Over S define the two endofunctors by

z	0_S	x	y	b	c	1_S
Rz	0_S	b	b	b	b	1_S
Tz	0_S	x	y	b	x	1_S.

Both preserve the order and the weak arrows. There are natural transformations $1 \leftarrow T \to R$ with weak components, and R, T restrict to the identity on $\{0_S, b, 1_S\}$. They exhibit the localized inclusion of the fibrant subcategory as an equivalence, at both the ordinary and the enhanced level.

The full intermediate subcategory on $\{0_S, b, c, 1_S\}$ is a diamond and has only identity weak arrows. Its fibrant subcategory is the chain $\{0_S, b, 1_S\}$, also with only identity weak arrows. The inclusion fails essential surjectivity at c, so (c) fails. Finally the only weak arrow with source C is 1_C, whereas $c : C \to S$ is not intrinsic. It has no required factorization. The program verifies the category laws, all sixty-six cospans and their universal properties, every intrinsic test, every test in (b), and both displayed deformation certificates. Its complete output is `code/BC/verification_output.json`. □

C.4 [Proved]. Why an object bound is not a finite category search bound.

There are infinitely many inequivalent finite categories with pullbacks and one object. A finite terminating exhaustive search through six objects therefore needs an additional arrow bound or a structural reduction. Every finite category with pullbacks has only monomorphisms, but this does not bound its number of arrows.

Proof. The one-object categories of cyclic groups C_m, for arbitrary positive integers m, have pullbacks: a cospan of invertible arrows has a pullback obtained using either inverse. They are pairwise inequivalent. To prove the structural assertion, every slice of a finite category with pullbacks is finite and has terminal object and binary products. For any objects X, Y in such a slice, the product powers Y^n give $|\operatorname{Hom}(X, Y^n)| = |\operatorname{Hom}(X, Y)|^n$. There is a common finite bound on all hom-set cardinalities, so $|\operatorname{Hom}(X, Y)| \leq 1$. Each slice is therefore a preorder. If $fh = fk$, the two maps h, k define parallel maps in the slice over the codomain of f, and must coincide. Thus f is monic. In particular finite endomorphism monoids are groups, but the preceding family shows that their orders remain unbounded. □

C.5 [Proved]. A direct verification of the perfect path replacement.

For each small S there is a strictly functorial coaugmented replacement $R_S : \mathrm{Cat}\,/S \to \mathrm{Cat}\,/S$ whose values are W_∞-smooth and proper and whose coaugmentation has underlying map in W_∞.

Proof. Let J be the integer fence, with an arrow from each even integer to each adjacent odd integer. Let $\mathcal{P}(S)$ have the eventually constant functors $J \to S$ and all natural transformations. The left and right endpoint functors are σ, β. These are defined on morphisms because naturality on identity tails forces their components to be eventually constant too.

We check the incoming and outgoing lifting categories of $(\sigma, \beta) : \mathcal{P}(S) \to S \times S$ directly. For an outgoing lift from γ with prescribed endpoint arrows $\ell \to \ell'$ and $r \to r'$, choose an odd left boundary a and an even right boundary b, far enough out that γ is already constant there. Use γ on $[a+1, b]$, use ℓ' on the left tail and r' on the right tail, and use the prescribed arrows at the joins $a+1 \to a$ and $b \to b+1$. The transformation from γ is the identity on the middle interval and the prescribed maps on the tails. In the full stage of lifts stationary outside these boundaries this object is initial: the map to another lift is its given transformation on the middle interval and the identity on the two tails. Naturality at the joins follows from the other lift's two naturality squares; uniqueness follows from the identity middle components of the universal lift. For incoming lifts choose an even left boundary and an odd right boundary and reverse this construction. It supplies a terminal object in each stage. Every finite simplex of either lifting category occurs in one stage, and enlarging the boundaries is filtered. Filtered colimits of simplicial sets preserve weak equivalences. Both full lifting categories are therefore weakly contractible. The lifting-category criterion for smoothness and properness proves the asserted two properties of the endpoint functor (Cisinski, 2019b, Theorem 4.4.36(ii'), and its opposite).

For $f : A \to S$ set

$$R_S(f) = (A \times_{S,\sigma} \mathcal{P}(S) \xrightarrow{\beta} S), \qquad \eta_f(a) = (a, \mathrm{const}_{f(a)}).$$

The map to $A \times S$ is a base change of the endpoint functor, and the projection $A \times S \to S$ is smooth and proper. These properties are stable under composition and base change, so $R_S(f)$ is perfect.

There is also a direct proof that the projection $R_S(f) \to A$ is weak. Restrict to a finite fence interval containing the support of a path and extend constantly outside it. Deleting its rightmost leaf, and then extending constantly from the neighbor, gives a retraction of finite diagram categories. The edge to the deleted leaf gives a natural transformation in one of the

two directions between this composite and the identity. Repeat to contract the finite fence to its leftmost vertex. Throughout, that vertex is $f(a)$ and the transformations are the identity on a. Thus the finite-stage projection to A is a category-homotopy equivalence, with the constant-path section. Passing to the filtered union proves the full projection weak. Its section η_f is weak by two-out-of-three. For $u : A \to A'$ over S, send (a, γ) to $(u(a), \gamma)$. These formulas preserve composition and the coaugmentation strictly. This supplies the promised functoriality and completes an independent verification of the portion of BISITE B.19 used here. □

C.6 [Proved]. All intermediate perfect subcategories compute the same localization.

Let W be any fundamental localizer. For either $\mathcal{P}_S = \mathrm{Parf}_{W_\infty} S$ or $\mathcal{P}_S = \mathrm{Parf}_W S$, and every full $\mathcal{P}_S \subseteq \mathcal{H} \subseteq \mathrm{Cat}\,/S$, both inclusions induce equivalences after localization at W_S, ordinarily and as ∞-categories. This applies in particular to all W_n and every homology localizer.

Proof. The minimal-localizer theorem gives $W_\infty \subseteq W$ (Cisinski, 2006, Theorem 6.1.18). The lifting tests also give $\mathrm{Parf}_{W_\infty} \subseteq \mathrm{Parf}_W$. C.5 therefore provides a functor R_S taking every object into either chosen full subcategory, and a natural transformation $\eta : 1 \to R_S$ with components in W_S. Naturality and two-out-of-three show that R_S preserves W_S. For an object of $\mathcal{H}$, both ends of η lie in $\mathcal{H}$, so the transformation is internal to it by fullness. The same holds in $\mathcal{P}_S$. After localization, these transformations are natural equivalences and make R_S an inverse to each inclusion. No assertion that all perfect maps are intrinsic for W enters this argument. In particular the failure of that intrinsic assertion for homology localizers does not obstruct the present conclusion. □

C.7 [Proved]. The precise conclusion at XII ms. 240–241.

The left factorization hypothesis at XII ms. 240 implies the three fibre axioms as printed at ms. 215–218. For arbitrary $g : T \to S$, pullback on intrinsic fibrant objects defines a derived reindexing functor on the localized slices. For a strict pullback $Y = X_1 \times_S X_2$ with $X_1 \to S$ intrinsic, both displayed relative homotopy-fibre identities at ms. 241 hold. These assertions do not require every terminal map to be intrinsic. The weaker existence condition in footnote 128, that intrinsic maps be pullbackable, suffices.

Proof. C.1 and C.2 prove the three axioms. Pullback along g takes intrinsic fibrations to intrinsic fibrations and preserves their weak arrows by C.1, so it descends to their localizations. C.2 identifies these with the localized full slices. The canonical isomorphisms between iterated pullbacks provide the coherence for successive base changes.

The first identity is computed by $X_1 \to S$ itself, which is already fibrant. For the second, choose a factorization $X_2 \xrightarrow{w} \bar{X}_2 \to S$ with $w \in W$ and the second map intrinsic. The comparison $X_1 \times_S X_2 \to X_1 \times_S \bar{X}_2$ belongs to W by the definition of the intrinsic map $X_1 \to S$. It therefore identifies the raw pullback with the replacement calculation in the localized slice over X_1, exactly as required. This also proves independence of replacement through the equivalences in C.2.

Inspecting these proofs, every pullback used is along an intrinsic map, or along a composite or a base change of one. Those maps need only be pullbackable; arbitrary pullbacks elsewhere in $\mathcal{M}$ are unnecessary. The terminal object needed in the localization argument is 1_S in the slice. Thus the caution in footnote 130 about all objects being globally fibrant is unnecessary for the stated relative assertions. The word "cofibrations" in footnote 129 has the opposite variance from the displayed factorization and should read "fibrations." The two mislabelled arrows at ms. 241 are $\bar{p}_2$, not p_2 and $\bar{q}_2$, respectively, as follows from their domains and codomains. These are mathematical typing corrections; no additional palaeographical claim is made. □

C.8 [Open]. The remaining minimum problem.

The least number of objects of a finite category with pullbacks admitting (a) and (b) but not (c) has not been determined here. C.3 proves an upper bound of five. The unbounded-arrow classification required to exclude all smaller finite categories has not been supplied. No exhaustive enumeration through six objects is claimed.

D Intrinsic fibrations and coefficient localizers

D.i Truncation and the boundary homomorphism

D.1 [Known]. The proper-localizer criterion, including its local form.

For $n \geq 0$, put $W_n = \{u : \tau_{\leq n}Bu \text{ is an equivalence}\}$. A functor $p : E \to B$ belongs to Fib_{W_n} if and only if every strict pullback square along $X \to B$ becomes a homotopy pullback after applying $\tau_{\leq n}B(-)$. Equivalently, it suffices to require this for all X having a terminal object. Another equivalent test is that, for every triangle $X \to Y \to B$ with X having a terminal object and BY n-connected, the functor $E \times_B X \to E \times_B Y$ belongs to W_n. For $p \in \mathrm{Liss}_{W_n}$ or $p \in \mathrm{Prop}_{W_n}$, it suffices to test the squares along the objects $b : * \to B$.

Proof. These are, respectively, Proposition 6.2.3(i)–(iv) and Proposition 6.2.4 of Cisinski (2006). Their hypothesis is that the fundamental localizer is proper. Section 9.2.1 of that source establishes properness of W_n, and Corollary 9.2.15 identifies its maps with the stated homotopy-group test. The associated localization is the category of n-types; its homotopy pullbacks are the pullbacks appearing here. This gives a criterion for arbitrary functors, without silently replacing strict pullbacks by homotopy pullbacks in spaces. In particular,

$$\mathrm{Fib}_{W_n} \cap \mathrm{Liss}_{W_n} = \{p \in \mathrm{Liss}_{W_n} : \tau_{\leq n}BE_b \simeq \mathrm{fib}_b(\tau_{\leq n}BE \to \tau_{\leq n}BB)\},$$

with the analogous equality for Prop_{W_n}. The equivalences in this formula are the canonical fibre comparison maps, for every b; abstract equivalence of their endpoints is insufficient. □

D.2 [Proved]. Lead 1 is true.

Suppose $p : E \to B$ is a small-category functor such that, for every $X \to B$, the canonical map $B(E \times_B X) \to BE \times^h_{BB} BX$ is a weak equivalence. For $n \geq 1$, it belongs to Fib_{W_n} if and only if every connecting homomorphism

$$\partial_b : \pi_{n+1}(BB, b) \longrightarrow \pi_n(BE_b, x)$$

is zero, for every b and every vertex $x \in E_b$. For $n = 0$, the criterion is that the monodromy action of $\pi_1(BB, b)$ on $\pi_0(BE_b)$ is trivial. Empty fibres and components outside the image of BE are allowed.

Proof. Let $F = BE_b$, choose $x \in F$, and replace $BE \to BB$ by a Serre fibration to write its long exact sequence. This replacement is used only to compute its homotopy fibre; the hypothesis identifies that fibre with the specified strict fibre. For $n \geq 1$, the canonical map

$$\tau_{\leq n}F \longrightarrow \mathrm{fib}_b(\tau_{\leq n}BE \to \tau_{\leq n}BB) \tag{D.2.1}$$

is an isomorphism on all homotopy groups below n, including the component sets and the actions in degree zero. In degree n its map is the quotient $\pi_nF \to \pi_nF/\operatorname{im}\partial_b$. For $n = 1$ the image is the normal subgroup specified by the exact homotopy sequence, so this statement is a statement

about groups, not about abelianization. Thus (D.2.1) is an equivalence if and only if $\partial_b = 0$. Necessity follows at once from D.1 applied to $X = *$. In degree zero, the fibre of $\pi_0 BE \to \pi_0 BB$ over a component is the orbit set of $\pi_0 F$ under $\pi_1 BB$. The quotient map to this orbit set is bijective exactly when every orbit is a singleton. This proves necessity for $n = 0$.

For sufficiency let $v : X \to B$ be arbitrary and write $E_X = E \times_B X$. For every $x' \in X$, its homotopy fibre is $BE_{v(x')}$. Naturality of the connecting homomorphism identifies the boundary for $E_X \to X$ with the composite of the boundary for $E \to B$ and $\pi_{n+1}(Bv)$. It is therefore zero. Apply (D.2.1) to both fibrations and compare the fibres of

$$\tau_{\leq n} BE_X \longrightarrow \tau_{\leq n} BE \times^h_{\tau_{\leq n} BB} \tau_{\leq n} BX$$

over $\tau_{\leq n} BX$. These fibre maps are equivalences. A map of spaces over a fixed base that is an equivalence on every homotopy fibre is an equivalence, proving the square criterion in D.1. For $n = 0$, monodromy after pullback is the restriction of the original action, so it remains trivial and the same argument works with component sets. On a component with empty homotopy fibre, the total pullback is empty on both sides. Every component meeting a nonempty fibre is treated with one basepoint in each of its fibre components. No surjectivity assumption on $\pi_0 BE \to \pi_0 BB$ was used. □

D.3 [Proved]. An elementary complete criterion at W_0.

For an arbitrary functor $p : E \to B$, put $E_j = E \times_B B_j$ for each connected component B_j of B. Then $p \in \mathrm{Fib}_{W_0}$ precisely when (i) every $\pi_0(E_b) \to \pi_0(E_j)$ is bijective, and (ii) every arrow $\beta : b \to b'$ in B_j has a lift in every prescribed component of E_j.

Proof. Pull back the W_0-map $* \to B_j$ to obtain (i). Pull back the two inclusions $* \to [1]$ along the base change of p by $\beta : [1] \to B$. By (i), each prescribed component has exactly one connected fibre over either endpoint. These fibres must be connected to each other in the total pullback. A zigzag connecting the layers must contain an arrow over β, proving (ii). Conversely, over a connected component X_a of any $X \to B_j$, label objects of $E \times_B X_a$ by their component in E_j. Condition (i) says each labelled fibre is nonempty and connected. Condition (ii) connects these fibres along each arrow of X_a. A zigzag in X_a therefore connects precisely the objects with the same label. This proves, naturally in X, $\pi_0(E \times_B X) = \pi_0 E \times_{\pi_0 B} \pi_0 X$. This formula preserves every W_0-map and proves the result, including empty categories. □

D.4 [Refuted]. Intrinsic fibrations need not be a right lifting class.

Let W be a fundamental localizer with $W_\infty \subseteq W \subseteq W_0$. Then Fib_W is not the class of maps with the right lifting property against any class of functors, and in particular against no set. This applies to every W_n, W_∞, nonzero ordinary homology, and the nontrivial coefficient localizers considered here.

Proof. For each integer $r \geq 1$, let C_r be the fence poset on $0, \ldots, r$, with a strict order relation between adjacent vertices directed from the even vertex to the odd vertex and no other strict relations. Its nerve is an interval. Let $A_r = C_r^{\triangleright}$, obtained by adjoining a terminal object ∞_r, and let $p_r : A_r \to [1]$ take C_r to 0 and ∞_r to 1. For any $X \to [1]$, the pullback is the Grothendieck construction of the diagram on X that has fibre C_r at colour 0 and $*$ at colour 1. The map of this diagram to the constant point diagram is objectwise a weak equivalence. The homotopy-colimit comparison for the Grothendieck construction gives $A_r \times_{[1]} X \to X$ in W_∞. Thus every base change of p_r is a weak equivalence, and two-out-of-three shows $p_r \in \mathrm{Fib}_W$.

Form the countable product $p : \prod_r A_r \to \prod_r [1]$. Both its domain and codomain have terminal objects. The inclusion of the all-zero object $* \to \prod_r [1]$ belongs to W_∞. Its pullback along p is

$\prod_r C_r \to \prod_r A_r$. The first category is disconnected: the objects $(0)_r$ and $(r)_r$ cannot be joined by a finite zigzag. A zigzag with k arrows changes each coordinate by at most k, whereas their rth coordinates differ by r. Choosing $r > k$ is a contradiction. The second category is connected, so this pullback is not in W_0, hence not in W.

Any right lifting class is closed under products: solve a lifting square in each coordinate and take the product of the lifts. This contradicts the displayed family. All categories and compositions have been specified, and the program checks the individual fences and their distance bound; disconnectedness of the infinite product is proved by the finite-zigzag argument, not inferred from finite computations. □

D.ii Locally constant derived coefficients

D.5 [Known]. The coefficient tests and their variance.

For a nonzero commutative ring k, a functor $u : A \to B$ lies in $W^{\mathrm{lc}}_{D(k)}$ exactly when it is bijective on components, induces isomorphisms on π_1 at all basepoints, and induces k-homology equivalences of the simply connected universal covers of corresponding components. Consequently $W^{\mathrm{lc}}_{D(\mathbb{Z})} = W_\infty$. For $k = \mathbb{Q}$ the higher test is $\pi_i(Bu) \otimes \mathbb{Q}$ being an isomorphism for every $i \geq 2$. For sets and groupoids as coherent coefficients, the locally constant classes are W_1 and W_2, respectively.

Proof. Evaluation at a basepoint of a connected space Z identifies $\mathrm{Fun}(Z, D(k))$ with modules over the specified loop algebra $C_*(\Omega Z; k)$. Evaluation is conservative and preserves colimits; its left adjoint has monad $C_*(\Omega Z; k) \otimes_k -$. The monadic comparison follows from the bar resolution, or Lurie (2017, Theorem 4.7.3.5). Under this comparison restriction along a based map is restriction of scalars along its loop-algebra map. Restriction is an equivalence only if the unit on the regular source module is an equivalence, hence only if that algebra map is a quasi-isomorphism; conversely a quasi-isomorphism induces an equivalence of module categories by the two-sided bar resolution. The degree-zero map is $k[\pi_1 A] \to k[\pi_1 B]$. Since this is the specified map on bases, it is invertible only if the group map is bijective. Each loop-space component then reduces to the identity component, the loop space of the universal cover. The bar and cobar constructions on simply connected spaces show that a map of their loop chains is a k-homology equivalence if and only if the original map is one: use the tensor-length filtration; after desuspending reduced chains all generators have positive degree, so the filtration is finite in each total degree. Tensoring bounded-below free chain complexes preserves the requisite quasi-isomorphisms. This gives the asserted universal-cover criterion. With $k = \mathbb{Z}$, homological Whitehead on the simply connected covers proves the equality with W_∞. The rational identification is also precisely Rivera et al. (2021, Theorem 3). For r-truncated coefficients, restriction depends only on $\tau_{\leq r+1} Z$: the cores of all diagram categories $\mathrm{Fun}([m], \mathcal{S}_{\leq r})$ are $(r+1)$-types. Conversely, evaluating the unit on the free object at a basepoint detects $\tau_{\leq r} \Omega Z$. This proves $W^{\mathrm{lc}}_{\mathcal{S}_{\leq r}} = W_{r+1}$. The zero ring is excluded because then all coefficient categories are terminal. □

D.6 [Proved]. Rational local coefficients require only the degree-two boundary.

Under the homotopy-pullback hypothesis of D.2, $p : E \to B$ belongs to $\mathrm{Fib}_{W^{\mathrm{lc}}_{D(\mathbb{Q})}}$ if and only if every boundary $\pi_2(BB, b) \to \pi_1(BE_b, x)$ is zero. There is no vanishing condition on the higher rational boundaries. In particular the stronger proposed test in Part D(3) is false.

Proof. Assume these boundaries vanish. For a $W^{\mathrm{lc}}_{D(\mathbb{Q})}$-map $u : X \to Y$ over any base change of B, compare the homotopy fibration sequences with common fibre F. The isomorphism on fundamental groups of $X \to Y$ identifies the monodromy orbit sets, hence the components

of the two total spaces. On each such component the vanishing boundary gives an injection $\pi_1 F \to \pi_1 E_X$ and an exact extension by the stabilizer subgroup of $\pi_1 X$ acting on the chosen fibre component. The two extensions have isomorphic kernels and quotients, so the total fundamental groups are isomorphic. Tensor the higher long exact sequences with $\mathbb{Q}$. For degrees at least two all groups in the relevant five-term segments are abelian, and the zero degree-two boundary permits the sequence to end with $\pi_2 E_X \to \pi_2 X \to 0$. Exactness of rational tensoring and the five lemma give isomorphisms on every higher rational homotopy group. D.5 proves sufficiency.

Conversely take $z \in \pi_2(BB, b)$ with boundary $a \neq 1$ in $\pi_1 F$. Pull back the fibration to a based sphere representing z, and compare its pullback along a degree-m sphere map. A degree-m map of S^2, for $m \neq 0$, belongs to $W^{\mathrm{lc}}_{D(\mathbb{Q})}$. On the chosen total-space components the induced fundamental-group map is

$$\pi_1 F/\langle a^m \rangle \longrightarrow \pi_1 F/\langle a \rangle.$$

Both displayed cyclic subgroups are normal, by the exact sequence of the pulled-back fibration over S^2. If a has finite order $d > 1$, choose $m = d$; if it has infinite order, choose $m = 2$. The map then has a nontrivial kernel and is not a coefficient equivalence. We justify the strict categorical witness. The weak equivalence $NB \to \mathrm{Ex}^\infty NB$, together with right properness of simplicial sets, gives a Quillen equivalence of the slice model categories. Thus the sphere triangle in spaces over BB has a representative $K' \to K \to NB$ in simplicial sets: represent the target by a fibrant object of the slice and then represent the map to it. Apply the category-of-simplices construction to this strict triangle. Last-vertex evaluation gives functors $\Delta/K' \to \Delta/K \to B$, and the natural weak equivalences $B(\Delta/K) \simeq |K|$ identify their realized triangle with the chosen triangle in the slice of spaces. Since its first map is a rational local-coefficient equivalence, so is $\Delta/K' \to \Delta/K$. The homotopy-pullback hypothesis on p identifies the strict categorical pullbacks of this triangle with the homotopy pullbacks used above. This transfers the nontrivial fundamental-group kernel to a categorical witness. Thus intrinsic membership forces $a = 1$.

To exhibit the unnecessary higher condition, take a categorical homotopy-pullback representative of the path fibration over S^3. For example factor $* \to K$, where K is the face poset of the boundary of a 4-simplex, into a Thomason weak equivalence followed by a Thomason fibration. Right properness makes its strict pullbacks homotopy pullbacks. Its total category is weakly contractible and its fibre has type ΩS^3. Its degree-two boundary is zero, while its boundary $\pi_3 S^3 \to \pi_2 \Omega S^3$ is an isomorphism $\mathbb{Z} \to \mathbb{Z}$. The preceding equivalence places this map in the stated intrinsic class. For definiteness the factorization is constructed as follows. Starting from $P_0 = *$ over K, form P_{r+1} by attaching one copy of $c\,\mathrm{Sd}^2\,\Delta^m$ for every commutative lifting square from $c\,\mathrm{Sd}^2\,\Lambda^m_k$ into P_r over K, for all $m \geq 1$ and $0 \leq k \leq m$, and set $P = \mathrm{colim}_{r<\omega} P_r$. Each attachment is the categorical pushout: freely adjoin its objects and arrows and impose precisely its categorical relations and the relations of the attaching functor. This specifies all objects, arrows, and composition of P as equivalence classes of finite words. Finite presentability of the attaching domains gives the right lifting property; Thomason's generating-trivial-cofibration theorem gives $* \to P \in W_\infty$. Thus this is a completely specified infinite-category example, not a finite category certificate. □

D.7 [Open]. The remaining coefficient classification.

For arbitrary functors without the hypothesis of D.2, a comparable explicit fibre criterion for $\mathrm{Fib}_{W^{\mathrm{lc}}_{D(\mathbb{Q})}}$ *is not established here. Nor do we establish a complete characterization of* $\mathrm{Fib}_{W^{\mathrm{lc}}_{D(\mathbb{F}_p)}}$, *even among the homotopy-pullback functors of D.2. The integral, set, and groupoid cases reduce exactly to* W_∞, W_1, W_2 *by D.5. Merely declaring higher boundary maps zero after tensoring with a field is not a proof of the unresolved mod-p case: integral fundamental groups are retained by that coefficient localizer.*

D.iii Homology and covering maps

D.8 [Refuted]. A connected intrinsic homology covering need not be trivial.

Let p be a prime. Any categorical covering whose monodromy factors through a finite p-group belongs to $\mathrm{Fib}_{W_{H\mathbb{F}_p}}$. *In particular the regular covering $EC_p \to BC_p$ is a connected nontrivial covering in this class. Thus the proposed assertion for every nonzero spectrum in Part D(4) is false.*

Proof. Write G for the finite p-group and S for the fibre set. The permutation module $V = \mathbb{F}_p[S]$ has a finite filtration with trivial G-action on its successive quotients. Here is a proof independent of a semisimplicity assertion. The augmentation ideal J of $\mathbb{F}_p[G]$ is nilpotent. Induct on $|G|$: choose a central subgroup $C_p \subseteq G$, let z generate it, and use $(z-1)^p = 0$. The quotient algebra by $(z-1)$ is $\mathbb{F}_p[G/C_p]$. If its augmentation ideal has nilpotence exponent d, then $J^d \subseteq (z-1)$ and $J^{dp} = 0$. The filtration $V \supseteq JV \supseteq \cdots$ is therefore finite, also for infinite S, and G acts trivially on every quotient.

After any base change $X \to B$, the covering is classified by the restricted permutation action. Its homology is the homology of X with this local coefficient module. A mod-p homology equivalence $Y \to X$ induces an isomorphism on homology with any constant $\mathbb{F}_p$-vector-space coefficient, since tensoring over a field is exact. Apply the long exact sequences of homology with local coefficients successively to the finite filtration. They prove the same assertion with coefficient V, hence for the pulled-back coverings. This proves the universal intrinsic condition, component by component. For the concrete finite example, BC_p has one object and arrows $\mathbb{Z}/p$; EC_p has objects $\mathbb{Z}/p$ and one arrow $s \to t$, mapped to $t - s$. Composition is addition of differences. Every base arrow has a unique lift from any source and to any target. The total category is indiscrete and contractible; the fibre has p objects. The program verifies all category laws and both unique-lift conditions for $p = 2, 3, 5, 7$, including the nilpotence matrix identity. The proof applies to every prime. □

D.9 [Open]. The remaining homology classification.

A complete classification for arbitrary nonzero spectra E, arbitrary coverings, and all W_∞-smooth and proper functors is not proved here. D.8 shows that the answer must depend on the coefficient theory and on monodromy. D.12 below gives complete covering classifications for ordinary integral and rational homology. Product projections and unions of components give intrinsic examples. Neither those results nor boundary homomorphisms alone give the classification for every spectrum. No finite computation in this return claims to test all base changes in the definition of Fib_{W_E}.

D.iv The eight inherited finite maps

D.10 [Proved]. BISITE B.15 has no changed entries.

*For the maps defined below, the entries are as follows; Q*lc *denotes $W^{\mathrm{lc}}_{D(\mathbb{Q})}$.*

Map	*Smooth, both*	*Proper, both*	*Intrinsic* W_1	*Intrinsic* Qlc
D.3: $p : E \to B$	*yes*	*yes*	*yes*	*yes*
D.3: $q : B \to [1]$	*no*	*yes*	*no*	*no*
D.3: qp	*no*	*yes*	*no*	*no*
D.4: $p : E \to B$	*yes*	*yes*	*no*	*no*
D.4: $q : B \to [1]$	*no*	*yes*	*yes*	*no*
D.4: qp	*no*	*yes*	*no*	*no*
D.5: $H \to K$	*yes*	*yes*	*no*	*no*
D.5: $q : R \to [1]$	*no*	*yes*	*no*	*no*

Proof. We specify all categories. For D.3, B has objects a, b, endomorphisms 1_a and C_2 at b, and two arrows $a \to b$ labelled by C_2, with postcomposition given by addition. Put $L(a) = L(b) = C_2$, and let a labelled arrow act by addition. Set $E = \int_B L$ and let q record the two colours. Then p is a covering. Coverings preserve the two classes: the base map identifies fundamental groups and their permutation actions, while the universal covers of corresponding total components identify with the universal covers of the base components. D.5 proves preservation of both tests after every base change. The incoming lift categories for q and qp include two-point discrete categories; their outgoing lifts have initial objects. Pulling $\{0\} \to [1]$ back along q gives $* \to B \simeq BC_2$, failing on π_1. Along qp it gives a discrete two-point fibre into the connected contractible E, failing on π_0.

Let K have objects the nonempty proper subsets of $\{0, 1, 2, 3\}$, with inclusion arrows. For $x \subseteq y \subseteq z$ let $c(x, y, z) \in C_2$ be 1 only when $(x, y, z) = (\{0\}, \{0, 1\}, \{0, 1, 2\})$. Let H have the objects of K and arrows $(x \subseteq y, m)$, $m \in C_2$, with $(y \subseteq z, n)(x \subseteq y, m) = (x \subseteq z, m+n+c(x, y, z))$. There are no three strict composable inclusions, so the normalized cocycle identity holds. The projection is a bifibration, by unique solution for the group label of every cartesian or cocartesian factorization. Its fibre is BC_2. The chosen cocycle evaluates nontrivially on the triangulated sphere BK, so the connecting map $\pi_2 BK \to C_2$ is onto. Equivalently, collapse a spanning tree in the edge presentation: all group generators become one central involution; the triangle relations outside the marked triangle form a disk and kill all horizontal generators; the marked triangle kills the involution. Thus $\pi_1 BH = 0$. In $H \times_K H$ the same presentation leaves $C_2^2/\langle(1, 1)\rangle = C_2$. The accompanying program constructs all arrows and relations and performs explicit Tietze eliminations to these groups. Thus $* \to K$ tests failure for W_1, and $H \to K$ itself tests failure for rational local coefficients: it is a rational higher-homotopy equivalence with trivial fundamental groups, but its self-pullback has fundamental group C_2. This also proves that the nonzero boundary in D.6 is detected integrally.

For D.4 adjoin a to K, with $B(a, x) = K(\{0\}, x)$, no arrows into a except its identity, and the forced composition. Let $r : B \to K$ take a to $\{0\}$ and otherwise be the identity; set $E = B \times_K H$. The inclusion $K \to B$ is right adjoint to r, and the same adjunction pulls back to $H \to E$, so $BB \simeq S^2$ and $BE \simeq BH$. The map p is a bifibration and inherits both failures from $H \to K$. The maps q and qp are opfibrations. An object of K not containing 0 has no incoming lift from colour 0, proving nonsmoothness. The fibre of qp at 0 is BC_2, and its inclusion into E fails on π_1. The map q is universally a W_1-equivalence: its fibre diagram consists of a point and the simply connected K, and groupoid completion of its Grothendieck construction contracts each fibre to its chosen section object. The unique paths in the completed fibre make this contraction natural over any base change. Two-out-of-three gives intrinsic W_1-membership. Its fibre inclusion $* \to B \simeq S^2$ is not a rational higher-homotopy equivalence, excluding the other membership.

Finally R has b, a, endomorphism group C_2 at b, identity at a, and a unique arrow $b \to a$ fixed by precomposition with C_2. It has terminal object a, is an opfibration over $[1]$, and has incoming lift category BC_2 at a. The inclusion of its colour-zero fibre fails on fundamental groups. These

checks prove all table entries. The independently written program verifies functoriality, every outgoing and incoming lifting category, and the fundamental-group presentations; it does not infer universal preservation merely from a finite list of pullbacks. □

D.v The axioms of XVI, section 4

D.11 [Proved]. The requested fibration-axiom table.

Let $\mathcal{F}_W$ denote Grothendieck fibrations with W-equivalent transport functors. We use the printed meanings: (7a) is $\mathcal{F}_W \subseteq \mathrm{Fib}_W$; (7b) says a W-map between two members of $\mathcal{F}_W$ over a common base is universally in W; (7c) says $\mathcal{F}_W \cap W$ is universally in W; and (7F) is existence of W-then-Fib_W factorizations. Then

W	*(7a)*	*(7b)*	*(7c)*	*(7F)*	$\mathrm{Liss}_W \cap \mathrm{Prop}_W \subseteq \mathrm{Fib}_W$
W_0	*no*	*no*	*no*	*yes*	*no*
W_1	*no*	*no*	*no*	*yes*	*no*
W_n, $n \geq 2$	*no*	*no*	*no*	*yes*	*no*
$W_{\mathbb{Q}}$	*no*	*no*	*no*	*no*	*no*
$W^{\mathrm{lc}}_{D(\mathbb{Q})}$	*no*	*no*	*no*	*no*	*no*
W_∞	*yes*	*yes*	*yes*	*yes*	*yes*

Proof. For W_0, the regular C_2-covering has connected base and total category and a disconnected fibre. It is a bifibration, is itself in W_0, and is not universally in W_0. For $W_{\mathbb{Q}}$ the same covering is a rational homology equivalence, since averaging contracts the positive rational group homology of C_2. The same fibre detects failure. For W_1 and $W^{\mathrm{lc}}_{D(\mathbb{Q})}$, the finite bifibration $H \to K$ of D.10 is itself in the respective class but not universally in it. These examples each refute (7c) and (7b), the latter by comparison with id_B over B, as well as (7a) and the last column.

For general finite n, let K_n be the face poset of the boundary of an $(n+2)$-simplex, so $BK_n \simeq S^{n+1}$. The smooth and proper categorical path replacement established in Part C has contractible total category and fibre ΩS^{n+1}. Its projection and $* \to K_n$ belong to W_n, but its fibre is not n-connected. This proves the last-column failure. To obtain the required Grothendieck fibration for (7c), start with a Thomason-fibration replacement $P \to K_n$ of $* \to K_n$ and form the contravariant comma diagram $b \mapsto b\backslash P$. Its Grothendieck construction $Q \to K_n$ is a Grothendieck fibration. The forgetful functor $Q \to P$ has the right adjoint $e \mapsto (p(e), e, 1)$ and hence is a weak equivalence. Right properness identifies each fibre $b\backslash P$ with the homotopy fibre, and identifies all its transports as weak equivalences. Thus $Q \to K_n$ lies in $\mathcal{F}_{W_n} \cap W_n$ and has the same non-W_n fibre. This refutes (7c). Comparison with id_{K_n} refutes (7b). Pulling the W_n-map $* \to K_n$ back refutes (7a).

For W_∞, Quillen's Theorem B applied to a Grothendieck fibration with weakly equivalent transports identifies all strict base changes as homotopy pullbacks. This proves (7a). A weak equivalence between the total spaces of two such homotopy fibrations over the same base is a weak equivalence on all homotopy fibres; after any further base change it remains so, proving (7b) and (7c). Smoothness together with properness gives the same homotopy-pullback property, by the fibre-to-comma comparisons on both sides of an arrow. Thomason's right proper model structure supplies (7F). For W_n, properness in Cisinski (2006, Section 9.2.1), together with Theorem 5.2.15 and Example 6.1.14 of that source, supplies (7F).

We justify the two negative (7F) entries. For an accessible fundamental localizer W, existence of the stated factorizations is equivalent to properness of W. For the less immediate direction, let W_{asph} be the fundamental localizer generated by its aspherical projections. If factorizations exist, a W-map between intrinsic fibrations over a common base remains in W after every base

change: factor the base map into W followed by an intrinsic fibration and apply two-out-of-three to the resulting square. Now factor the graph of any $f : X \to Y$ in W as $X \xrightarrow{i} P \xrightarrow{q} X \times Y$, with $i \in W$ and $q \in \mathrm{Fib}_W$. Product projections belong to Fib_W, so both $r = \mathrm{pr}_X\, q$ and $p = \mathrm{pr}_Y\, q$ are intrinsic and in W. They are therefore universally in W. Pulling p back to Y/y shows P/y is W-aspherical, hence W_{asph}-aspherical. Absolute Theorem A gives $p \in W_{\mathrm{asph}}$, and likewise $r \in W_{\mathrm{asph}}$. From $ri = 1$ and $pi = f$, it follows that $f \in W_{\mathrm{asph}}$. Thus $W = W_{\mathrm{asph}}$; Proposition 6.1.8 and Theorem 6.1.11 of Cisinski (2006) identify this with properness when W is accessible.

Here is the required accessibility argument for the two rational localizers. Choose an uncountable regular cardinal κ greater than $|\mathbb{Q}|$. For a weak arrow $u : A \to B$ start with any subfunctor of size less than κ. For rational homology, enlarge it by choosing finite chain witnesses for surjectivity and injectivity on every homology group. For rational local coefficients, first choose finite zigzags witnessing bijectivity on components, then do the same for the rational chain complexes of every based loop space of $\mathrm{Ex}^\infty NA \to \mathrm{Ex}^\infty NB$. These are the exact tests of D.5. Each simplex in the Kan replacement, each loop simplex, and each chain has finite support in the original nerves. There are fewer than κ witnesses at each stage. Closing their arrows under composition and repeating countably often produces a weak subfunctor of size less than κ. The weak subfunctors so obtained form a κ-filtered system whose union is u: any fewer than κ subfunctors have an upper bound by the same enlargement. The two tests commute with filtered colimits, and every fundamental localizer is closed under filtered colimits by Cisinski (2006, Corollary 4.2.22). Consequently representatives of the small weak subfunctors generate each localizer as a fundamental localizer. This proves the needed set-generation, without inferring it merely from closure under filtered colimits.

To see they are not proper, use Lemma 9.4.5 of Cisinski (2006): a proper localizer contained in W_r makes a weak equivalence with local Kan target a strong r-equivalence, in particular surjective on π_{r+1}. For $W_{\mathbb{Q}}$ take $K(\mathbb{Z}, 1) \to K(\mathbb{Q}, 1)$ and $r = 0$. For $W^{\mathrm{lc}}_{D(\mathbb{Q})}$ take $K(\mathbb{Z}, 2) \to K(\mathbb{Q}, 2)$ and $r = 1$. These are equivalences for the indicated localizers, but their displayed homotopy-group maps are not onto. Their targets are local: the homotopy groups of maps into $K(\mathbb{Q}, j)$ are rational cohomology groups in degrees at most j, invariant under the corresponding equivalences. In the second case rational local coefficient equivalences imply ordinary rational homology equivalences by the universal-cover spectral sequence. This verifies the local-target hypothesis, and finishes the table. □

D.vi Complete covering tests for ordinary integral and rational homology

D.12 [Proved]. Trivial monodromy is necessary and sufficient for integral and rational homology.

Let $p : E \to B$ be a categorical covering, that is, a discrete fibration and discrete opfibration. For either $k = \mathbb{Z}$ or $k = \mathbb{Q}$, it belongs to $\mathrm{Fib}_{W_{Hk}}$ *if and only if its monodromy on every fibre set is trivial. Equivalently, over each connected component B_0 it is isomorphic to a product $B_0 \times S_0 \to B_0$, where S_0 may be empty or infinite. In particular a nonempty connected covering of a connected base is intrinsic for either of these two homology localizers exactly when it is isomorphic over the base to its identity covering. This assertion is specific to these coefficients; D.8 gives the contrary mod-p behavior.*

Proof. A categorical covering defines a set-valued local system on the groupoid completion of B. Conversely the unique source and target lifts reconstruct the covering from this local system. It is a product on a connected component precisely when every based loop acts trivially; choose paths from a basepoint to give a compatible trivialization. A homology equivalence with either indicated nonzero coefficient ring is a bijection on components and is a homology equivalence on each component. Products with a fixed discrete set, hence arbitrary disjoint unions of copies,

preserve such equivalences. This proves sufficiency after every base change, including empty and infinite fibres.

For necessity choose a loop in BB acting by a nonidentity permutation P of the fibre set S. First let $k = \mathbb{Q}$. Pull the covering back to a circle representing this loop, and compare it with its pullback by the degree-m map of the circle. Every nonzero degree is a rational homology equivalence. The components of these two coverings are the orbits of P and P^m, respectively. If P has a finite orbit of length $d > 1$, take $m = d$; this orbit splits into d distinct P^m-orbits. If it has an infinite orbit, take $m = 2$; that orbit splits into two. The induced map on zeroth rational homology has a nonzero kernel. Thus the pullback is not a rational homology equivalence.

For $k = \mathbb{Z}$ we use a different witness. Fix $n \geq 2$. In the universal cover of $S^1 \vee S^n$, the Hurewicz isomorphism identifies π_n with $\mathbb{Z}[t, t^{-1}]$, with t acting by the circle deck transformation. Attach one $(n+1)$-cell along the class $2t - 1$, and call the resulting CW space X. Collapsing the n-sphere to the basepoint and extending across the attached cell gives a map $u : X \to S^1$ with the original circle as section. The extension exists because $\pi_n S^1 = 0$. Its relative cellular chain complex has two terms, with differential the augmentation of $2t - 1$, namely 1. Hence the inclusion $S^1 \to X$ and its retraction u are integral homology equivalences.

Pull the covering over S^1 back along u. Relative to the lifted circle its two-term cellular chain complex has differential

$$2P - 1 : \mathbb{Z}[S] \longrightarrow \mathbb{Z}[S].$$

Decompose S into permutation orbits. On an orbit of length d, its cokernel is $\mathbb{Z}/(2^d - 1)$: the relations $2e_{i+1} = e_i$ reduce to one generator and $(2^d - 1)e_0 = 0$. It is nonzero for $d > 1$. On an infinite orbit the cokernel is $\mathbb{Z}[t, t^{-1}]/(2t - 1) \cong \mathbb{Z}[1/2]$, also nonzero. Therefore the relative homology of the pulled pair is nonzero whenever P is nonidentity. Since the lifted projection has the lifted circle as a section, it cannot be an integral homology equivalence.

These space-level witnesses supply actual small-category tests over the original B, rather than tests over a replacement base alone. Apply the slice realization construction in D.6 to each triangle over BB, and then the category-of-simplices functor followed by last-vertex evaluation. A covering has the homotopy-pullback property: its unique transport lifts give the homotopy-fibre comparison, or this follows by integrating its set-valued local system. Thus its strict categorical pullbacks realize the two coverings in the argument. The first categorical map is a W_{Hk}-equivalence and its pullback is not, proving necessity. The space X is completely specified by the indicated attaching homotopy class; any representative gives the same relative chain complex, and the slice construction may use its full singular complex, so no unstated finiteness claim is involved.

The program `code/root/verify_coverings.py` checks the integral matrices on cycles of lengths one through twelve by Smith normal form, verifies augmentation one, and records determinants $2^d - 1$. This is a check of the displayed chain calculation; the argument above, including infinite orbits and every base change, proves the universal assertion. □

E The full path categories

E.i The three-object marked chain

E.1 [Proved]. All alternating fixed-shape path spaces for the marked chain.

Let $P = [2]$ and let W consist of its identities and $0 \to 2$. For $n \geq 2$, let $F_n(a, b)$ be the full poset of paths of type $(-, +)^n$ from a to b, with backward arrows in W and arbitrary endpoint-fixed vertical arrows. Every $F_n(a, b)$ is contractible, except $F_n(1, 1)$, which is the disjoint union of an isolated point and a contractible nonempty poset. These two components are preserved by identity

padding. The corresponding assertions hold integrally in every homological degree and for all homotopy groups, not only through a finite range.

Proof. An object is a sequence $(x_0, \dots, x_{2n})$ with $x_0 = a$, $x_{2n} = b$. At every negative edge the pair (x_{2j}, x_{2j+1}) is one of $(0,0), (1,1), (2,2), (2,0)$; at every positive edge $x_{2j+1} \le x_{2j+2}$. Morphisms are pointwise inequalities.

If $a = 0$, the path $(0, 0, \dots, 0, b)$ is least. If $a = 2$, the path $(2, 0, \dots, 0, b)$ is least. For $a = 1, b = 2$, the path $(1, 1, 2, \dots, 2)$ is greatest. For $a = 1, b = 0$ and $n \ge 2$, the path $(1, 1, 2, \dots, 2, 0, 0)$ is greatest. Each satisfies all edge conditions, and comparison coordinate by coordinate proves its stated universal property.

It remains to treat $(a, b) = (1, 1)$. Any nonconstant path must contain both a 2 and a 0: it can leave 1 only by a positive move to 2, and it can return below 2 only through the allowed negative move $2 \to 0$. Such a path is incomparable with the constant path, which is therefore an isolated component. Let Q_n be the remaining full subposet. If $p \in Q_n$, its first zero coordinate $k(p)$ is odd and at least 3. Define $R(p)$ by replacing coordinates $2, \dots, k(p) - 1$ by 2 and leaving every other coordinate unchanged. This is an admissible path, since its modified prefix is $1, 1, 2, \dots, 2, 0$, followed by the original suffix.

We have $p \le R(p)$. If $p \le q$, then $k(p) \le k(q)$, because every zero of q is a zero of p. Before $k(p)$ both modified paths have the same maximal interior value. Between $k(p)$ and $k(q)$, the coordinates of $R(q)$ are 2; from $k(q)$ onward the comparison is the original comparison $p \le q$. Hence R is monotone. Its image is exactly the principal upper set of

$$u_n = (1, 1, 2, 0, \dots, 0, 1).$$

Indeed a path above u_n has coordinate 2 equal to 2, and must remain 2 until its first zero; it is therefore fixed by R. Conversely every $R(p)$ lies above u_n. Thus R is a closure operator with image having initial object u_n, and NQ_n is contractible. Padding a path by a final identity pair preserves whether it is constant. It therefore preserves the two displayed components. This proves all claims, including the eventual stability in n. □

E.2 [Proved]. The comparison J is an equivalence for the marked chain.

For $([2], \{\mathrm{id}, 0 \to 2\})$, the specified comparison from weak-vertical paths to full paths is a weak equivalence for every pair of endpoints. Its infinity-localization is equivalent to the ordinary walking-retraction category, with objects A, B, arrows $i : A \to B$, $r : B \to A$, and $ri = 1_A$. Its mapping spaces are discrete; $\mathrm{End}(B) = \{1_B, ir\}$*, and every other hom-set is a singleton. The objects $0, 2$ map to A and 1 maps to B.*

Proof. We first specify the passage from E.1 to the all-type category. Let Ty have sign words as objects and monotone endpoint-preserving type maps as arrows. Enrich each hom-set to the poset of such maps and endpoint-fixed transformations. We check its contraction explicitly. If the target word is empty, this poset is terminal. If the first source run has sign opposite to the first target run, every type map is constant on that source run, so deleting it is an isomorphism of hom-posets. Otherwise let the first source and target runs have vertex intervals $[0, a]$ and $[0, b]$ with common sign ε. The map $h(j) = \max(j, b)$ collapses the first target run and fixes the remaining vertices. For a type map F, put $RF(0) = 0$ and $RF(i) = hF(i)$ for $i > 0$. No map crosses the first target turning vertex during its first source run. Consequently RF has that first run equal to $(0, b, \dots, b)$; it is a monotone endpoint-preserving functor. Away from vertex 0, apply h to transformations. If $\varepsilon = +$, the pointwise arrows give $1 \Rightarrow R$; if $\varepsilon = -$, they give $R \Rightarrow 1$. At both endpoints these components are identities, and on the first source edge the required square commutes because the target is thin. The image is the full subposet with the prescribed

first run, on which R is the identity. Deleting the two prescribed runs reduces the number of target runs and completes the induction. Each type map preserves backward edges as a string of backward edges or an identity; thus this argument does not introduce unmarked backward edges. Precomposition of full path categories is an action of these hom-posets; arbitrary vertical components are exactly what makes this enriched action defined. Passing to classifying spaces therefore extends the diagram coherently to the reachability preorder of types, since the enriched mapping spaces are empty or contractible.

The functor from $\mathrm{Ty}^{\mathrm{op}}$ to this preorder is homotopy final. At a type σ, its comma category is the opposite of the full subcategory of types admitting a map to σ. The functor $\tau \mapsto \sigma * \tau$ has transformations to τ and to σ by block collapse, so this comma category is contractible. The alternating sequence $t_n = (-,+)^n$ is also final in the preorder: every finite sign word is obtained from some t_n by collapsing unused edges and subdividing retained runs, and the permissible values of n form a nonempty final segment of $\mathbb{N}$. Thus the canonical Grothendieck homotopy-colimit comparison gives

$$B\,\mathrm{Ch}_W(P;a,b) \simeq \operatorname*{hocolim}_n BF_n(a,b).$$

This is the enriched-type argument of PATHS C.P2–C.P3, with its comma contractions and the necessary arbitrary-vertical action retained. E.1 shows that the right side is homotopy discrete, with two components exactly at $(a,b) = (1,1)$.

We recall why the specified map J has a retraction on homotopy types, so that no arbitrary comparison replaces it. Let $\mathcal{L}$ be the infinity-localization. For any type τ, invert its backward edges; its interval becomes the ordinal whose length is the number of forward edges. The universal localization property identifies the corresponding relative diagram category in $\mathcal{L}$ with the ordinal diagram category. Endpoint composition then gives a functor to $\mathrm{Map}_{\mathcal{L}}(a,b)$. These equivalences are natural in type maps, so their inverses can be taken as one inverse equivalence of diagrams. Groupoid completion and the type homotopy colimit give a map $Q : B\,\mathrm{Ch}_W(P;a,b) \to \mathrm{Map}_{\mathcal{L}}(a,b)$. On weak-vertical paths QJ is the canonical hammock comparison, a weak equivalence. The restricted comparison is Dwyer–Kan's hammock mapping-space construction; its Grothendieck form follows from the Reedy-cofibrant diagram of fixed-shape nerves, whose latching inclusions insert identity edges. This is PATHS B.2 and B.6 with the actual comparison maps retained.

Consequently J is split injective on every higher homotopy group. Since its target has no higher homotopy, neither does its source. Both component sets are the morphisms of the ordinary localization: composing paths after formally inverting W gives that identification, and elementary composition, identity insertion, and cancellation of a marked backtrack give exactly its relations. Thus J is also bijective on components and is a weak equivalence. An infinity-category with discrete mapping spaces is equivalent to the nerve of its homotopy category. Identifying 0 with 2 using the marked composite leaves exactly i, r with $ri = 1$, which is the displayed walking retraction. □

The program `code/AE/path_chain.py` constructs these posets independently for lengths $2, 4, 6, 8, 10, 12$, checks all pointwise comparisons, and records elementary beat-point-removal certificates. At length 6, the object counts, with row a and column b, are

$$\begin{pmatrix} 6 & 9 & 13 \\ 4 & 6 & 9 \\ 9 & 13 & 19 \end{pmatrix},$$

and the component counts are 1 in every entry except $(1,1)$, which has 2. The proof, rather than numerical stabilization, establishes the unbounded limit. These are the cofinal fixed-shape categories, not the full subcategory containing every word of length at most six; the latter finite computation is not claimed here.

E.ii The finite catalogue and its limits

E.3 [Proved]. Complete relative-poset catalogue through four objects.

There are respectively $1, 3, 14, 102$ *isomorphism classes of pairs* (P, W) *on* $1, 2, 3, 4$ *objects, where* P *is a poset and* W *contains identities and satisfies two-out-of-three. The attached catalogue lists every pair and also the unique empty relative poset, for* 121 *types in all. The program computes all endpoint full-path posets of types* $(-,+)^n$*,* $n = 1, 2, 3$*, giving* 5313 *rows. In* 5307 *rows all connected components are certified contractible by beat-point removal. Exactly six rows have a non-discrete remaining core, all for* $P = [3]$ *and* $W = \{\mathrm{id}, 0 \to 2, 1 \to 3\}$*. These are bounded calculations, not a proof of the all-type comparison for all* 120 *pairs.*

Proof. Every finite poset has a linear extension. The program therefore enumerates each subset of pairs (i, j) with $i < j$, retains exactly the transitive relations, and identifies relations under every permutation of the object set. It then enumerates every subset of the nonidentity arrows and retains exactly those for which no composable triple has exactly two of its three arrows marked. This is precisely the two-out-of-three condition, including composition closure. A second canonicalization under all permutations identifies the relative pairs. Thus the finite catalogue is exhaustive.

For every endpoint pair and every stated type, a recursive traversal lists exactly the legal vertex sequences. Pointwise order gives exactly its arrows, because a poset has at most one arrow of each type and every naturality square commutes automatically. Each deleted beat point has either a greatest strict predecessor or a least strict successor; sending it to that object and fixing all others is a deformation retraction of nerves. The recorded sequence of these deletions therefore certifies the stated contractible components. The six exceptional cores are retained in full in the output. Four are the four-point circle. The other two have 10 vertices, 25 edges, and 16 triangles. Exact integral Smith forms give vanishing reduced homology, and the spanning-tree fundamental-group presentation has 16 generators which the recorded Tietze eliminations remove completely. They are therefore simply connected and acyclic, hence contractible by Hurewicz and Whitehead. The final table `relative_posets_homotopy_table.csv` records integral homology through degree three and the fundamental group at every basepoint for all 5313 rows. Stopping the beat-point algorithm alone was not used to conclude this. □

E.4 [Proved]. Transient higher homotopy in a four-object calculation.

For $P = [3]$ *marked only at identities,* $0 \to 2$*, and* $1 \to 3$*, the full fixed-shape path poset* $F_2(1, 2)$ *of type* $(-,+)^2$ *has classifying space* S^1*, whereas the infinity-localization of* (P, W) *is the terminal infinity-category. Thus a fixed finite path shape need not already give the localization mapping space, even in this small example.*

Proof. The path poset has exactly four objects

$$(1, 1, 1, 1, 2), \quad (1, 1, 3, 1, 2), \quad (1, 1, 2, 0, 2), \quad (1, 1, 2, 2, 2).$$

The first and third are minimal and each is below both of the other two; there are no other nonidentity comparisons. Its nerve is the four-edge circle. Write the successive generating arrows of $[3]$ as f, g, h. In the localization, gf and hg are equivalences. Hence g has both a left and a right homotopy inverse and is an equivalence; then so are f and h. The localization is therefore the groupoid completion of $[3]$, which is contractible. The program verifies the four objects, all four comparisons, and the exact finite-poset description. □

E.5 [Refuted]. No universal finite length bound for all four-object cases.

The full path category for every relative poset on at most four objects is not computed by a finite all-word length cutoff, even at the level of components.

Proof. Take the four-object poset with two minimal objects a_0, a_1, two maximal objects b_0, b_1, and all four relations $a_i < b_j$, and mark every arrow. Its nerve is a four-edge circle. Since all arrows are marked, the path comparison is the identity and its endomorphism space at any object is ΩS^1, with components indexed by $\mathbb{Z}$. At any finite word-length cutoff, there are finitely many path objects, so its nerve has finitely many components. Its inclusion into the full path category cannot be a weak equivalence. The order relation and its circle nerve are explicitly checked by the finite catalogue. This does not obstruct computing a limit by a sequence of cutoffs; it refutes treating one such cutoff as a proof of that limit. □

E.6 [Open]. The general comparison and the additive and pointed refinements.

The specified map J for arbitrary two-out-of-three relative categories, its all-endpoint assertion for the remaining four-object relative-poset cases, and the literal all-vertical and ind-stage transfers in PATHS C.9–C.10 remain unresolved here. E.7–E.8 below give the additive and pointed refinements of the corrected localized hom-spaces. E.2 settles the particular three-object example that PATHS B.8 singled out. E.3 does not settle the unbounded four-object assertion. The standard weak-vertical definition computes the localization and has strict concatenation, but a general additive or smash-product refinement requires additional coherent structure; none is deduced solely from the computations above.

E.iii Additive and pointed refinements of the corrected model

E.7 [Proved]. The corrected additive refinement under direct-sum stability.

Let k be a commutative ring in the fixed universe, let C be a small additive k-linear category, and let W be a wide subcategory closed under finite direct sums of arrows. The infinity-localization $\mathcal{L} = C[W^{-1}]$ has a coherent enrichment in the connective derived infinity-category of k-modules, written $D^{\leq 0}(k)$ in cohomological grading. Its underlying mapping spaces are the localization mapping spaces, and its composition is k-bilinear. Consequently the corrected weak-vertical path categories admit composition-compatible $D^{\leq 0}(k)$-refinements whose underlying spaces are their nerves. No model-category or suspension–loop hypothesis is required for this assertion about the corrected model.

Proof. We first check precisely which algebraic structure survives localization. Localization commutes with finite products of relative categories. Indeed a functor $C \times D \to E$ inverts $(W, 1)$ and $(1, V)$ exactly when its curried functor takes values in the full subcategory $\mathrm{Fun}_V(D, E)$ and inverts W there. Applying the universal property twice gives the product assertion, including natural transformations and their higher coherences.

The direct-sum functor $\oplus : C \times C \to C$ is relative by the hypothesis, and the diagonal functor is relative. Its ordinary adjunctions $\oplus \dashv \Delta \dashv \oplus$ descend to the localizations: the functors descend, and full faithfulness of restriction along a localization extends their units, counits, and triangle identities. Similarly the two adjunctions defining the zero object descend. Hence $\mathcal{L}$ has the finite biproducts induced from C.

Let $\mathcal{F}_k$ be the small category of finite free k-modules with all matrices, monoidal under tensor product. The ordinary k-linear category C has an action

$$\mathcal{F}_k \times C \longrightarrow C, \qquad (k^n, a) \longmapsto a^{\oplus n}.$$

A matrix acts by its scalar multiples of identity arrows. Matrix multiplication and the distributivity isomorphisms supply its associative and unital module coherence. For every $V \in \mathcal{F}_k$, the functor $V \otimes -$ preserves W, by finite-direct-sum stability. The product-localization argument therefore gives a coherent action of $\mathcal{F}_k$ on $\mathcal{L}$, preserving finite direct sums in both variables.

For $a, b \in \mathcal{L}$ define the presheaf

$$H(a,b) : \mathcal{F}_k^{\mathrm{op}} \longrightarrow \mathcal{S}, \qquad H(a,b)(V) = \mathrm{Map}_{\mathcal{L}}(V \otimes a, b).$$

It takes finite direct sums in $\mathcal{F}_k$ to products, because the action takes them to coproducts in $\mathcal{L}$. The infinity-category of these product-preserving presheaves is $D^{\leq 0}(k)$. To see the precise input, $\mathcal{F}_k^{\mathrm{op}}$ is the algebraic theory of k-modules. Its strict simplicial algebras are simplicial k-modules. Badzioch's rigidification theorem, Theorems 1.3 and 6.4, identifies homotopy product-preserving diagrams with those strict algebras up to weak equivalence; Dold–Kan identifies the latter with connective chain complexes up to quasi-isomorphism. Thus $H(a,b)$ is an actual object of the claimed derived infinity-category, not only an abelian group structure on π_0. Evaluation at k gives exactly $\mathrm{Map}_{\mathcal{L}}(a,b)$.

We also construct composition, since objectwise rigidification alone would not suffice. For $U, V \in \mathcal{F}_k$, composition and the action give the natural pairing

$$\mathrm{Map}(U \otimes b, c) \times \mathrm{Map}(V \otimes a, b) \longrightarrow \mathrm{Map}((U \otimes_k V) \otimes a, c), \quad (f,g) \longmapsto f \circ (U \otimes g).$$

This defines, by Day convolution, a morphism

$$H(b,c) \otimes_k^{\mathbf{L}} H(a,b) \longrightarrow H(a,c).$$

Here the tensor product on product-preserving presheaves is the reflected Day tensor. The reflection is monoidal: its generating localization maps are $y(U) \amalg y(V) \to y(U \oplus V)$ and $\varnothing \to y(0)$; Day tensoring with a representable $y(T)$ carries them to maps of exactly the same form, because $T \otimes_k -$ preserves finite direct sums. Tensoring with an arbitrary presheaf follows by colimits. Thus the pairing has a uniquely induced morphism after this reflection, coherently for all iterated compositions. The reflected tensor agrees with derived tensor because it sends representables associated with U, V to the representable associated with $U \otimes_k V$ and preserves sifted colimits separately. Every simplicial k-module is a sifted colimit of finite free modules, as follows from its usual simplicial free resolution and expressing free modules as filtered colimits of their finite free submodules. These two properties determine the tensor product and the displayed morphism. Equivalently, the matrix operations show directly that the pairing is linear separately in both variables, which is precisely the relation imposed by the reflected Day tensor.

Associativity is the action associativity together with associativity of composition in $\mathcal{L}$; all higher compatibility diagrams are images of those same module-action diagrams. The identity of a gives the unit $k \to H(a,a)$, and its unit diagrams are the action unit diagrams. This proves a coherent enriched structure. The canonical composition-compatible hammock comparison identifies the underlying spaces of these hom-objects with the nerves of corrected weak-vertical path categories. Transporting the enriched structure along that specified comparison gives the claimed refinements; no equivalence for arbitrary vertical arrows is used. □

E.8 [Proved]. The corrected pointed refinement after localization.

Let C be a small category with a zero object 0, and let W be any wide subcategory. Its infinity-localization is pointed. Its localization mapping spaces, and therefore the corrected weak-vertical path spaces with the zero-arrow basepoints, admit associative unital composition through the smash product, coherently in all variables. This assertion concerns the localized hom-spaces; it does not assert a strictly absorbing operation on every prescribed finite path stage.

Proof. The two functors $\{0\} \to C$ and $C \to \{0\}$ are relative, and the former is both left and right adjoint to the latter. The adjunction descent argument in E.7 therefore makes the image of 0 both initial and terminal in $\mathcal{L} = C[W^{-1}]$. In particular the spaces $\mathrm{Map}_{\mathcal{L}}(0, a)$ and $\mathrm{Map}_{\mathcal{L}}(a, 0)$ are contractible for every a.

The zero map $a \to b$ is the composite through 0. On the wedge

$$(\mathrm{Map}(b, c) \times \{0\}) \cup (\{0\} \times \mathrm{Map}(a, b)),$$

composition factors through 0. Its contraction to the zero map is induced by the contractible incoming and outgoing mapping spaces of 0. On the intersection the two contractions agree coherently, since both are the same composite through the zero object. The universal property of the homotopy pushout defining smash therefore gives

$$\mathrm{Map}(b, c) \wedge \mathrm{Map}(a, b) \longrightarrow \mathrm{Map}(a, c).$$

For any finite composable sequence, the same construction contracts the entire fat wedge: if several inputs are zero, collapse all intervening maps through 0. The spaces of choices made through 0, and their compatibility extensions made through 0, are contractible, because the relevant mapping spaces to and from 0 are contractible. This is a statement about these factorizations through the zero object; it does not assert contractibility of the space of all nullhomotopies in an arbitrary mapping space. Associativity of composition supplies the face compatibilities, and identities supply degeneracies. This gives all the coherences of an enrichment over pointed spaces with smash tensor and unit S^0. The hammock comparison sends the one-edge zero-arrow path to the specified zero map and respects concatenation, so it identifies the corrected path hom-spaces with this pointed enrichment. □

The original all-vertical comparison and the literal ind-stage assertion remain separate. In particular, a weak equivalence from the colimit of based path stages to a pointed localization hom-space does not provide a uniform later stage for every required absorption homotopy on an entire infinite stage. E.8 does not supply that missing uniformity and is not recorded as a proof of the full ind-pointed assertion in PATHS C.10.

F Total asphericity and ordinary colimits

F.1 [Proved]. The action criterion used in this part.

Here total asphericity means the condition of XII ms. 88 tested by all ordinary final images; we do not substitute the different diagonal-asphericity terminology used elsewhere in the literature. For a nonempty connected small category A, total W_∞-asphericity is equivalent to every connected component of $\mathrm{El}_A(H)$ being weakly contractible, for every covariant $H : A \to \mathrm{Set}$. It is also equivalent to the same condition with integral acyclicity in place of weak contractibility. Exactness of $\mathrm{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ implies these conditions.

Proof. If A is totally integrally aspherical, it is integrally acyclic by the identity test. For a connected nonempty category of elements $E = \mathrm{El}_A(H)$, adjoin an object v to A with arrows $v \to a$ the elements of $H(a)$, and composition induced by H. The inclusion $A \to A_H$ is final: its comma at v is E, and the other commas have initial objects. Thus BA_H is acyclic. Its nerve is the homotopy pushout $BA \leftarrow BE \to *$. The reduced homology sequence gives acyclicity of BE. For a component of an arbitrary action, restrict H to that component.

This property passes to every component E of an action category. Indeed for $K : E \to \mathrm{Set}$ the action $L(a) = \coprod_{h \in H_E(a)} K(a, h)$ on A has category of elements exactly $\mathrm{El}_E(K)$, with all its morphisms. Now take $G = \pi_1 BE$. For a cyclic subgroup $C \leq G$, the action on cosets in the

groupoid completion of E defines a connected covering category with fundamental group C. Its first integral homology must vanish by the inherited property, but that homology is C. All cyclic subgroups are therefore trivial. Hence BE is simply connected and acyclic, and homological Whitehead makes it weakly contractible.

Conversely every comma $b\backslash A$ of an ordinary final functor $A \to B$ is a connected nonempty action category, for the action $a \mapsto B(b, f(a))$. The assumption and Quillen's Theorem A make $BA \to BB$ a weak equivalence. The singleton action makes BA contractible. Thus every final image is contractible, as required. Finally, for any $F : \mathrm{El}_A(H) \to \mathrm{Ab}$, the functor $(\Sigma_H F)(a) = \bigoplus_{h \in H(a)} F(a, h)$ is exact and has $\operatorname{colim}_A \Sigma_H F = \operatorname{colim}_{\mathrm{El}_A(H)} F$. Exactness on A implies exactness on its action categories and on their components. Apply this to the usual representable bar resolution of the constant integral module; its derived colimits are integral nerve homology. This proves the last assertion, and also specifies the variance of every action used. □

F.2 [Proved]. What totality forces before the affine gap.

A totally W_∞-aspherical category has common successors for every finite family of objects. For every original parallel pair $a, b : p \to q$, there are an object r and $t \in \mathbb{Z}[A(q, r)]$ of augmentation one such that $ta = tb$.

Proof. If two objects have no common successor, their upward closures U, V are disjoint cosieves. Start with the constant two-point action and identify the two points exactly on $U \cup V$. Its element category is the double of A glued along $U \amalg V$. Nerves form the corresponding homotopy pushout. Since BA is contractible and both U, V are connected, the Mayer–Vietoris sequence gives a nonzero first homology group of the double, contrary to F.1. Induction gives finite common successors. For the parallel pair form the quotient set action H of $A(p, -)$ generated by $a \sim b$. In the exact sequence

$$0 \longrightarrow I \longrightarrow \mathbb{Z}[A(p, -)] \longrightarrow \mathbb{Z}[H] \longrightarrow 0,$$

I is generated by $\delta = a - b$ at q. The middle diagram is projective, and F.1 gives $L_1 \operatorname{colim} \mathbb{Z}[H] = 0$. The image of $\operatorname{colim} I$ in $\operatorname{colim} \mathbb{Z}[A(p, -)] = \mathbb{Z}$ is zero, so $\operatorname{colim} I = 0$. Write the element δ as a finite sum of the relations $I(v_j)x_j - x_j$ in the direct-sum presentation of this colimit. Choose a common successor r for their finitely many objects and arrows $d_s : s \to r$. Write $x_j = \sum_l n_{jl} c_{jl} \delta$, with $c_{jl} : q \to \mathrm{dom}(v_j)$. Then

$$t = d_q - \sum_{j,l} n_{jl} (d_{\mathrm{cod}(v_j)} v_j - d_{\mathrm{dom}(v_j)}) c_{jl}$$

has augmentation one and satisfies $t\delta = 0$. All terms are arrows $q \to r$, so this identity is in the specified free abelian group. □

F.3 [Proved]. The requested injection and free-category examples.

Let A_S be the full subcategory of Δ_{inj} on $[n]$, $n \in S \subseteq \mathbb{N}$. It is totally W_∞-aspherical exactly when S is a singleton or unbounded. In these cases its integral colimits are exact. The category of finite sets and injections, with or without the empty set, is not totally W_∞-aspherical. The one-object category of injective self-maps of $\mathbb{N}$ is not totally W_∞-aspherical. For the free category on any small directed graph, total W_∞-asphericity is equivalent to filteredness; local finiteness is unnecessary.

Proof. We first give the affine cones needed for unbounded S. For each $n \geq 0$ and $k \geq 1$, there is a homogeneous integral polynomial $C_{n,k}(z_0, \ldots, z_{n+1})$ of value one at $(1, \ldots, 1)$ such that $y_i^k D_i C_{n,k}$

is independent of i, where D_i identifies z_i, z_{i+1} and relabels the variables as $y_0, \dots, y_n$. For $n = 0$ take 1. Inductively set $Q = C_{n-1,2k}$ and prescribe $D_i C_{n,k} = Q \prod_{l \neq i} y_l^k$. The prescriptions agree on pairwise intersections: for adjacent merged pairs this is immediate; for separated pairs the equality is precisely $w_q^{2k} D_q Q = w_p^{2k} D_p Q$ from the induction hypothesis, multiplied by the common remaining factors. In coordinates $x_i = z_i - z_{i+1}$ and $u = z_{n+1}$, these are compatible restrictions to the coordinate hyperplanes $x_i = 0$. If Φ_T is their common restriction to $x_i = 0$, $i \in T$, lift it to the full polynomial ring by making it independent of the omitted coordinates x_i. Then inclusion–exclusion $C = \sum_{\varnothing \neq T} (-1)^{|T|+1} \Phi_T$ extends them. Substitution back into z preserves integrality and homogeneity. Its degree is $k(2^{n+1} - n - 2)$.

An injection $g : [n] \to [M]$ corresponds to its gap monomial

$$z_0^{g(0)} z_1^{g(1)-g(0)-1} \cdots z_n^{g(n)-g(n-1)-1} z_{n+1}^{M-g(n)}.$$

Deleting $g(i)$ corresponds to $P \mapsto y_i D_i P$. Thus $C_{n,1}$ gives an integral affine arrow $t : [n] \to [2^{n+1} - 2]$ for $n \geq 1$ equalizing all cofaces. More explicitly, after r deletions the common gap polynomial is

$$\Big(\prod_j z_j^{2^r - 1} \Big) C_{n-r,2^r}.$$

One further deletion identifies two exponents and contributes one additional gap, so its varying factor is $y_i^{2^r} D_i C_{n-r,2^r}$, independent of i by construction. The formula therefore follows by induction, for every deletion sequence. Every injection into $[n]$ is such a sequence of cofaces, so t equalizes every parallel pair. If S is unbounded, postcompose with an injection into some $[N]$, $N \in S$, above this target. Hence the category $\mathrm{Aff}_{\mathbb{Z}}(A_S)$ is filtered: it has common successors and the same t equalizes every parallel affine pair, because both have total coefficient one. Extend any abelian diagram linearly to this affine category. Its cocones are unchanged, and filtered colimits are exact. F.1 proves totality. The polynomial identities and all parallel injections are checked independently for $n \leq 3$ by the program; the induction proves all n. A singleton S gives the terminal category. If S is finite with at least two elements and maximum N, there are distinct injections $[m] \to [N]$ from some smaller $m \in S$, whereas the only arrow out of $[N]$ is its identity. They cannot satisfy F.2. The empty category is not total. This proves the complete stated classification.

For finite sets let $H(X)$ be the set of two-element subsets of X. The category of elements consists of (X, T), $|T| = 2$, with injections carrying T onto the target distinguished subset. The functor $(X, T) \mapsto (T, T)$ retracts it onto the groupoid of two-element sets. The inclusions $T \hookrightarrow X$ give a natural transformation from the inclusion of this retract to the identity. Thus the entire element category has type BC_2, and is connected with $H_1 = \mathbb{Z}/2$. This contradicts F.1. The finite restriction through cardinality three is separately checked by its complete edge-path presentation and Tietze eliminations.

For the injection monoid, let $s(n) = n + 1$. An affine equalizer of 1 and s would be an invariant finite-support integral signed measure of mass one for $f \mapsto fs$ on injective self-maps. Such a self-map has no periodic point, since $f(n + r) = f(n)$ is impossible for $r > 0$. An invariant finite-support signed measure is supported on finite cycles: each point in its finite support must have a predecessor in that support, and counting predecessors forces the self-map to permute the support. No such measure exists. F.2 excludes totality.

In a free category, no distinct parallel paths admit an affine equalizer. If their lengths differ, the terms of maximal length in the product with the longer path cannot cancel. If their lengths agree, concatenation with the two different initial words has disjoint images in the path basis. Thus F.2 forces the free category to have at most one arrow between any two objects. Its common-successor property then makes it filtered. Conversely filtered categories have exact integral colimits and hence are total by F.1. This proves the last assertion in its stated generality. □

F.4 [Proved]. Weakly terminal objects and the finite-object case.

A totally W_∞-aspherical category with a weakly terminal object is filtered. Consequently the universal Cat-colimit property in Part F(3) implies filteredness for every category with a weakly terminal object, and for every nonempty connected category with finitely many objects.

Proof. We include the monoid step because signed coefficients cannot be replaced by positive ones. Suppose a monoid M has, for every x, a, an affine t with $tx = txa$. The measure tx, of mass one and supported in Mx, is invariant under right multiplication by a. Its support is a union of finite periodic orbits and its coefficients are constant on each orbit, by the support-counting argument in F.3. Choose the least attainable positive period r in Mx. If a prime ℓ divides $r > 1$, choose $z \in Mx$ of period r and apply the hypothesis to z, a. The resulting invariant measure on Mz has mass one, so one of its orbit lengths s is not divisible by ℓ. But an element of that orbit is also fixed by a^r, since it lies in Mz. Its period divides $\gcd(r, s) < r$, a contradiction. Thus some y satisfies $yxa = yx$. Apply this with $x = 1$ and then with $x = y$ to equalize any given pair of elements of M. Hence M is filtered.

If q is weakly terminal in A, postcompose the affine equalizers supplied by F.2 back to q. The preceding lemma makes $\mathrm{End}_A(q)$ filtered. For $a, b : p \to r$, choose $d : r \to q$ and an affine endomorphism $t = \sum n_i c_i$ of q equalizing da, db. Filteredness of the endomorphism monoid gives z with $zc_i = z$ for every i. Since $\sum n_i = 1$, the genuine arrow $zd : r \to q$ equalizes a, b. Together with common successors this proves filteredness.

Now let I be nonempty and connected, and assume every diagram $I \to \mathrm{Cat}$ has its ordinary colimit computed by the canonical Grothendieck-construction comparison. Apply this to discrete-valued diagrams $H : I \to \mathrm{Set}$. It says that every component of $\mathrm{El}_I(H)$ is contractible. The constant singleton diagram also makes BI contractible. F.1 proves totality, so the preceding result applies when I has a weakly terminal object. In the finite-object case common successors provide such an object. Conversely filtered shapes satisfy the universal comparison: every finite list of objects, arrows, and relations in an ordinary category colimit is represented at a common stage, and every comparison comma category is filtered. Quillen's Theorem A proves the specified comparison is a weak equivalence. □

F.5 [Open]. The exact connected converse.

For a nonempty connected small category A with no weakly terminal object, we do not settle whether total W_∞-asphericity forces exactness of $\mathrm{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$. *The exact gap is whether the hypotheses of F.2 equalize every parallel pair in* $\mathrm{Aff}_{\mathbb{Z}}(A)$*, not merely every original parallel pair. F.3 resolves the listed injection and free-category candidates but does not remove this gap for a general countable category. Exact integral colimits remain a proved sufficient condition, and the affine-filtered criterion is the precise algebraic reformulation. The original claims of equality with filteredness are false by the unbounded A_S examples.*

F.6 [Open]. Total presheaves and composition of dwyerienne sieves.

We do not settle whether every totally $W_{\mathbb{Z}}$-aspherical presheaf is totally W_∞-aspherical, or equivalently the composition assertion specified in TOTASPH F.27. The proof of F.1 does not transfer: for a presheaf H on A, totality tests only those covariant actions on its element category pulled back from A, whereas the cyclic-cover argument needs arbitrary actions on that element category. The search over all monoids of order at most six and all presheaves of cardinality at most six was not performed in this return. The smaller monoid catalogue in Part A is not such a search and cannot certify this assertion. No acyclic-group example satisfying the required universal restricted-action test is provided here.

F.7 [Open]. Universal Cat-colimit shapes without a weakly terminal object.

For a connected countable category I without a weakly terminal object, and hence also in general, the implication from the universal ordinary-Cat-colimit comparison to filteredness remains unresolved here. F.4 gives a necessary totality condition and the full weakly-terminal case. The unbounded injective-ordinal examples do not refute this implication: the diagram $[n] \mapsto B\mathbb{N}^{n+1}$ has categorical colimit $B\mathbb{N} \simeq S^1$, whereas its Grothendieck construction has integral homology $\mathbb{Z}$ in every degree. On all of Δ_{inj} its degree-k fibre homology, for $k \geq 1$, is the projective representable $\mathbb{Z}[\Delta_{\mathrm{inj}}([k-1], -)]$; the homotopy-colimit spectral sequence collapses. On an arbitrary unbounded full subcategory A_S, the restricted diagram need not be representable. Instead its higher derived colimits vanish by exactness in F.3, and its colimit is $\mathbb{Z}$: any two marked k-element ordered subsets admit embeddings into a common larger ordinal identifying their marks, then into an object of A_S. The categorical colimit is still $B\mathbb{N}$, since the same embeddings identify all coordinate generators and introduce precisely their commutative-monoid relation. Thus all these shapes fail the premise. This prevents importing the nonfiltered total example as an answer to the different Cat-colimit question.

G Spaces over a base: two decisive tests

G.i The proposed interval realization does not work

G.1 (Refuted). The proposed assignment

$$R(F) = \operatorname*{hocolim}_{I \in \mathcal{I}(\mathbb{R})} F(I) \times I \longrightarrow \mathbb{R},$$

where $\mathcal{I}(\mathbb{R})$ consists of bounded nonempty open intervals and arrows are inclusions, need not have open-inverse-image homotopy cosheaf F. This failure already occurs for a represented cosheaf.

Proof. Let $F(U) = *$ if $0 \in U$, and $F(U) = \varnothing$ otherwise. This is the open-inverse-image cosheaf of the continuous map $\{0\} \to \mathbb{R}$. For an explicit verification of codescent, if $0 \notin U$, every term of the covering Cech diagram is empty. If $0 \in U$, the nonempty terms are indexed by tuples of members of the cover which contain 0; their Cech nerve is contractible. Its homotopy colimit is $*$.

In the usual topological bar realization of the displayed homotopy colimit, for any $x \in \mathbb{R}$ choose a bounded interval I containing both 0 and x. The summand $F(I) \times I$ of the space of zero-simplices then supplies a point over x. Geometric realization cannot remove this point. Taking $U = (1, 2)$ shows that $R(F) \times_{\mathbb{R}} U$ is nonempty, whereas $F(U) = \varnothing$. These spaces cannot be weakly equivalent. The same obstruction holds for a derived homotopy colimit in the slice, since its structure maps include the indicated summands. This disproves the proposed formula; it does not disprove essential surjectivity of the comparison functor. □

G.ii Open-test preservation exceeds shape submersions

G.2 (Proved). The continuous maps whose strict pullback preserves all open-test weak equivalences include maps which are not shape submersions in the sense of Volpe (2025, Definition 3.23). In particular, the projection from the Cantor space $K = \{0, 1\}^{\mathbb{N}}$ to a point is such a map.

Proof. Let $u : X \to X'$ be a weak homotopy equivalence. For every open $U \subset K$, the restricted pullback is $u \times 1_U : X \times U \to X' \times U$. The singular-complex functor preserves products, and products of Kan complexes preserve weak equivalences. Thus this is a weak equivalence, including when U is empty.

No nonempty open subspace V of K is essential. Indeed, if the constant-sheaf functor $\Delta : \mathcal{S} \to \mathrm{Shv}(V)$ had a left adjoint, it would preserve products. Put $D_n = \{0, 1\}$. The coordinate functions $V \to D_n$ give a section of $\prod_n \Delta D_n$. A section of $\Delta(\prod_n D_n)$ is locally constant as a function to the discrete set $\prod_n D_n$. Its image under the comparison to $\prod_n \Delta D_n$ would be the coordinate family only if it were the inclusion $V \to K$ on underlying sets. That inclusion is not locally constant: K has no isolated points, and the same holds for every nonempty open V. Hence Δ does not preserve this countable product. This contradiction proves the claim about essentiality. A shape submersion to a point requires a basis of essential neighbourhoods, so $K \to *$ is not one. □

G.3 (Open). G.1 refutes the specific proposed realization, not the comparison $\mathrm{Top}/\mathbb{R}[W_{\mathrm{open}}^{-1}] \to \mathrm{CoShv}(\mathbb{R}; \mathcal{S})$. Neither a replacement realization nor full faithfulness is proved here. Consequently the requested extension through intervals, circles, graphs, manifolds, CW complexes and the listed non-locally contractible spaces is also not settled. The dual raw-pullback recognition theorem in G(3), an exact characterization of all pullback-preserving maps in G(4), and the complete identification of the raw-space theories with the stable duality in G(5) remain open in this audit. G.2 refutes the proposed equality with Volpe's shape submersions. G.5 proves the requested inclusion of all local product projections in the preserving class.

G.4 (Proved). For a locally compact Hausdorff space Y and a stable infinity-category $\mathcal{V}$ admitting small limits and colimits, Lurie's covariant Verdier duality identifies $\mathrm{Shv}(Y; \mathcal{V})$ with $\mathrm{CoShv}(Y; \mathcal{V})$, by $A \mapsto (U \mapsto \Gamma_c(U, A))$ (Lurie, 2017, Theorem 5.5.5.1 and Proposition 5.5.5.10). This known theorem does not by itself provide either raw-space localization in Part G. Its displayed construction fails to be an equivalence for pointed-space coefficients.

Proof of the unstable obstruction. Take $Y = \mathbb{R}$ and the nonzero constant sheaf $A = S^0$ of pointed discrete spaces. For a compact $K \subset U$, every connected component of the open set U meets $U \setminus K$: a nonempty open interval cannot be a compact subset of $\mathbb{R}$. A locally constant section which is the basepoint outside K is therefore the basepoint everywhere. All section spaces here are discrete, so their pointed homotopy fibre

$$\Gamma_K(U, A) = \mathrm{fib}(A(U) \to A(U \setminus K))$$

is contractible. Its filtered colimit over compact K is contractible. Thus $\Gamma_c(-, A)$ is the zero cosheaf although A is not zero. The compact-support transform is not conservative and cannot be an equivalence in this unstable setting. □

The attached parametrized-spaces theorem uses a different homotopy theory: for a locally compact CW base it embeds the usual parametrized homotopy theory as the locally homotopically constant homotopy sheaves (Shulman, 2008). It does not assert that open-inverse-image tests present every cosheaf. Its comparison employs spaces of sections and a fibrant replacement; substituting raw open inverse images changes the problem. This separates what is available in that source from the unresolved realization in G.3.

G.5 (Proved). Every product projection $p : V \times U \to U$ preserves the open-test class under strict pullback, with no local-contractibility assumption on either factor. More generally, the preserving class contains every map $p : Z \to Y$ admitting an open cover $\{Z_a\}$ such that $p(Z_a) = U_a$ is open in Y and $Z_a \cong V_a \times U_a$ over U_a. In particular it contains all shape submersions in Volpe's sense, and G.2 makes that containment strict.

Proof. We use open-cover homotopy codescent. If $T = \bigcup_a T_a$ is an arbitrary open cover and Π denotes weak homotopy type, the augmented Cech diagram satisfies

$$\operatorname*{colim}_{[n]\in\Delta^{\mathrm{op}}} \coprod_{(a_0,\dots,a_n)} \Pi(T_{a_0} \cap \cdots \cap T_{a_n}) \simeq \Pi(T). \tag{G.5.1}$$

No numerability or separation hypothesis is needed. Indeed the singular simplices lying in some cover member form a simplicial subset weakly equivalent to $\mathrm{Sing}(T)$: subdivide finite sphere and disk representatives until their simplices lie in cover members. For each such singular simplex, the nonempty set of cover members containing its image has a contractible Cech nerve. Applying this degreewise proves (G.5.1). These are precisely the small-simplex and covering-sieve statements of Lurie (2017, Lemma A.3.3 and Proposition A.3.2).

Let $u : X \to X'$ over U be an open-test equivalence, and write $X_A = X \times_U A$ for an open $A \subset U$. For an arbitrary open $W \subset V \times U$, choose an open rectangle cover $W = \bigcup_a (V_a \times U_a)$. For every finite tuple of indices, put $V_J = \bigcap_{a\in J} V_a$ and $U_J = \bigcap_{a\in J} U_a$. The corresponding intersections in the two pulled-back spaces are exactly $V_J \times X_{U_J}$ and $V_J \times X'_{U_J}$. Their comparison is a weak equivalence: $X_{U_J} \to X'_{U_J}$ is one by hypothesis, and product with any space preserves weak homotopy equivalences, as follows from $\mathrm{Sing}(V_J \times X_{U_J}) = \mathrm{Sing}(V_J) \times \mathrm{Sing}(X_{U_J})$. Thus the two Cech diagrams are termwise equivalent. Taking their colimits in spaces and applying (G.5.1) proves the open test over W. This proves the product-projection assertion.

Finally preservation is local on the source Z of the base-change map. Given any open $W \subset Z$, cover it by $W \cap Z_a$. An open-test equivalence over Y restricts to one over U_a, since U_a is open in Y. Every finite intersection of these opens lies in one chart Z_a, where the product argument just proved applies to arbitrary open subsets. Consequently the pulled-back comparison is a weak equivalence on every finite intersection. Formula (G.5.1) then gives the equivalence over W. Since W was arbitrary, this establishes the assertion for the locally product map. The case $K \to *$ used in G.2 is the product projection with $V = K$. □

H A second proof of the finite-poset realization theorem

H.1 (Proved). Let $f : P \to Q$ be a proper functor of finite posets, meaning that

$$L(p, q) = \{r \in P : r \geq p,\ f(r) = q\}$$

is weakly contractible whenever $f(p) \leq q$. The geometric direct image along $g = |Nf|$ preserves least-vertex-constructible sheaves and agrees with categorical right Kan extension, for space, spectrum, and derived-module coefficients.

The proof below replaces BISITE D.11's initiality induction on the poset of fibre cells by normalization of barycentric coordinates, followed by geometric proper base change.

Proof. We first record the finite-poset coefficient calculation used below. If A is a finite poset and $D : A^{\mathrm{op}} \to \mathcal{V}$, then

$$\Gamma(|NA|, \nu_A^* D) \simeq \lim_{A^{\mathrm{op}}} D. \tag{H.1.1}$$

Here is a proof independent of BISITE D.7. Refine to the ordinary open-simplex stratification of the order complex. Cellular sheaf descent computes global sections as the limit over the face poset $\mathcal{C}(A)$ of nonempty chains, ordered by inclusion. The coefficient on a face κ is $D(\min \kappa)$, and for $\kappa \subseteq \kappa'$ the restriction is the map $D(\min \kappa) \to D(\min \kappa')$. The cellular-descent description follows by the open-star cover: a star has the specified stalk at its central face, intersections give the face incidences, and sheaf descent gives their limit. Equivalently, the exit mapping space between two open simplices is contractible when the first is a face of the second and empty otherwise; straight-line contraction inside each simplex proves this description. The functor

$\min : \mathcal{C}(A) \to A^{\mathrm{op}}$ is initial. Indeed its comma category over a is the face poset of $A_{\geq a}$, whose nerve is the barycentric subdivision of the contractible complex $|N(A_{\geq a})|$. Initiality proves (H.1.1), with its canonical limit cone, for every complete coefficient category.

We next prove a topological lemma. Suppose $h : R \to [n]$ is a map of finite posets satisfying the displayed outgoing-lift condition. Choose $\lambda_i > 0$ with $\sum_i \lambda_i = 1$, and let $Z_\lambda \subset |NR|$ be the inverse image of the point with these barycentric coordinates. Set $R_0 = h^{-1}(0)$. Deleting the coordinates at positive levels and dividing those at level zero by λ_0 defines a continuous map

$$r_\lambda : Z_\lambda \longrightarrow |NR_0|.$$

If R_0 is empty, both spaces are empty. Otherwise this map has weakly contractible fibres and is a weak homotopy equivalence.

We prove this assertion by induction on n. For $n = 0$ it is the identity. Suppose $n > 0$ and take $z \in |NR_0|$ with positive support the chain κ_0. Put $p = \max \kappa_0$ and

$$U_p = \{u \in R : u \geq p,\ h(u) > 0\}.$$

Give U_p its level map to $\{1, \ldots, n\} \cong [n-1]$. This map satisfies the same lifting condition, since a further upper lift of $u \geq p$ remains in U_p. Its first fibre is $L(p, 1)$ and is weakly contractible. A direct coordinate calculation identifies $r_\lambda^{-1}(z)$ with the geometric fibre of this level map at weights $\lambda_i/(1-\lambda_0)$ for $i > 0$: the coordinates on κ_0 are fixed as $\lambda_0 z$, and the remaining coordinates range over chains of U_p. The induction hypothesis makes this fibre weakly equivalent to $|NL(p, 1)|$. It is therefore weakly contractible.

All these spaces are finite polyhedra (the source is a finite polytopal complex). The map r_λ is proper. Proper base change for sheaves of spaces (Lurie, 2009, Corollary 7.3.1.18) identifies the stalk of $r_{\lambda *} r_\lambda^* A$ at z, for any sheaf A, with the sections of the constant sheaf A_z on $r_\lambda^{-1}(z)$. On a finite polyhedron T those sections are $\mathrm{Map}(\Pi T, A_z)$. They equal A_z when T is weakly contractible. Thus the unit $A \to r_{\lambda *} r_\lambda^* A$ is an equivalence on every stalk, and hence an equivalence: sheaves of spaces on finite polyhedra are hypercomplete, by finite homotopy dimension. Applying this to each constant sheaf with value K and then taking global sections gives

$$\mathrm{Map}(\Pi |NR_0|, K) \simeq \mathrm{Map}(\Pi Z_\lambda, K) \quad (K \in \mathcal{S}).$$

Yoneda proves that r_λ is a weak equivalence and closes the induction. This proof also shows the stronger unit statement for every sheaf. Spectrum and derived-module versions follow by testing on all shifts of the sphere or the rank-one free module; these compact generators detect equivalences and commute with stalk formation.

Now fix $x \in |NQ|$ with support $q_0 < \cdots < q_n$. Apply the lemma to the restricted poset $f^{-1}\{q_0, \ldots, q_n\}$. We obtain $r : g^{-1}(x) \to |NP_{q_0}|$ and the unit equivalence just proved. If $F : P^{\mathrm{op}} \to \mathcal{V}$ is a categorical coefficient, its least-vertex sheaf restricted to $g^{-1}(x)$ is precisely the inverse image along r of the least-vertex sheaf attached to $F|_{P_{q_0}}$. To check this without any fibre-cell initiality argument, let $\nu_P : |NP| \to P$ be the minimum-vertex map to the downset Alexandrov topology. The associated sheaf is $\nu_P^* F$. The equality of minimum vertices gives the strict equality

$$\nu_P|_{g^{-1}(x)} = (P_{q_0} \hookrightarrow P)\, \nu_{P_{q_0}}\, r.$$

Functoriality of inverse image gives the claimed identification. Consequently

$$\Gamma(g^{-1}(x), F) \simeq \Gamma(|NP_{q_0}|, F|_{P_{q_0}}) \simeq \lim_{P_{q_0}^{\mathrm{op}}} F.$$

The categorical Kan value at q_0 is $\lim_{(P_{\leq q_0})^{\mathrm{op}}} F$. The inclusion $P_{q_0}^{\mathrm{op}} \to (P_{\leq q_0})^{\mathrm{op}}$ is initial: its comma category at p is $L(p, q_0)^{\mathrm{op}}$. Quillen's Theorem A therefore identifies this limit with the

one just computed. This is the single-level categorical fibre formula; the multi-level geometric comparison has already been proved by the different argument above.

Finally g is proper. The equality $\nu_Q g = f\nu_P$ supplies the canonical mate $\nu_Q^* f_* F \to g_* \nu_P^* F$. At x, proper base change identifies this mate with restriction of the Kan limit cone to the fibre. Under the normalization just constructed it is the initial-functor comparison in the preceding paragraph, and is an equivalence. Stalkwise detection proves the claimed sheaf equivalence. Its source is least-vertex constructible; this proves preservation and identifies the actual adjunction mate, hence its units and counits. □

H.2 (Refuted). The unmodified finite-poset assertion cannot be extended to all finite categories with nonidentity idempotents while retaining one least-vertex stratum for each object and all categorical presheaf coefficients. The smallest possible endomorphism example already fails the coefficient comparison.

Proof. Let C have one object and two arrows $1, e$, with $e^2 = e$. The projection $p : C \to *$ is categorically proper: every category over the terminal category is proper, since its outgoing lifting categories have initial objects. Moreover BC is contractible. Indeed the zero e gives a natural transformation from the identity functor of C to the constant functor at its object.

The least-vertex partition of BC has one stratum, the whole space. A constructible sheaf is therefore a local system, and since BC is contractible its global sections equal its stalk. Take the categorical set coefficient $F(*) = \{0, 1\}$ with $F(e)$ the constant map to 0. Its categorical right Kan extension to the point is its set of fixed elements, namely $\{0\}$. The one-stratum locally constant sheaf having the stated stalk $\{0, 1\}$ has two global sections. These cannot agree. For a field k, the same example takes $F(*) = k^2$ and $F(e)$ the projection onto the first summand; the categorical limit is k, whereas the geometric constant-sheaf sections are k^2. There is no derived correction: $k[e]/(e^2 - e) \cong k \times k$, and the invariant-summand functor is exact.

The defect occurs before asking for a geometric properness theorem: the category C is not the exit category of its one-stratum realization. Restricting coefficients to those which invert e removes this example, but is a substantive alteration of the asserted extension. One object and two arrows is minimal for the presence of a nonidentity endomorphism. The program `code/GHIJ/small_counterexamples.py` checks the category laws, the outgoing lifting category, the contracting zero transformation and the two-element action. □

H.i All outgoing lift categories for forgetting one mark

We use Jansen's exit variance. Objects of $\mathcal{C}_{g,n}$ are essential multicurves σ, and

$$\mathrm{Hom}(\sigma, \tau) = \{g : g\sigma \supseteq \tau\}/\Delta_\sigma.$$

Here $\Delta_\sigma \cong \mathbb{Z}^\sigma$ is the multitwist subgroup, acting on representatives on the right. Thus adding a node points opposite to an exit arrow. This is precisely Jansen (2023, Construction 3.18); his Proposition 3.23 identifies the opposite with the category of degenerations. His Corollary 6.6 and Theorem 6.7 give the exit/exodromy interpretation. The printed inequality in the second assertion of Theorem 6.7 is a sign error; the stability hypothesis throughout is $2g - 2 + n > 0$.

Let $Q : \mathcal{C}_{g,n+1} \to \mathcal{C}_{g,n}$ forget the last mark and stabilize, and put $F = Q^{\mathrm{op}}$ in entrance variance. Write $q\sigma$ for the surviving multicurve, identifying repeated images, and ϕ for the forgetful mapping-class homomorphism.

H.3 (Proved). Fix σ upstairs and an exit arrow $\beta : q\sigma \to \eta$ downstairs. Its outgoing lifting category for Q is either weakly contractible or equivalent to $B\mathbb{Z}$. It is equivalent to $B\mathbb{Z}$ exactly

when forgetting the mark contracts a rational bridge of σ and β smooths the node produced by that contraction. This classification holds for every stable (g, n).

Proof. Choose a representative b of β and a lift $\widetilde{b}$ through the forgetful mapping-class homomorphism. Conjugating targets by $\widetilde{b}^{-1}$ reduces to $\beta = [1] : \tau \to \eta$ with $\tau = q\sigma$ and $\eta \subseteq \tau$. The forgetful homomorphism is surjective by the Birman exact sequence (Kent et al., 2009, Section 2; the three-punctured sphere case is immediate).

An object of the outgoing lifting category is $\alpha = [g] : \sigma \to \rho$ with $q\rho = \eta$ and $\phi(g) \in \Delta_\tau$. Each curve of τ has at least one preimage in σ, and a twist on such a preimage maps to its twist. Therefore $\Delta_\sigma \to \Delta_\tau$ is surjective. Multiplying g on the right by a suitable element of Δ_σ leaves $[g]$ unchanged and makes $\phi(g) = 1$. The vertical isomorphism $[g^{-1}] : \rho \to g^{-1}\rho$ then identifies this lift with a normalized lift $[1] : \sigma \to s$, where $s \subseteq \sigma$ and $qs = \eta$.

Set

$$H = \{d \in \Delta_\sigma : \phi(d) \in \Delta_\eta\}.$$

The full category of normalized lifts has the exact presentation

$$\mathrm{Ob} = \{s \subseteq \sigma : qs = \eta\}, \qquad \mathrm{Hom}(s, t) = \begin{cases} H/\Delta_s, & s \supseteq t, \\ \varnothing, & \text{otherwise}, \end{cases} \tag{H.3.1}$$

with composition induced by addition in the free abelian group Δ_σ. Indeed the triangle condition is $[h] \circ [1] = [1]$ as arrows with source σ, so its representative must belong to Δ_σ. Such twists fix every submulticurve of σ, and the remaining verticality condition is exactly $\phi(h) \in \Delta_\eta$. Quotienting a morphism with source s divides by Δ_s. This proves (H.3.1) on objects, morphisms and composition, not only on components.

Only the component carrying the forgotten mark can become unstable. If it does, it is a rational component with two remaining special points. Contracting it has exactly one of two effects: a bubble with a retained mark deletes one curve, or a bridge identifies two curves. No second component becomes unstable, since the numbers of special points of adjacent components are unchanged by stabilization. Thus the twist map is one of the following three coordinate maps:

$$\mathbb{Z}^\sigma \to \mathbb{Z}^\tau : \quad \text{a bijection of bases}, \quad \text{deletion of one coordinate}, \quad (a_1, a_2, \ldots) \mapsto (a_1 + a_2, \ldots).$$

In the first case there is one normalized object and its automorphism group is trivial. In the deletion case let $s_{\max}$ contain all preimages of η and the deleted curve. Then $\Delta_{s_{\max}} = H$, so (H.3.1) makes it an initial object. In the identification case, if the duplicate image belongs to η, take both duplicate curves in $s_{\max}$; again $\Delta_{s_{\max}} = H$ and the object is initial. If that image does not belong to η, there is a unique normalized object s, while

$$H/\Delta_s \cong \ker(\mathbb{Z}^2 \xrightarrow{(a,b)\mapsto a+b} \mathbb{Z}) \cong \mathbb{Z}.$$

Its generator is the difference of the two bridge twists. This is exactly the stated $B\mathbb{Z}$ case. The reasoning is independent of the genus and of automorphism groups of the other components, since the triangle condition eliminated those groups before the computation. □

H.4 (Proved). The stable-split enumeration gives $4, 26, 236$ boundary strata for $\overline{\mathcal{M}}_{0,4}$, $\overline{\mathcal{M}}_{0,5}$ and $\overline{\mathcal{M}}_{0,6}$ respectively. For forgetting $5 \to 4$, all 47 normalized stratum/smoothing cases comprise 44 contractible lifting categories and 3 of type $B\mathbb{Z}$. For forgetting $6 \to 5$, all 766 such cases comprise 696 contractible categories and 70 of type $B\mathbb{Z}$.

Proof. The program `code/GHIJ/marked_curve_lifts.py` enumerates each stable bipartition of the labelled set, quotients by swapping its two sides, and enumerates every pairwise compatible family. Compatibility is checked by emptiness of one of the four intersections of sides. Such families are exactly the edge splits of stable labelled genus-zero trees: one constructs the tree successively by the compatible splits, and contracting all its edges recovers the unique one-vertex tree. A family has at most $n-3$ splits, so the enumeration is finite and complete.

Deleting the last label computes $q\sigma$ and detects deleted and repeated images. Every subset $\eta \subseteq q\sigma$ is enumerated. The program constructs all objects of (H.3.1) and the rank of every nonempty morphism group; these groups have no torsion because the twist constraints are disjoint coordinate sums with coefficient one. It then either exhibits the initial object or checks that the category has one object and automorphism group $\mathbb{Z}$. Its reduced chain model is respectively zero or the usual circle complex, giving integral homology through degree three and the same Betti numbers over $\mathbb{F}_2$ and $\mathbb{Q}$. The CSV and full group presentations are supplied. These are finite presentations of the infinite lifting categories; they are not falsely labelled finite quotient categories. Arbitrary mapping-class representatives reduce to these cases by H.3. □

H.5 (Proved). For every stable (g,n) the entrance forgetful functor $F : T_{g,n+1} \to T_{g,n}$ is W_∞-proper. It is smooth only for $(g,n) = (0,3)$.

Proof. We give an algebraic proof of the contractibility needed for properness, using graph-of-groups normal forms instead of lifted geodesics. In exit variance, fix σ and an incoming arrow $\beta : \eta \to q\sigma$. Lifting a representative of β and then normalizing the arrow upstairs identifies its incoming lifting category with the poset

$$\mathcal{P}(\eta;\sigma) = \{\rho \supseteq \sigma : q\rho = \eta\}, \qquad \text{ordered by reverse inclusion.} \tag{H.5.1}$$

Here is the morphism check. A normalized object is $[1] : \rho \to \sigma$. A triangle from this object to $[1] : \rho' \to \sigma$ has representative in Δ_ρ, since both composites are the identity coset with source ρ. Every such representative fixes ρ, and is the same coset; hence there is exactly one triangle when $\rho \supseteq \rho'$ and none otherwise. All normalization isomorphisms are vertical because their downstairs representatives belong to Δ_η.

First take $\sigma = \varnothing$ and a nonempty η. Cut the punctured surface S along representatives of η. Van Kampen gives a graph of groups with vertex groups $\pi_1(S_v)$ for the complementary components and edge groups $\mathbb{Z}$ for the cut curves. The edge injections are injective: each cut curve is essential and each boundary loop in a negative-Euler-characteristic surface is nontrivial. With $G = \pi_1(S)$, form the coset graph whose vertices are $G/\pi_1(S_v)$ and whose edges are the corresponding $G/\mathbb{Z}$ with the two incidence maps. The normal-form theorem for this graph of groups says this graph is a tree. Concretely, a maximal tree in the finite dual graph reduces the presentation to amalgamated products and HNN extensions; an edge path with no immediate reversal gives a reduced word, and the amalgam/HNN normal form prevents a nonempty such word from closing at its initial coset.

The nonperipheral part of (H.5.1) is the opposite face poset of this coset tree. This description can be checked by point pushing: a lift with one curve over each member of η places the extra point in a complementary component, and paths moving it there are identified precisely modulo the corresponding vertex group. A lift with a duplicated curve places it in the annulus between the two preimages; paths are identified modulo that curve's cyclic group. Deleting either of the duplicate curves gives the two incidence maps. These assertions follow from isotopy extension and the point-pushing exact sequence; they are also the coset description underlying Kent et al. (2009, Section 7). They do not require a choice of hyperbolic metric or geodesic representatives.

A curve which becomes peripheral encloses the extra point and one retained puncture. At most one can occur, and it cannot coexist with a duplicated-image annulus. Deleting it attaches

one terminal edge to the barycentrically subdivided coset tree: the other curves lift uniquely outside its twice-punctured disc. Thus all such possibilities add leaves, without adding a cycle. The nerve remains a tree and is contractible. If η is empty, the empty multicurve is terminal in (H.5.1), giving contractibility directly.

For general σ, cut along its curves. On every component not containing the forgotten point, missing curves of $\eta \setminus q\sigma$ lift uniquely. In the component which does contain that point, if it stays stable, the problem is exactly the preceding one with retained boundary circles treated as punctures. An added curve which becomes parallel to a fixed boundary is one of the peripheral leaves just described. If the component destabilizes, it is a pair of pants and admits no essential refinement; keeping σ already fixes the bubble or bridge, so the remaining lift poset is a singleton. This decomposes (H.5.1) into one of the contractible posets above and singleton factors. All incoming lifts of Q are therefore contractible, proving properness of F.

For $(g, n) = (0, 3)$ the target entrance category is terminal and every functor to it is smooth. Every other stable moduli space has a boundary node. Inserting a rational bridge carrying the forgotten mark at that node gives the duplicate-curve case in H.3; smoothing that node produces the lifting category $B\mathbb{Z}$. Thus F is not smooth in every other case. □

H.6 (Proved). For the universal stable curve $\overline{\mathcal{M}}_{g,n+1} \to \overline{\mathcal{M}}_{g,n}$, geometric direct image of constructible coefficients agrees with right Kan extension on the entrance categories, for compactly generated presentable coefficients. Point-fibre base change and base change by inclusions of closed unions of boundary strata agree with categorical proper base change. In positive genus these are assertions about the moduli stack, not its coarse space.

Proof. The forgetful map is the proper representable universal stable curve. Over one stratum, normalize the nodal fibres. Its components are proper smooth families of marked smooth curves with disjoint node and marking sections. Ehresmann local triviality, applied together with these sections, trivializes each such family on a sufficiently small contractible analytic chart of the stratum. Gluing the section pairs along the fixed dual graph gives a trivialization of the stratified universal curve over that chart. For positive genus one uses an orbifold chart and retains the finite stabilizer action; the argument descends along that chart. Proper base change now says that direct-image stalks on the stratum are sections of one fixed stratified fibre with locally constant transport. They are constructible.

The exit functor induced by the geometric map is Q. On monodromy it forgets the point, giving ϕ; on a degeneration it erases the peripheral curve or identifies the two bridge curves, giving q. These generate the morphisms in Jansen's Construction 3.18. The relation given by a twist in the source maps into the twist group of its stabilized image, so the map on representatives descends and agrees with composition. The natural exodromy equivalence therefore identifies inverse image with restriction along Q, or along F in entrance variance. The now-proved preservation of constructibility allows both right adjoints to restrict to these full constructible categories. Their contractible uniqueness identifies the geometric direct image with the categorical right Kan extension, including its unit and counit.

For an actual point fibre, the stratified fibre is the stable curve with its marked and nodal points distinguished. Its open component strata are punctured surfaces of negative Euler characteristic and hence are $K(\pi, 1)$'s. Each puncture or node branch has cyclic link group injecting into the appropriate surface group. Thus its exit mapping spaces are the discrete coset sets of these cyclic subgroups, with the expected surface groups as open-stratum automorphisms. Point pushing identifies these groups and cosets with the strict fibre of Q; the bubble and bridge points have trivial automorphisms after their twists are divided out. This gives the exit-category comparison for the fibre itself, including its mapping sets, rather than merely its total classifying space.

A closed union of boundary strata is a replete full cosieve in entrance variance. Its inverse image is defined by the identical object condition after applying F, and the morphisms are full; the resulting square is a strict categorical pullback. This statement concerns the actual boundary inclusion; in positive genus a clutching normalization need not be an inclusion and is not silently substituted for it. In these point and closed-boundary squares the geometric and categorical comparisons are mates of the same restriction identity under the identified adjunctions. They agree, and are invertible by geometric proper base change and H.5.

For compactly generated presentable coefficients, test sheaves on the compact objects. Such tests preserve limits and filtered colimits, hence commute with stalks, and are conservative. Consequently the space-valued proper base-change and local triviality arguments give the asserted coefficient version. No inversion of stabilizer orders is required for this stack statement. □

H.7 (Proved). The program `code/GHIJ/finite_poset_direct_image.py` independently enumerates all maps between one representative of every poset isomorphism class with at most five objects, including the empty poset. The counts of posets of sizes $0, 1, 2, 3, 4, 5$ are $1, 1, 2, 5, 16, 63$. It examines $1,349,916$ monotone maps and recognizes $96,372$ proper maps. Properness is certified by explicit simplicial collapses of every nonempty outgoing lifting order complex; noncontractible cases are rejected by homology over $\mathbb{Q}$ or $\mathbb{F}_2$. The program aborts rather than classifying an acyclic but uncertified complex as contractible. No such undecided case occurs in this range.

For every proper map and every nonempty target simplex, it computes the categorical stalk and the actual geometric-fibre cellular cohomology, for the constant coefficient and for each principal-cosieve indicator. This gives $5,536,094$ stalk/coefficient cases, each computed over both $\mathbb{Q}$ and $\mathbb{F}_2$, with no discrepancy. A fibre cell is indexed by a source chain meeting each level of the target simplex; it is a product of simplices, of dimension $|\kappa| - |\sigma|$. Its signed boundary deletes one vertex while keeping every level nonempty, using the product-boundary sign. The code verifies that consecutive boundary matrices multiply to zero. Exact rational ranks use integer elimination with gcd reduction, and binary ranks use bit arithmetic. For a principal cosieve, extension from the corresponding closed subcomplex reduces both computations to its intersection with the relevant fibre or lower inverse image.

The supplied per-map CSV records all tests as successful; the JSON gives the totals by source and target size. These numerical comparisons check cohomology, not a replacement proof of the canonical mate; the latter is established in H.1.

H.8 (Proved). By the independently reconstructed proof in D.10, the table agrees with every entry of BISITE B.15. In the following table the smooth and proper columns each apply to both W_1 and $W^{\mathrm{lc}}_{D(\mathbb{Q})}$; the last two columns distinguish the intrinsic tests. Thus there are 48 scalar comparisons and no changes. Map labels refer to GALOIS D.3–D.5, as in BISITE.

map	smooth, both	proper, both	intrinsic W_1	intrinsic $\mathbb{Q}$lc
D.3 : p	yes	yes	yes	yes
D.3 : q	no	yes	no	no
D.3 : qp	no	yes	no	no
D.4 : p	yes	yes	no	no
D.4 : q	no	yes	yes	no
D.4 : qp	no	yes	no	no
D.5 : $H \to K$	yes	yes	no	no
D.5 : $q : R \to [1]$	no	yes	no	no

The proofs, reconstructed categories and program for these entries are in Part D. No finite collection of pullback tests is used as a substitute for the universal intrinsic condition.

Audit of the inherited proofs. BISITE D.11's fibre-cell initiality induction is complete once its stated cellular-exodromy and finite-dimensional hypercompleteness results are supplied; no contradictory case was found. H.1 gives a different geometric argument. BISITE F.15 agrees with Jansen after switching exit to entrance variance; its correction of the printed stability inequality is necessary. F.16's normalized lifting-poset and geodesic-tree proof is complete using its cited point-pushing identification; H.5 replaces the geodesic step by the amalgam/HNN normal-form argument. F.17 correctly restricts its geometric comparison to point fibres and the indicated actual closed boundary inclusions; it does not prove arbitrary geometric pullback compatibility. H.6 retains that scope in positive genus.

I The maximum diagonal-stable smooth class

I.1 (Proved). Let $p : E \to B$ be a functor of ordinary small categories, and form its strict diagonal $\Delta_p : E \to E \times_B E$. Then

$$p \text{ and } \Delta_p \text{ are } W_\infty\text{-smooth} \quad \Longleftrightarrow \quad p \text{ is a discrete fibration.}$$

Consequently the discrete fibrations form the largest subclass of Liss_{W_∞} closed under diagonals. They are also closed under identities, composition and strict pullback.

Proof. For smoothness use the incoming lifting category: given $x \in E$ and $a : b \to p(x)$, its objects are arrows $h : y \to x$ with $p(y) = b$ and $p(h) = a$, and its arrows are the commuting triangles over 1_b. If p is smooth, this category is weakly contractible, and in particular nonempty.

Take any two such lifts $h : y \to x$ and $k : z \to x$. They give an arrow $(h, k) : (y, z) \to (x, x)$ in the strict pullback. A lift of this arrow through Δ_p would be an arrow $l : w \to x$ satisfying $(w, w) = (y, z)$ and $(l, l) = (h, k)$. Thus its incoming lifting category is empty unless $y = z$ and $h = k$. Smoothness of Δ_p excludes emptiness. Existence and uniqueness of each incoming lift is exactly the discrete-fibration condition.

Conversely suppose p is a discrete fibration. Its incoming lifting categories are singletons. For an arrow (h, k) into a diagonal object, the discrete-fibration uniqueness gives $h = k$ and $y = z$; the diagonal lifting category is again a singleton. Therefore both maps are smooth. The usual unique-lift argument proves stability under composition and strict pullback. Any diagonal-stable subclass of the smooth maps satisfies the left side of the equivalence for each of its maps and is therefore contained in this class. □

I.2 (Proved). In the presheaf Kan formalism with space coefficients, a discrete fibration $p : E \to B$ has universally equivalent left and right Kan extensions if and only if p is an isomorphism. In particular the stable classification by finite discrete bifibrations does not carry over to unpointed spaces.

Proof. Pull back to any object $b \in B$. The fibre is a discrete set I. Left and right Kan extension to a point are respectively the coproduct and product functors $\mathcal{S}^I \to \mathcal{S}$. On the family constantly equal to $*$ they yield the discrete set I and $*$, respectively. Hence equivalence forces I to contain exactly one element. In a discrete fibration with exactly one object over each object of B, every arrow of B has exactly one lift with its specified target. This makes p bijective on objects and on all hom sets, and therefore an isomorphism. Conversely an isomorphism has equal inverse, left-Kan and right-Kan functors, universally under strict pullback. □

I.3 (Proved). Let $\mathcal{V}$ be a presentable closed symmetric monoidal infinity-category with tensor product preserving small colimits in each variable. The assignment $\Phi(S) = \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{V})$

extends to a lax symmetric monoidal correspondence functor with smooth right legs, sending $X \xleftarrow{a} M \xrightarrow{b} Y$ to $b_! a^*$. This independently completes the coherence argument in BISITE F.10.

Proof. We use the precise universal property of spans in Cnossen et al. (2025, Corollary 2.6(2)): for an infinity-category C and a pullback-stable wide subcategory E, restriction from the span $(\infty, 2)$-category with arbitrary left legs and right legs in E is an equivalence onto contravariant functors on C whose restrictions along E have left adjoints and satisfy Beck–Chevalley. The equivalence includes adjointable transformations and their higher coherences. No diagonal or cancellation hypothesis occurs in this statement.

Take $C = N\,\mathrm{Cat}$, so its pullbacks are the ordinary strict categorical pullbacks. The incoming-lift characterization of smoothness shows directly that smooth maps form a wide subcategory closed under pullback and composition. Restriction of presheaves is a coherent functor on C^{op}, and all left Kan adjoints exist by presentability. If $b : M \to Y$ is smooth, the inclusion $M_y^{\mathrm{op}} \to (y\backslash b)^{\mathrm{op}}$ is final: its relevant comma categories are exactly the incoming lifting categories. Hence

$$(b_! A)(y) \simeq \operatorname*{colim}_{M_y^{\mathrm{op}}} A.$$

In every strict pullback square the fibre categories on the two sides are identical. Applying this formula identifies the canonical Beck–Chevalley mate as an equivalence. All hypotheses of the quoted universal property now hold. It produces the correspondence functor, and its value on a span is the stated composite.

For completeness the monoidal enhancement is also covered by this universal property, and does not require choosing separate associativity isomorphisms. For each k, external tensor is the natural transformation

$$\prod_{i=1}^{k} \Phi(S_i) \longrightarrow \Phi\left(\prod_{i=1}^{k} S_i\right), \qquad (A_i) \longmapsto \boxtimes_i A_i.$$

It is left-adjointable in every smooth variable. Indeed Fubini for colimits and separate cocontinuity identify, with their canonical units,

$$(\textstyle\prod_i b_i)_!(\boxtimes_i A_i) \simeq \boxtimes_i (b_i)_! A_i.$$

Apply the transformation part of the span universal property to C^k, with right class E^k. Products commute with strict pullbacks, so both sides define functors on the corresponding span category. The displayed transformation has a unique coherent extension. The original associativity, symmetry and unit coherences of external tensor lift through the fully faithful restriction equivalence, including all higher relations. These extensions are exactly a lax symmetric monoidal structure.

Right Kan adjoints to restrictions exist, $b_!$ has right adjoint b^*, and pointwise tensor is closed by presentability and separate cocontinuity. These are the remaining adjoints in the 2022 definition of a six-functor formalism. The projection formula can also be checked directly by the fibre formula: restriction of b^*M to M_y is constant at $M(y)$, and tensoring with this object commutes with the displayed colimit. □

I.4 (Proved). The preceding proof does not follow from Mann (2022, Proposition A.5.10) by taking I to be all smooth maps and P to be the isomorphisms. That specialization violates the suitable-decomposition cancellation requirement. Indeed $p : [1] \to *$ and $pg : * \to *$, where g selects $1 \in [1]$, are smooth, but g is not smooth: the incoming arrow $0 \to 1$ has no lift. Thus $p, pg \in I$ does not imply $g \in I$. The span universal property used in I.3 is the needed replacement. Contractible uniqueness of individual adjoints, without this extension statement or an explicit unfurling construction, by itself does not state the necessary coherence theorem. The program

`code/GHIJ/small_counterexamples.py` enumerates the incoming lifts for p, pg, and g, verifying this finite cancellation counterexample.

I.5 (Proved). Opposition of the base categories transports I.3 to a correspondence formalism with proper right legs and values $\Psi(S) = \mathrm{Fun}(NS, \mathcal{V})$. The maximum diagonal-stable proper class is the discrete opfibrations. All allowed maps of either Kan formalism satisfy $f^! = f^*$, so they are cohomologically etale with trivial twist. These assertions follow by applying $S \mapsto S^{\mathrm{op}}$ to I.1 and I.3; incoming lifting categories become opposites of outgoing lifting categories, and opposition preserves weak contractibility.

I.6 (Proved). For pointed-space coefficients, the analogous universally ambidextrous discrete fibrations are exactly the inclusions of unions of connected components, up to isomorphism over the base.

Proof. At a point fibre I, left and right Kan extension are wedge and product. Applied to S^0 in every coordinate, their numbers of components are $1 + |I|$ and $2^{|I|}$. They differ when $|I| \geq 2$, including infinite cardinalities by Cantor's theorem. Thus each fibre is empty or a singleton. Pull back along a base arrow $0 \to 1$. A transition from a nonempty fibre over 0 to an empty one over 1 gives the discrete fibration $\{0\} \to [1]$. On a nonzero pointed coefficient its left Kan value at 1 is zero and its right Kan value is that coefficient. The reverse emptiness pattern is excluded already by the existence of the discrete-fibration reindexing function. Consequently the support is a union of components, and all transitions there are bijections. Conversely extension by zero on such a union is both Kan extensions, universally after pullback. □

I.7 (Open). No proof or counterexample is supplied here for the reversible accessible-localizer classification in Part I(2), including its interval between integral homology equivalences and W_0. For Part I(3), the unresolved assertion is an assignment $\mathbb{D} \mapsto \mathbb{D}'$ on all derivators satisfying Der 1–4, natural in exact morphisms, with $W_{\mathbb{D}'} = W_{\mathbb{D}}^{\mathrm{lc}}$; we also do not supply the alternative obstruction by two derivators equivalent on finite posets but with different locally constant towers. The weakest sufficient enhancement and an iteration theorem in that generality remain unresolved. I.9 below proves a specified localized-space realization and its finite iteration for the enhanced categories of spectra and $D(\mathbb{F}_p)$, including both second-stage calculations. It does not resolve the bare-derivator assertion or the reversible-localizer classification.

I.8 (Proved). For spectra or $D(k)$, with k a nonzero field, the universally ambidextrous discrete fibrations are precisely the finite discrete bifibrations: their fibres are finite sets, and all reindexing functions along base arrows are bijections. Together with I.2 and I.6 this gives the unstable and stable proper-map classifications within the maximum diagonal-stable class. Every map in that class is cohomologically smooth and etale, with trivial dualizing twist, because $f_!$ is left adjoint to f^* and hence $f^! = f^*$.

Proof. At a point fibre I, ambidexterity equates coproduct and product. An infinite product cannot preserve all coproducts: take the family of diagrams with the coefficient unit in one coordinate and zero in every other coordinate. The comparison is the canonical map from the direct sum to the product of the nonzero groups $\mathbb{Z}$ or k, and it is not surjective for an infinite index set. Compactness of the sphere or of k detects this failure. Thus every fibre is finite.

Pull back along an arrow $0 \to 1$, and let $r : I_1 \to I_0$ be its reindexing map. If $i \in I_0$ has no preimage, put a nonzero coefficient M at i and zero elsewhere. Its left Kan value at 1 is zero, while its right Kan value contains M as the sole nonzero factor. If i has $m \geq 2$ preimages, use the same coefficient. Its right Kan value is the homotopy limit of the star consisting of

m arrows $0 \to M$, namely $(\Omega M)^{m-1}$; its left Kan value is again zero. Taking M to be the suspension of the coefficient unit rules out this case. Hence every r is bijective. Conversely a finite discrete bifibration computes both Kan extensions fibrewise, and the canonical finite biproduct equivalence identifies them naturally under the fibre-transport bijections and every strict base change. □

No invertibility of a prime is implicit in I.8. Its fibres have no isotropy groups. If one returns to the larger smooth class, the distinction matters: for $BC_p \to *$, trivial $\mathbb{F}_p$ coefficients have homotopy orbits with nonzero homology in every nonnegative degree and homotopy fixed points with nonzero cohomology in every nonnegative degree. They are not equivalent. The standard periodic resolution has differentials alternating $1 - g$ and $1 + g + \cdots + g^{p-1}$; both become zero on the trivial module. If p is invertible in a field, averaging splits invariants and the norm gives ambidexterity for a finite p-group. These group projections are not in the maximum diagonal-stable class: their diagonals fail smoothness whenever the group is nontrivial. Thus this diagnostic does not enlarge or qualify I.8.

I.i A specified enhanced second stage

I.9 [Proved]. The second stages for spectra and mod-p coefficients.

Fix a Grothendieck universe $\mathcal{U}$, and use a larger universe for categories of $\mathcal{U}$-small objects. Let $\mathcal{C}$ be either $\mathrm{Sp}_{\mathcal{U}}$ or $D(\mathbb{F}_p)$, with its indicated presentable infinity-category enhancement. We number the stages by

$$W_{\mathcal{C}}^{(0)} = W_{\mathbb{D}_{\mathcal{C}}}, \qquad W_{\mathcal{C}}^{(1)} = W_{\mathbb{D}_{\mathcal{C}}}^{\mathrm{lc}}, \qquad \mathcal{C}_1 = \mathcal{S}_{W_{\mathcal{C}}^{(1)}}, \qquad W_{\mathcal{C}}^{(2)} = W_{\mathbb{D}_{\mathcal{C}_1}}^{\mathrm{lc}}. \tag{I.9.1}$$

Here $\mathcal{S}_W$ is the accessible reflective localization of spaces at exactly the class W, and $\mathbb{D}_{\mathcal{E}}(A) = \mathrm{Ho}\,\mathrm{Fun}(NA^{\mathrm{op}}, \mathcal{E})$. The localizations in (I.9.1) exist and preserve finite products. Their constant-coefficient classes satisfy $W_{\mathbb{D}_{\mathcal{C}_1}} = W_{\mathcal{C}}^{(1)}$. For every map $u : X \to Y$ of small spaces,

$$u \in W_{\mathcal{C}}^{(2)} \quad \Longleftrightarrow \quad \pi_0(u) \text{ is bijective and } \Omega_x u \in W_{\mathcal{C}}^{(1)}. \tag{I.9.2}$$

The loop condition is tested at one basepoint in every component of X. Consequently

$$W_{\mathrm{Sp}}^{(1)} = W_{\mathrm{Sp}}^{(2)} = W_{\infty}, \tag{I.9.3}$$

and

$$u \in W_{D(\mathbb{F}_p)}^{(2)} \quad \Longleftrightarrow \quad \begin{cases} \pi_0(u) \text{ is bijective,} \\ \pi_i(u) \text{ is an integral isomorphism for } i = 1, 2, \\ H_*(u\langle 2\rangle; \mathbb{F}_p) \text{ is an isomorphism.} \end{cases} \tag{I.9.4}$$

For a pointed connected component, $X\langle 2\rangle$ means the homotopy fibre at its chosen basepoint of $X \to \tau_{\leq 2}X$. The map $u\langle 2\rangle : X\langle 2\rangle \to Y\langle 2\rangle$ is induced by u and the basepoints. The last two conditions of (I.9.4) are tested on every corresponding pointed component. There are no nilpotence or finite-type hypotheses. The second class is again accessible and product stable, so $\mathcal{C}_2 = \mathcal{S}_{W_{\mathcal{C}}^{(2)}}$ realizes it as an ordinary constant-coefficient class. This construction can be iterated at every finite stage for these two starting coefficients. It is a specified realization by localized spaces; functorial construction on all bare derivators and all exact morphisms remains the separate Open assertion of I.7.

Proof. We prove the action comparison first under an explicit localization hypothesis, and then construct the localizations required here.

The constant class of a product-preserving localization. Let $W \subseteq W_0$ be the exact class inverted by an accessible reflector $L : \mathcal{S} \to \mathcal{E} \subseteq \mathcal{S}$, and suppose that W is stable under product with every small space. Local objects are closed under limits. The unit maps give

$$X \times Y \longrightarrow LX \times Y \longrightarrow LX \times LY$$

in W, and $LX \times LY$ is local. Thus the specified comparison $L(X \times Y) \to LX \times LY$ is an equivalence. The point is local, so L preserves finite products. This is the meaning of Cartesian localization used here.

For any local space b and any space a, the mapping space $\mathrm{Map}(a, b)$ is local: the test against $w \in W$ is the test of $w \times a$ against b. In particular every mapping space between two objects of $\mathcal{E}$ is itself a local space. For constants with values $a, b \in \mathcal{E}$, the comparison attached to $u : X \to Y$ is the map on components of

$$\mathrm{Map}\big(Y, \mathrm{Map}_{\mathcal{E}}(a, b)\big) \longrightarrow \mathrm{Map}\big(X, \mathrm{Map}_{\mathcal{E}}(a, b)\big).$$

It is bijective when $u \in W$. Conversely, take $a = *$ and let b range over all local spaces. Bijectivity gives

$$\mathrm{Hom}_{\mathrm{Ho}\,\mathcal{E}}(LY, b) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Ho}\,\mathcal{E}}(LX, b).$$

Yoneda in the homotopy category makes the specified Lu an isomorphism, hence an equivalence. Therefore $W_{\mathbb{D}_{\mathcal{E}}} = W$. The category $\mathcal{E}$ is presentable, so its functor categories have pointwise limits and colimits and both Kan extensions. These give Der 1–4 for the displayed $\mathbb{D}_{\mathcal{E}}$. Local constancy means that every arrow of the shape is sent to an equivalence, so the corresponding full infinity-category is $\mathrm{Fun}(BA, \mathcal{E})$. Reversing paths identifies the presheaf convention with the space-valued convention used in the calculation.

The specified action monad and both implications. For a connected pointed space (X, x), evaluation

$$U_x : \mathrm{Fun}(X, \mathcal{E}) \longrightarrow \mathcal{E}$$

is conservative, because every point of X is connected to x by an invertible path. It preserves all colimits, which are computed pointwise. Its left adjoint $F_x = x_!$ sends a to the copower of a by the path spaces out of x. Consequently its monad is

$$T_X(a) = U_x F_x(a) = L(\Omega_x X \times a) \simeq L(\Omega_x X) \times a. \tag{I.9.5}$$

The multiplication is induced by concatenation of based loops, and the unit by the constant loop. The infinity-categorical Barr–Beck theorem applies: the left adjoint exists, conservativity was checked, and preservation of geometric realizations follows from pointwise colimits (Lurie, 2017, Theorem 4.7.3.5). Hence $\mathrm{Fun}(X, \mathcal{E})$ is the category of algebras for this particular monad.

Let $u : (X, x) \to (Y, y)$ be based, with both spaces connected, and let $R = u^*$. The identity $U_x R = U_y$ and composition of left Kan extensions give a canonical equivalence $u_! F_x \simeq F_y$. The unit of $u_! \dashv R$, evaluated on a free object, is therefore

$$F_x(a) \longrightarrow R F_y(a).$$

Applying U_x gives exactly the monad map

$$L(\Omega_x X \times a) \xrightarrow{L(\Omega_x u \times 1_a)} L(\Omega_y Y \times a). \tag{I.9.6}$$

These are the specified maps, not abstract equivalences between their endpoints. If R is an equivalence, this adjunction unit is an equivalence for every a. Taking $a = *$ proves that $L(\Omega_x u)$ is an equivalence, hence that $\Omega_x u \in W$. Conversely, if $\Omega_x u \in W$, product stability makes every

map (I.9.6) an equivalence. It is a map of monads because u preserves loop concatenation and the constant loop. Equivalence of these monads induces an equivalence of their algebra categories, and the natural Barr–Beck comparisons identify that restriction with u^*. This proves sufficiency. In particular an unspecified Morita equivalence is not used to infer equivalence of loop objects.

The same argument applies to the homotopy-category definition of W^{lc}. Restriction has a left Kan-extension adjoint. If the induced functor on homotopy categories is an equivalence, the unit and counit of that adjunction are isomorphisms there and therefore equivalences in the infinity-categories. The original restriction is consequently an equivalence. The converse is immediate.

Both $\varnothing$ and $*$ are local, since $W \subseteq W_0$. If u misses a target component, the map of local systems that is $\varnothing \to *$ on that component and the identity elsewhere becomes an equivalence after restriction, so restriction is not conservative. If two source components map into one target component, a source local system taking values $\varnothing$ and $*$ on those two components cannot be a restriction. Thus an equivalence of restriction functors forces a component bijection. Once components correspond, apply the connected calculation to each. This proves (I.9.2), including empty spaces and arbitrary component sets.

The first-stage coefficient tests. The preceding evaluation argument also works for any presentable coefficient category before localization. For spectra its monad is

$$a \longmapsto \Sigma^\infty_+ \Omega_x X \wedge a,$$

and for $D(k)$ it is $a \longmapsto C_*(\Omega_x X; k) \otimes^{\mathbf{L}}_k a$. The same free-object unit, tested respectively on the sphere and on k, proves that a specified restriction is an equivalence exactly when its specified loop-spectrum or loop-chain map is an equivalence. The zero object and a nonzero coefficient detect components as above.

For spectra, smashing a loop-spectrum equivalence with $H\mathbb{Z}$ gives an integral homology equivalence of loop spaces. Its degree-zero map is the free abelian group map on the specified map of component groups $\pi_1 X \to \pi_1 Y$. A map between free abelian groups induced by a function of bases is invertible only when that function is bijective. Thus the fundamental-group map is an isomorphism. Translation by loops identifies all components; restriction to the identity components gives an integral homology equivalence

$$\Omega\widetilde{X} \longrightarrow \Omega\widetilde{Y}.$$

The bar construction carries this to an integral homology equivalence of $\widetilde{X} \to \widetilde{Y}$. Indeed normalized chains on a bar construction have a tensor-length filtration with the suspension of the augmentation ideal in positive degrees; the associated tensor-power comparisons are quasi-isomorphisms and the filtration is finite in each total degree. All chain complexes here are bounded below and degreewise free over $\mathbb{Z}$. The simply connected universal covers therefore have a weak equivalence by homological Whitehead. Together with the component and fundamental-group isomorphisms this proves $u \in W_\infty$. The reverse implication follows by homotopy invariance. We have proved $W^{(1)}_{\mathrm{Sp}} = W_\infty$; its localized-space coefficient is all of $\mathcal{S}$. Formula (I.9.2) now gives $W^{(2)}_{\mathrm{Sp}} = W_\infty$.

For $k = \mathbb{F}_p$, the same degree-zero argument gives the integral fundamental-group isomorphism, since $H_0(\Omega X; k) = k[\pi_1 X]$ and the algebra map is the one induced by u. Translation reduces the remaining loop-homology condition to $\Omega\widetilde{u}$. For maps of simply connected spaces v, one has

$$H_*(v; k) \text{ is invertible} \quad \Longleftrightarrow \quad H_*(\Omega v; k) \text{ is invertible.} \tag{I.9.7}$$

For precision, use normalized chains on a 1-reduced simplicial model. In the cobar construction the desuspended reduced generators have positive degree, so the tensor-length comparison converges

in each degree; over a field tensor powers preserve quasi-isomorphisms. The cobar-to-loop-chain comparison identifies its conclusion with the forward direction of (I.9.7). The reverse direction is the bar argument just used. This is the classical Adams comparison; a primary modern statement and proof are Rivera (2022, Theorem 1), and the 1-reduced comparison is also made explicit in Rivera et al. (2022, Proposition 4 and proof of Theorem 41). The degreewise bounded filtration, rather than a finite-type assumption, is what is needed. This recovers D.5's first-stage criterion with all maps retained.

Existence and exactness of the chosen localizations. For either coefficient category, the class detected by restriction of local systems is closed under all small colimits of arrows of spaces: $\mathrm{Fun}(-,\mathcal{C})$ sends colimits of spaces to limits of infinity-categories, and a limit of equivalences is an equivalence. It also has two-out-of-three and contains equivalences. Thus it is strongly saturated in the sense relevant to localization. It is product stable by the canonical identity

$$\mathrm{Fun}(X \times Z, \mathcal{C}) \simeq \mathrm{Fun}\big(Z, \mathrm{Fun}(X, \mathcal{C})\big).$$

For spectra this class is already W_∞, so the localization is the identity. We supply set-generation for the mod-p class, rather than infer accessibility merely from its closure properties.

Choose an uncountable regular $\kappa \in \mathcal{U}$. Represent a map of spaces by a simplicial-set map $f : K \to L$ and use the functorial Kan replacement $Q = \mathrm{Ex}^\infty$. This replacement preserves subcomplexes and filtered unions, and each simplex in QK depends on finitely many simplices in K: at a finite Ex-stage it is a map from a finite subdivided simplex. For a vertex a, use the based-loop model

$$\Omega_a QK = \{(a,a)\} \times_{QK \times QK} (QK)^{\Delta^1}.$$

It too preserves filtered unions of pointed subcomplexes and has finite support in the original complex. A complex with fewer than κ simplices gives fewer than κ loop simplices and loop chains over $\mathbb{F}_p$.

Suppose $f \in W^{(1)}_{D(\mathbb{F}_p)}$, and start with a submap $K_0 \to L_0$ of size less than κ. For each old target vertex choose an ambient source vertex and a finite path connecting its image to that target vertex. Whenever two old source vertices have connected images in the old target, choose an ambient path connecting those source vertices. For each old source vertex a and every cycle z in the target loop chains, choose an ambient source cycle w and an ambient chain b satisfying $f_*w - z = db$. For every old source cycle w and old target chain b satisfying $f_*w = db$, choose an ambient source chain c with $dc = w$. The component bijection and loop-chain quasi-isomorphism of f provide all these witnesses. There are fewer than κ tests, and every witness has finite support in K and L. Adjoin those supports, their faces and degeneracies, and their images under f.

Repeat countably many times. Regular uncountability keeps the union $K_\infty \to L_\infty$ smaller than κ. Every component relation, cycle, or bounding chain in this union occurs at a finite stage, so the next stage supplies its required witness. The union is therefore again in $W^{(1)}_{D(\mathbb{F}_p)}$. Given fewer than κ such small weak submaps of f, apply the same procedure to their union to obtain an upper bound. Thus all these small weak submaps form a κ-filtered system whose union is f. Filtered unions of simplicial sets compute the corresponding colimits of spaces; this follows equally from the finite-support sphere and homotopy tests in Q.

There is a $\mathcal{U}$-small set of representatives of the simplicial-set maps of size less than κ in this class. The preceding construction expresses every weak map as a filtered colimit of these representatives, up to isomorphism. Since the class is strongly saturated and closed under such colimits, it is exactly the strongly saturated class generated by that set. Lurie (2009, Proposition 5.5.4.15) now gives an accessible reflective localization whose inverted maps are exactly $W^{(1)}_{D(\mathbb{F}_p)}$. The product-stability proof and the first paragraph make it Cartesian. This completes the existence hypotheses used in (I.9.2).

Evaluation of the second mod-p stage and iteration. Apply the first-stage criterion to $\Omega_x u$. Its component condition is exactly that $\pi_1(u)$ be an isomorphism, and its fundamental-group condition on the identity component is exactly that $\pi_2(u)$ be an isomorphism. Translation carries that condition to every loop component. The universal cover of the identity component of $\Omega_x X$ is canonically equivalent, as a space over $\Omega_x X$, to

$$\Omega(X\langle 2\rangle).$$

Indeed this map has simply connected source, lands in the identity component, and is an isomorphism on every homotopy group in degrees at least two; the fibre sequence of the Postnikov truncation gives exactly the universal-cover map. This construction is natural in the pointed map u. Apply (I.9.7) to the 2-connected covers to convert the loop-cover homology condition into the last line of (I.9.4). This proves the stated formula.

To realize this second class, the same bounded-submap proof uses the finite-support conditions on components and based loops together with the quasi-isomorphism test on $C_*(\Omega_x^2 QK; \mathbb{F}_p)$. Component translations show that these are exactly (I.9.2)'s conditions. The additional fundamental-group witnesses are finite loop and homotopy data, so the same regular cardinal and countable closure procedure apply. The second class is strongly saturated and product stable because it is itself a locally constant coefficient class for $\mathcal{C}_1$. Consequently its exact accessible Cartesian localization exists and realizes it as a constant class by the first paragraph.

More generally at each finite stage the mod-p criterion uses component bijectivity, the finitely many lower homotopy-group isomorphisms, and the quasi-isomorphism of chains on the corresponding iterated based-loop space. All representatives and witnesses still have finite support. Induction therefore constructs the subsequent localized-space coefficient categories; spectra remain at $\mathcal{S}$. The universal property makes these choices unique up to equivalence for each specified class. It does not furnish an assignment on unspecified bare derivators or a functor on all their exact morphisms. In (I.9.1), the phrase "second stage" thus refers precisely to the second locally constant step, after the first class has been realized by the chosen localized-space derivator. □

J Strict two-categorical indexing: the ordinary-base case

J.1 (Proved). Let S be an ordinary small category, regarded as a locally discrete strict 2-category. On $2\,\mathrm{Cat}\,/S$ define the presheaf fine class by requiring $s\backslash u$ to be a Street weak equivalence for every $s \in S$. Its infinity-localization is equivalent to $\mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S})$. The dual over-slice class gives $\mathrm{Fun}(NS, \mathcal{S})$. These equivalences are compatible with the Street-nerve comparisons to the contravariant and covariant simplicial theories over NS.

Proof. Because the base has only identity 2-cells, its lax under-slice agrees with the strict 2-comma object. Define

$$G(X \xrightarrow{p} S)(s) = s\backslash X, \qquad G(X) : S^{\mathrm{op}} \longrightarrow 2\,\mathrm{Cat},$$

with strictly functorial reindexing by precomposition. For a strict diagram $A : S^{\mathrm{op}} \to 2\,\mathrm{Cat}$, define its 2-categorical Grothendieck construction $E(A)$ as follows. Objects are (s, x) with $x \in A(s)$; a 1-cell $(s, x) \to (t, y)$ is $(a : s \to t, h : x \to A(a)y)$; a 2-cell between such arrows exists only when their base arrows agree, and is the corresponding 2-cell between their h components. Strict composition in A gives strict composition in $E(A)$.

For every s there is a strict 2-functor

$$R_s : s\backslash E(A) \longrightarrow A(s), \quad ((t, y), a : s \to t) \longmapsto A(a)y.$$

Its section takes x to $((s, x), 1_s)$. The arrows $(a, 1)$ give a strict 2-natural transformation from section followed by R_s to the identity. Thus R_s is a Street weak equivalence; a strict 2-natural

transformation supplies a simplicial homotopy after the usual cylinder comparison. The R_s assemble naturally to $GE(A) \to A$.

There is also a natural strict 2-functor over S,

$$j_X : X \longrightarrow EG(X), \qquad x \longmapsto (p(x), (x, 1_{p(x)})).$$

At s, the composite $R_s(s\backslash j_X)$ is the identity on $s\backslash X$. For an object $((t, (x, \alpha : t \to p(x))), \beta : s \to t)$ of $s\backslash EG(X)$, the arrow with base α and identity component gives a natural transformation to its image under $(s\backslash j_X)R_s$. Consequently $s\backslash j_X$ is a Street weak equivalence. Notice that all displayed equalities are strict because the base is an ordinary category; this is the precise place where that hypothesis enters.

By definition G preserves and detects the fine class. The natural equivalence $GE \to 1$ proves that E sends pointwise Street weak equivalences to fine equivalences. Together with $j : 1 \to EG$, these are inverse equivalences after infinity-localization. The projective model structure on strict S^{op}-diagrams in the combinatorial Thomason model category of strict 2-categories presents $\mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S})$: apply the projective diagram Quillen equivalence induced by $c_2\,\mathrm{Sd}^2 \dashv \mathrm{Ex}^2\,N_2$ (Ara and Maltsiniotis, 2014, Theorem 6.27 and Corollary 6.32). The standard strict-diagram comparison is Lurie (2009, Proposition 4.2.4.4), applied on the simplicial-set side of that Quillen equivalence.

To identify the comparison functor, observe the strict identity

$$N_2(s\backslash X) = N(S_{s/}) \times_{NS} N_2X.$$

It follows because the Street nerve preserves limits and the base comma construction is the indicated strict pullback. In the contravariant straightening theory over an ordinary nerve, the value at s is the weak homotopy type of this pullback with the representable left fibration $N(S_{s/})$. This is the co-Yoneda calculation for straightening: on a simplex $\Delta^n \to NS$ the pullback has one contractible component for each arrow $s \to \sigma(n)$, and the general calculation follows by the simplicial resolution and homotopy colimits. Thus the Street nerve has the same straightened diagram as the equivalence already constructed. The over-slice case follows by total 2-categorical duality and invariance of Street weak equivalences under that duality (Ara and Maltsiniotis, 2022, Theorem 6.6). □

J.2 (Open). The proof of J.1 does not establish the requested recognition for the walking noninvertible or invertible 2-cell, the one-object generating-2-cell examples, or a general strict 2-category. For those bases the strict comma identity used above fails: lax triangles contain nonidentity 2-cells, and reindexing must retain their coherent action. No relative strictification or straightening theorem with all the required hypotheses has been proved here.

J.3 (Open). The universally pullback-preserving classes in Part J(2), the complete boundary-piece tests, and the proposed lax-comma Der 4 base-change theorem in J(3) are not established here. The absolute model structure and Theorem A do not imply those claims. The verified statements available from the sources are Theorem A for the specified slices (Ara and Maltsiniotis, 2018, Theorem 5.2) and invariance under every cell duality (Ara and Maltsiniotis, 2022, Theorem 6.6). Reversing only 2-cells exchanges lax and oplax coherence; reversing all positive-dimensional cells exchanges the matched under/over fine theories. A full four-variance base-change table requires the still-missing recognition and lax-comma compatibility.

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